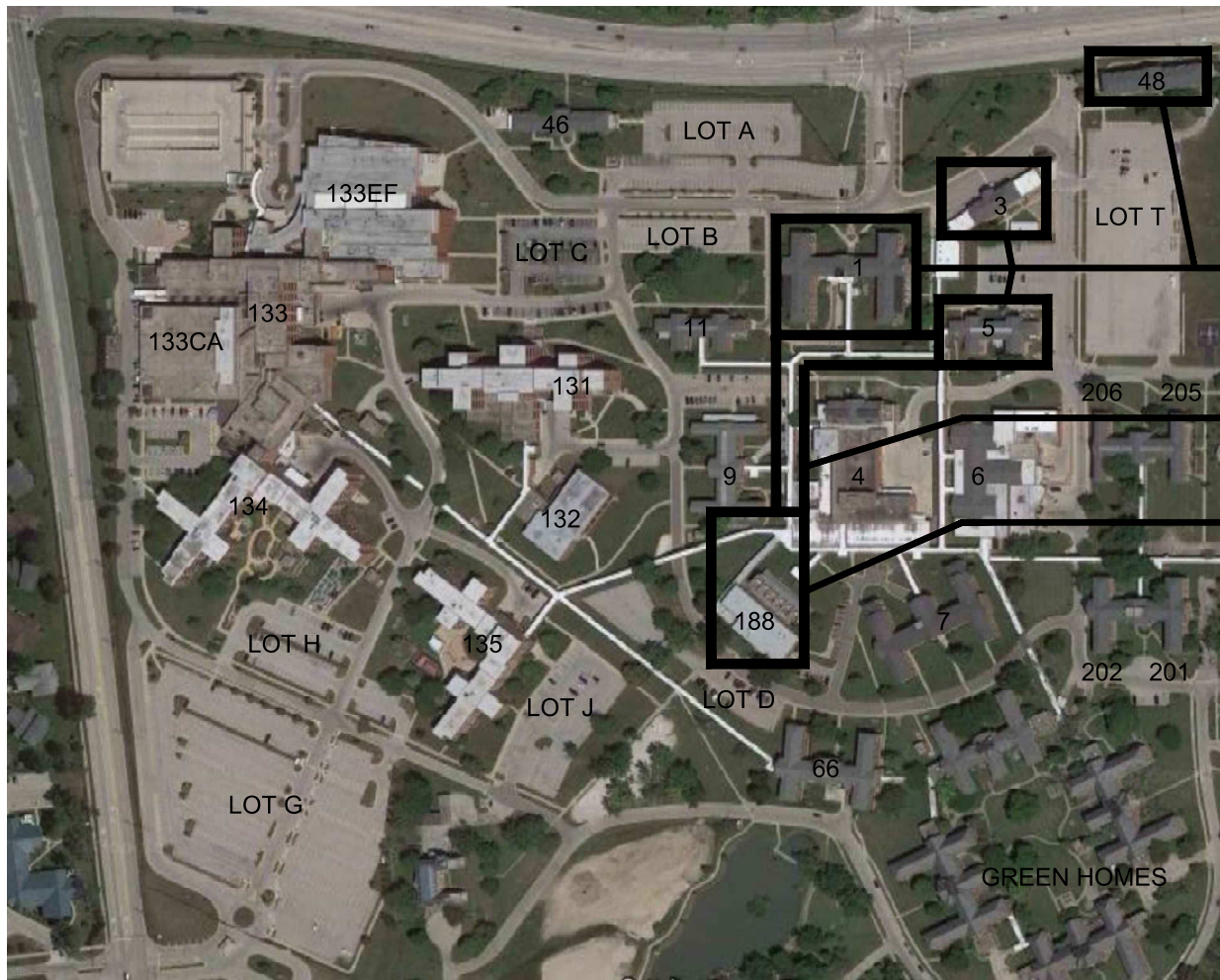


CHILLED WATER FROM B188 TO B3

CAPTAIN JAMES A. LOVELL FEDERAL HEALTH CARE CENTER (FHCC)

PROJECT 556-23-104 3001 GREEN BAY ROAD
NORTH CHICAGO, IL 60064

CAMPUS AERIAL PHOTO



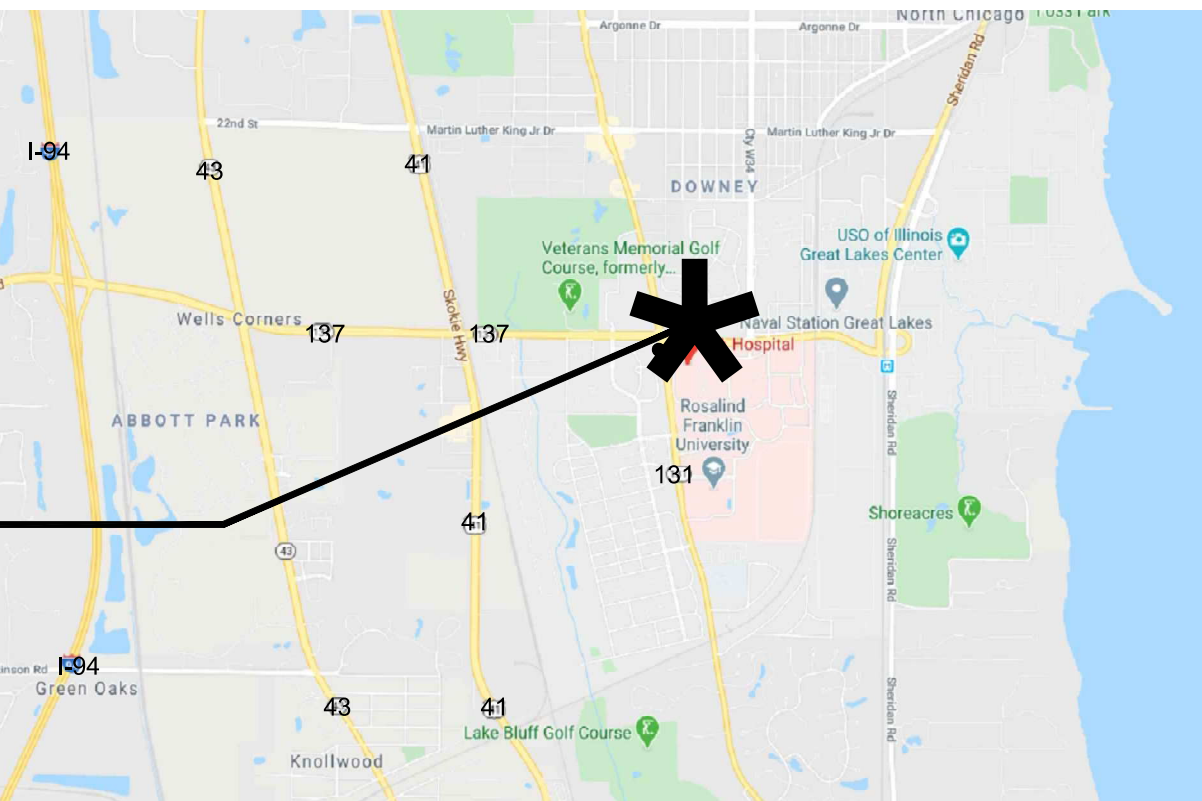
XX = BUILDING NUMBER

PROJECT AREA:
BUILDINGS 1, 3, 5 & 48
PROJECT AREA:
TRAMWAY TUNNELS
PROJECT AREA:
BUILDING 188

INDEX OF DRAWINGS

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CIVIL	
C101	BUILDING 3 AND 5 OVERALL CIVIL SITE PLAN
ARCHITECTURAL	
AS001	OVERALL ARCHITECTURAL SITE PLAN
AD401	ACCESS PIT DEMOLITION
AD402	DEMOLITION FOR EXIT CORRIDOR AT BUILDING 4
AE401	ACCESS PIT DETAILS
AE402	CORRIDOR PLAN AND SECTION
AE403	CORRIDOR SECTION BETWEEN BUILDINGS 188 AND 4
AE404	EXIT CORRIDOR PLAN AND SECTIONS AT BUILDING 4
AF101	FINISH PLAN AND DETAILS
MECHANICAL	
MP001	ABBREVIATIONS, LEGENDS, AND GENERAL NOTES
MS001	OVERALL MECHANICAL SITE PLAN
MP101	BUILDING 188 FLOOR PLAN
MP102	BUILDING 4 AND 9 FLOOR PLAN
MP103	BUILDING 3 AND 5 FLOOR PLAN
MP401	MECHANICAL SECTIONS AND SCHEDULES
MP501	MECHANICAL DETAILS
ELECTRICAL	
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ES001	OVERALL ELECTRICAL SITE PLAN
EP101	BUILDING 188 FLOOR PLAN
EP102	BUILDING 4 AND 9 FLOOR PLAN
EP103	BUILDING 3 AND 5 FLOOR PLAN
EP104	BUILDING 1 FLOOR PLAN
EE501	DETAILS

LOCATION MAP



PROJECT
SITE

PROJECT DESCRIPTION

THE SCOPE OF WORK IS:

- INSTALL NEW CHILLED WATER SUPPLY AND RETURN PIPING MAINS FROM BUILDING 188 TO BUILDING 3 WITH BRANCH PIPING SERVING BUILDINGS 1 AND 5.
- INSTALL NEW UNDERGROUND PIPING TO SERVE BUILDING 3 AND 48 (FOR FUTURE CONNECTION).

COORDINATION DRAWING REQUIREMENTS

THE CONTRACTOR SHALL PREPARE COORDINATION DRAWINGS. SUCH DRAWINGS SHALL BE ARRANGED AND DESIGNED THROUGH AND FROM COORDINATION WITH ALL OTHER MAJOR AND MINOR SUBCONTRACTORS. THE CONTRACTOR AT THEIR OWN EXPENSE SHALL PREPARE A THREE-DIMENSIONAL COORDINATION MODEL AND PRODUCE TWO-DIMENSIONAL COORDINATION DRAWINGS FROM MODEL, SPECIFICALLY CREATED BY THE CONTRACTOR FOR THIS PROJECT.

MODEL SHALL INCLUDE AND DRAWINGS SHALL SHOW, BUT NOT BE LIMITED TO: MECHANICAL EQUIPMENT; MECHANICAL PIPING; LIGHTING FIXTURES (WHEN CEILING MOUNTED); COLUMNS, BEAMS AND JOISTS (WITH ELEVATIONS MARKED); FIRE ALARM DEVICES; ELECTRICAL CONDUITS LARGER THAN 2-INCH DIAMETER; AND ELECTRICAL BUSWAY AND REQUIRED CLEARANCES. SECTIONS SHALL BE CUT THROUGH AREAS SHOWING MATERIALS AND SYSTEMS OF ALL CONTRACTS.

IF THERE ARE ANY COORDINATION ISSUES THAT CANNOT BE RESOLVED BY THE GC'S COORDINATION DRAWING PREPARER, THE GC SHALL SUBMIT RFI TO THE VA. ALL AREAS OF TRAMWAY UTILITY TUNNEL SHALL BE SHOWN INCLUDING STEAM, CONDENSATE PIPING, DRIP LEGS, CONDENSATE TRAPS, AND ELECTRICAL CONDUIT AND BOXES.

CONTRACTOR SHALL PERFORM CLASH DETECTION ON ALL MODELED ELEMENTS AND RESOLVE CLASHES AMONGST TRADES. CONTRACTOR SHALL OBTAIN SERVICES OF A GEOTECHNICAL CONSULTANT TO VERIFY SOIL BEARING IN UTILITY TUNNEL.

NOTE THAT ROUGH-IN / FABRICATION CANNOT BEGIN UNTIL COORDINATION MODEL AND DRAWINGS HAVE BEEN APPROVED. ANY MATERIAL, EQUIPMENT, SYSTEMS, ETC., PROCUREMENT OR INSTALLATION COMMENCED PRIOR TO APPROVAL IS TAKEN AT THE CONTRACTORS OWN RISK AND MAY HAVE TO BE MODIFIED, MOVED AND/OR RECONFIGURED AT NO INCREASE IN CONSTRUCTION TIME OR MONIES TO BE BURDENED BY THE GOVERNMENT. BOTH MODEL AND DRAWINGS ARE TO BE MADE A PART OF COORDINATION DRAWING SUBMISSION TO GOVERNMENT AND DESIGNER. ENSURE THAT ALL SUB-CONTRACTORS ARE RESPONSIBLE FOR COORDINATION ACROSS ENTIRE SET OF CONSTRUCTION DOCUMENTS.

SCHEDULE OF BID ITEMS

- ITEM 1: BASE BID: THE CONTRACTOR SHALL PROVIDE ALL NECESSARY TOOLS, LABOR, MATERIALS, EQUIPMENT, PERMITS, LICENSE, AND SUPERVISION FOR "AE DESIGN SERVICES FOR THE ADDITION OF CHILLED WATER PIPING FROM B188 TO B3," IN ACCORDANCE WITH THE DRAWINGS AND SPECIFICATION.
- ITEM 2: DEDUCT ALTERNATE NO.1: INCLUDES BASE BID MINUS ALL WORK ASSOCIATED WITH ROUTING 6" PIPING TO BASEMENT OF BUILDING 3, INCLUDING TEES, ISOLATION VALVES, BLIND FLANGES, BYPASS REDUCERS, BUILDING 3 PIPE SUPPORTS, CORE DRILLING OF PIPE TUNNEL/BUILDING 3 EXTERIOR WALL AND EQUIPMENT SLAB INSTALLATION. THE BRANCH PIPING SERVING BUILDING 5 SHALL REMAIN. IF DEDUCT IS TAKEN, INSTALL SHUT-OFF VALVES, BLANK-OFF PLATES AND BYPASS VALVE ASSEMBLY ON THE MAIN LINE. PROVIDE POWER AND CONTROL POINTS FOR BYPASS VALVE AND PRESSURE SENSOR.
- ITEM 3: DEDUCT ALTERNATE NO.2: INCLUDES DEDUCTIVE ALTERNATE NO.1 AND THE REMOVAL OF THE SUPPLY AND RETURN CHILLED WATER PIPING, SHUT-OFF VALVES AND BLANK-OFF PLATES SERVING BUILDING 5 AND 3 BACK TO THE CONNECTION POINT OF BUILDING 1. THE BRANCH PIPING SERVING BUILDING 1 SHALL REMAIN. IF DEDUCT IS TAKEN, INSTALL SHUT-OFF VALVES, BLANK-OFF PLATES AND BYPASS VALVE ASSEMBLY ON THE MAIN LINE. PROVIDE POWER AND CONTROL POINTS FOR BYPASS VALVE AND PRESSURE SENSOR.
- ITEM 4: DEDUCT ALTERNATE NO.3: SHALL INCLUDE DEDUCT ALTERNATE NO.2 MINUS THE INSTALLATION OF THE CHILLED WATER PIPING SERVING BUILDING 1 BACK TO THE POINT WHERE THE PIPING HAS A CHANGE IN DIRECTION FROM PLAN NORTH/SOUTH TO PLAN EAST/WEST. IF DEDUCT IS TAKEN, INSTALL SHUT-OFF VALVES, BLANK-OFF PLATES AND BYPASS VALVE ASSEMBLY ON THE MAIN LINE. PROVIDE POWER AND CONTROL POINTS FOR BYPASS VALVE AND PRESSURE SENSOR.

1	25% SD	05/25/2023
2	50% DD	07/07/2023
3	50% DD RESUBMISSION	08/29/2023
4	75% DD	05/17/2024
5	95% DD	06/18/2024
6	100% CD	07/26/2024
Revision Number	Revisions:	Date:

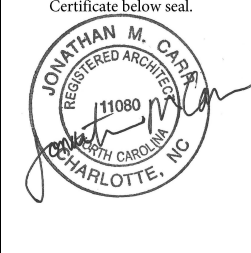
ARCHITECT/ENGINEER OF RECORD

A/E:
HARRELL DESIGN GROUP PC
10720 SIKES PLACE, SUITE 100
CHARLOTTE, NC 28227
704-814-1346
POC: LEE HARRELL PE, LEED AP, QCxP
PRESIDENT
HDG PROJECT: 22018



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OFFICE OF CONSTRUCTION AND FACILITIES MANAGEMENT

U.S. Department
of Veterans Affairs

DRAWING TITLE

COVER SHEET

APPROVED

100% CONSTRUCTION
DOCUMENTS - ISSUED
FOR CONSTRUCTION

FULLY SPRINKLERED

PROJECT TITLE

CHILLED WATER FROM B188 TO B3
CONTRACT NO. - 36C25219D0022
TASK ORDER NO. - 36C25223N0287

LOCATION

CAPTAIN JAMES A. LOVELL FEDERAL HEALTH CARE
CENTER (FHCC)

ISSUE DATE

JULY 26, 2024

CHECKED

JC

DRAWN

GG

PROJECT NUMBER

556-23-104

BUILDING NUMBER

DRAWING NUMBER

G1001

A

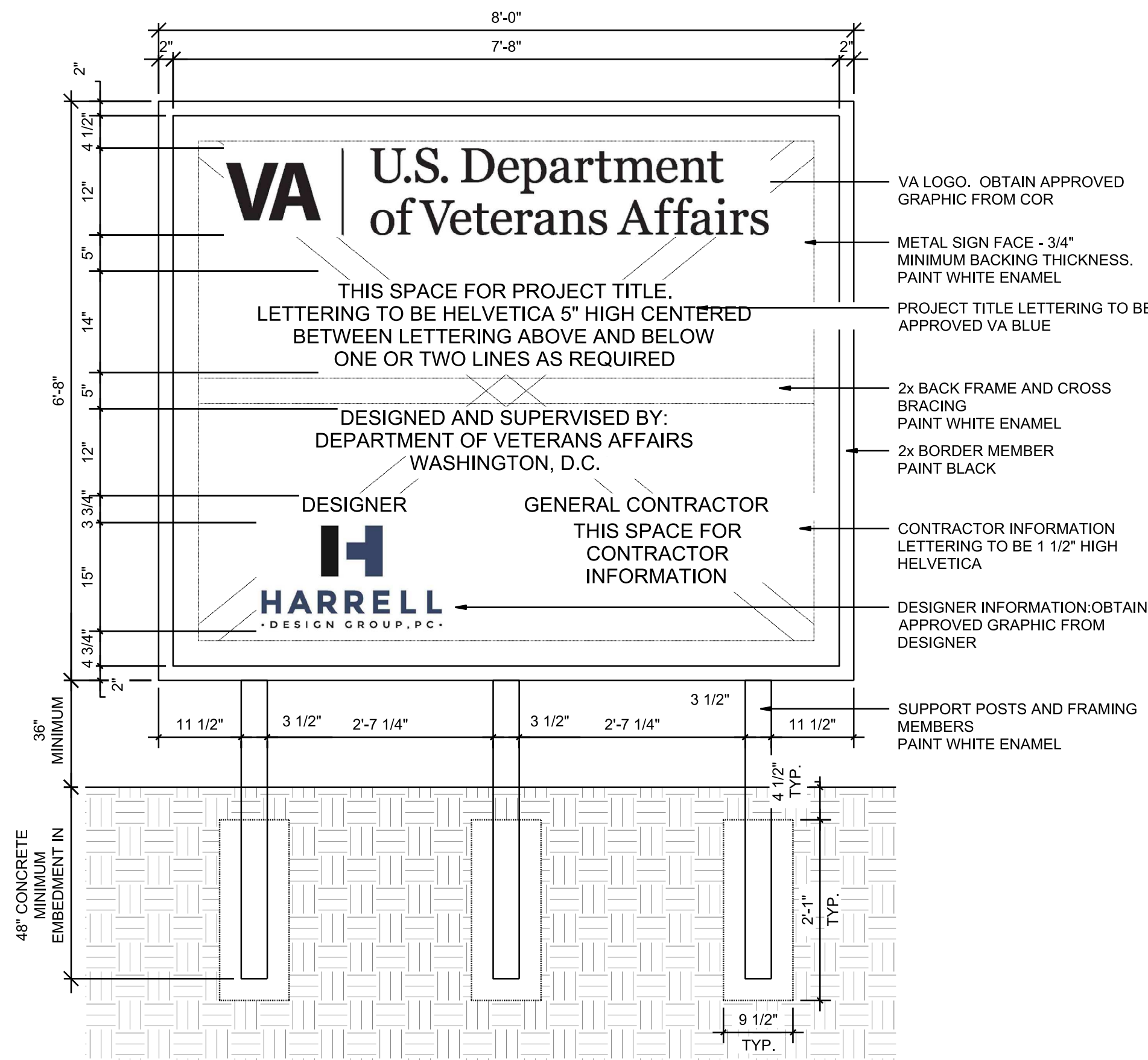
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C

D

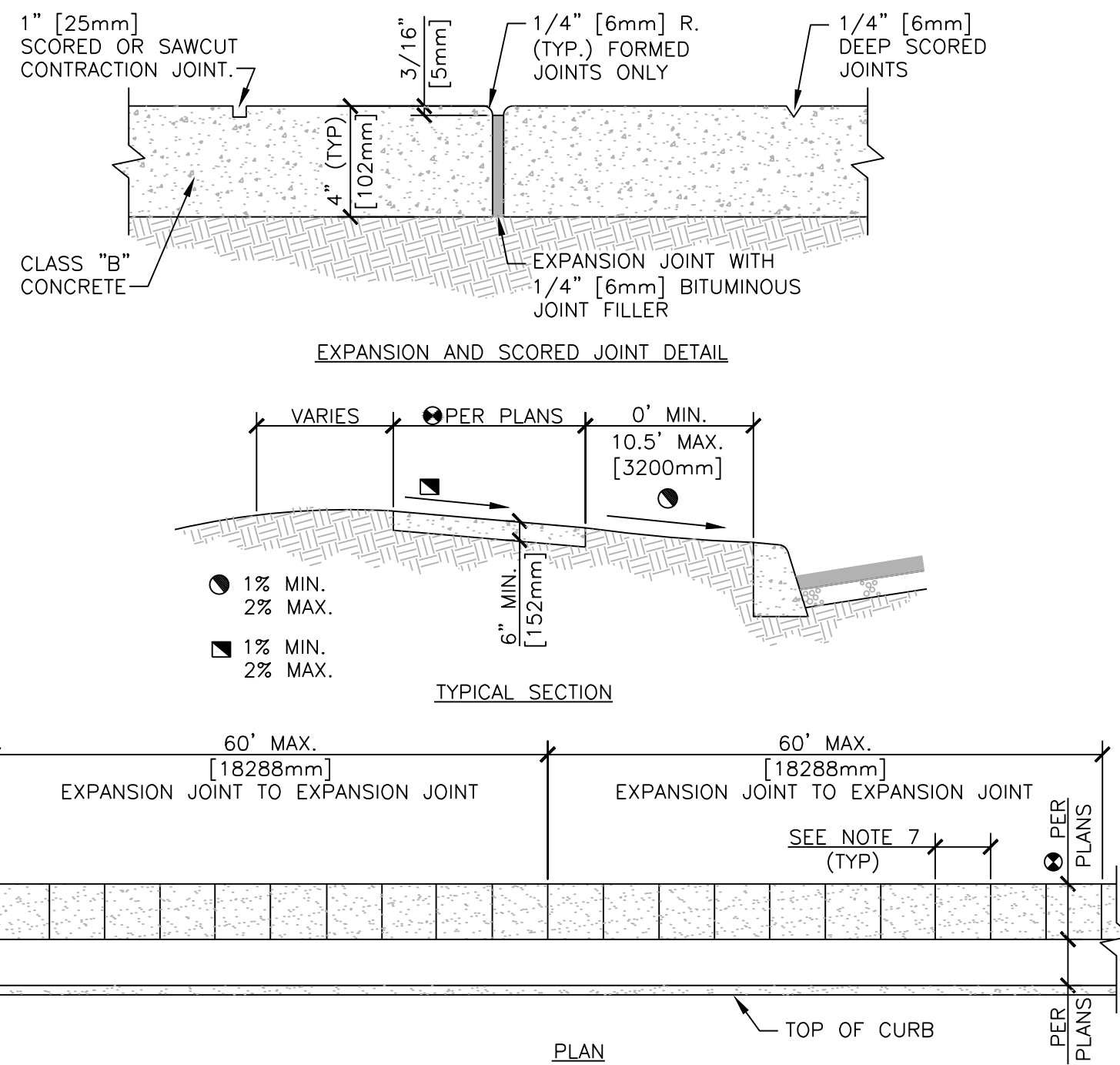
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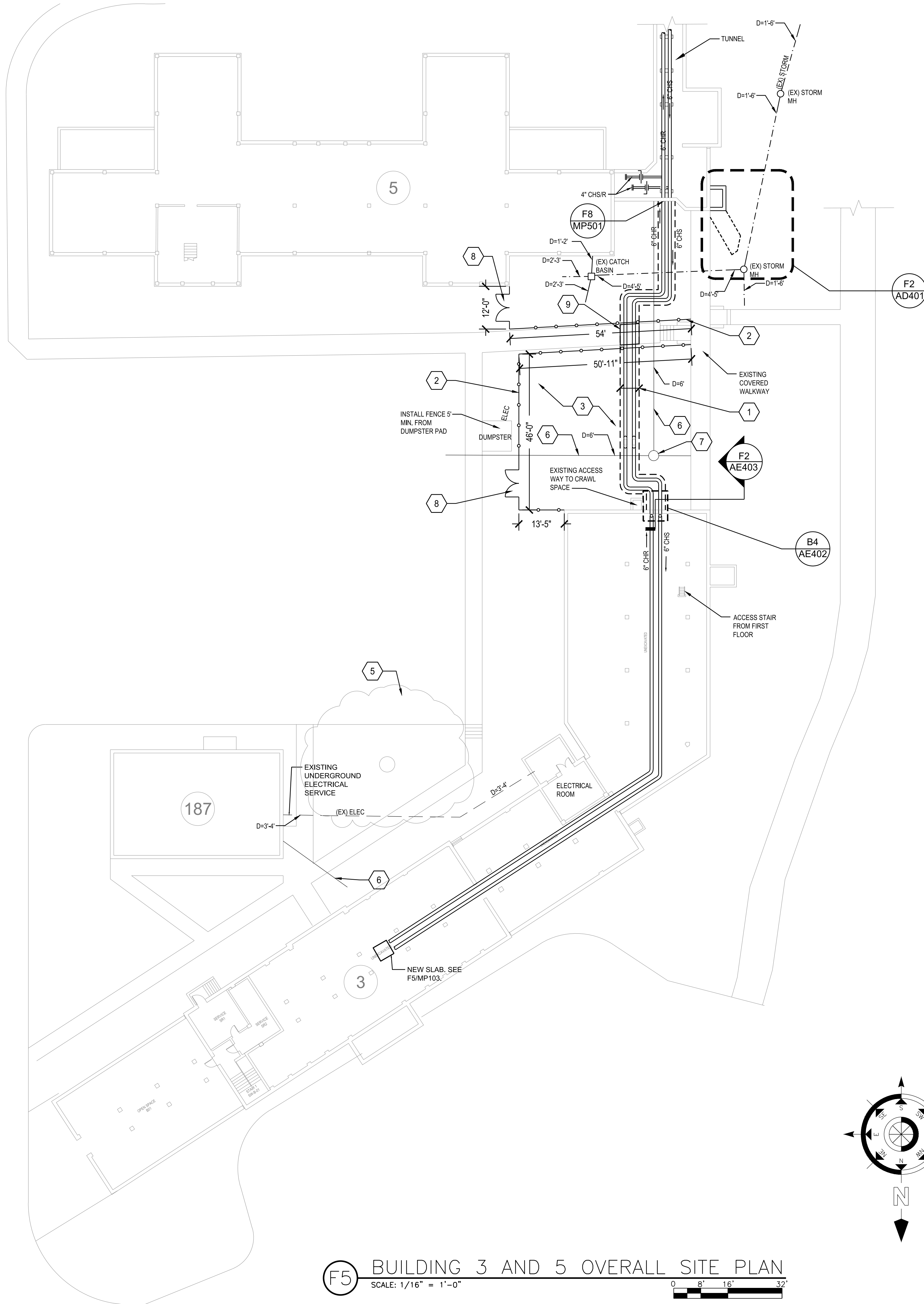
NOTE: LOCATE SIGN PER VA-COR

C1 PROJECT SIGN
SCALE: 3/4" = 1'-0"

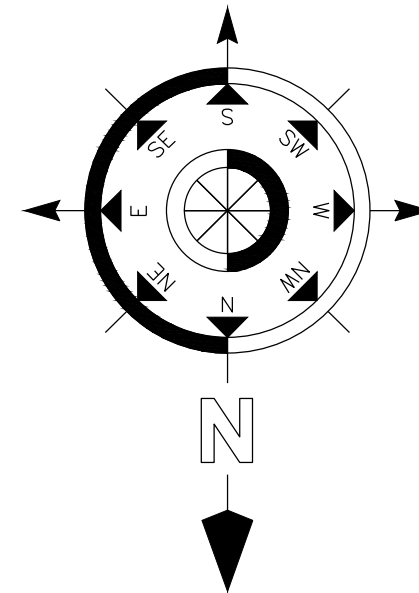


- NOTES:**
- EXPANSION JOINTS SHALL BE LOCATED WHERE SIDEWALK ABUTS CONCRETE DRIVEWAYS, CURB OR OTHER ADJACENT STRUCTURES.
 - ONE-HALF INCH BITUMINOUS JOINT FILLER SHALL BE INSTALLED AT EXPANSION JOINT LOCATIONS AND SHALL EXTEND THE FULL DEPTH OF THE CONCRETE.
 - 1" DEEP CONTRACTION JOINTS SHALL BE PLACED AT INTERVALS OF APPROXIMATELY 15' [4572mm] OR AT A SPACING THAT MATCHES THE ADJACENT CURB.
 - FORMED CONTRACTION JOINTS SHALL BE FINISHED WITH A TOOL HAVING A 1/4" [6mm] RADIUS.
 - SCORED JOINTS SHALL BE 1/4" [6mm] DEEP AND PLACED AT THE SPACING INDICATED FOR THE WIDTH OF SIDEWALK OR MATCH SCORED JOINTS OF ADJACENT CURB.
 - CONCRETE SHALL BE FINISHED BY MEANS OF A FLOAT, STEEL TROWELLED AND BROOMED WITH A FINE BRUSH IN A TRANSVERSE DIRECTION.
 - 1/4" DEEP SCORED JOINTS (TYP) SPACED AT 6' [1829mm] OR EQUAL TO SIDEWALK WIDTH.

F1 CONCRETE SIDEWALK DETAILS
SCALE: NONE



F5 BUILDING 3 AND 5 OVERALL SITE PLAN
SCALE: 1/16" = 1'-0"



GENERAL SHEET NOTES

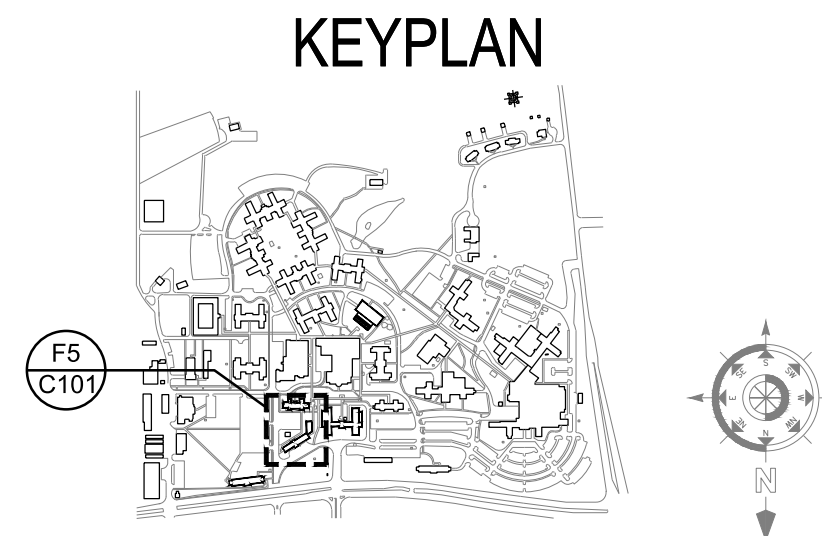
- NEW CHILLED WATER PIPING SHALL BE ROUTED WITHIN PIPING TUNNEL, UNLESS NOTED OTHERWISE.
- SOD AREA OF EXCAVATION FOR PIPE TRENCH AFTER IT IS BACKFILLED COMPACTED AND GRADED.
- PROVIDE BACKFILL AFTER COMPLETION OF WATERPROOFING REPAIRS TO EXISTING FOUNDATION WALLS. SEE SPECIFICATIONS FOR BACKFILL AND WATERPROOFING REQUIREMENTS.
- PROTECT EXISTING LIGHTNING PROTECTION GROUND RINGS WHERE ENCOUNTERED.

SHEET KEYNOTES

- ALL SOIL REMOVAL WITHIN DEFINED AREA OF 5 FEET WIDE SHALL BE PERFORM BY HAND EXCAVATION IN THIS AREA.
- CONSTRUCTION FENCING
- AFTER REMOVAL OF CONSTRUCTION MATERIALS, PROVIDE SOD IN FENCED AREAS.
- SOD PER SPECIFICATION 32 90 00.
- PROTECT EXISTING LARGE TREE FROM CONSTRUCTION OPERATIONS.
- EXISTING EHRM UNDERGROUND CONDUIT.
- EXISTING EHRM MANHOLE.
- PAIR 4'-0" TEMPORARY CONSTRUCTION GATES.
- REMOVE PORTION OF EXISTING CONCRETE SIDEWALK AND REPLACE WITH NEW COMPACTED SOIL AND SIDEWALK AFTER COMPLETION OF PIPE INSTALLATION AND BACKFILL.

SHEET LEGEND

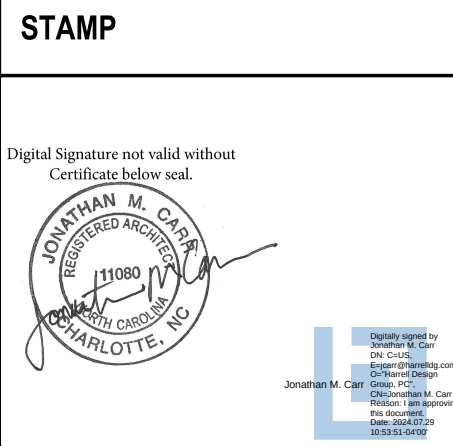
- D=2'-3" DEPTH IN GROUND
--- AREA OF EXCAVATION
--- EXISTING UNDERGROUND STORM WATER PIPE
--- EXISTING UNDERGROUND ELECTRICAL CONDUIT
--- TEMPORARY CONSTRUCTION FENCE
○ EXISTING LIGHT POLE



1	50% DD RESUBMISSION	08/29/2023
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ARCHITECT/ENGINEER OF RECORD
A/E:
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10720 SIKES PLACE, SUITE 100
CHARLOTTE, NC 28227
704-814-1346
POC: LEE HARRELL PE, LEED AP, QCxP
PRESIDENT
HDG PROJECT: 22018

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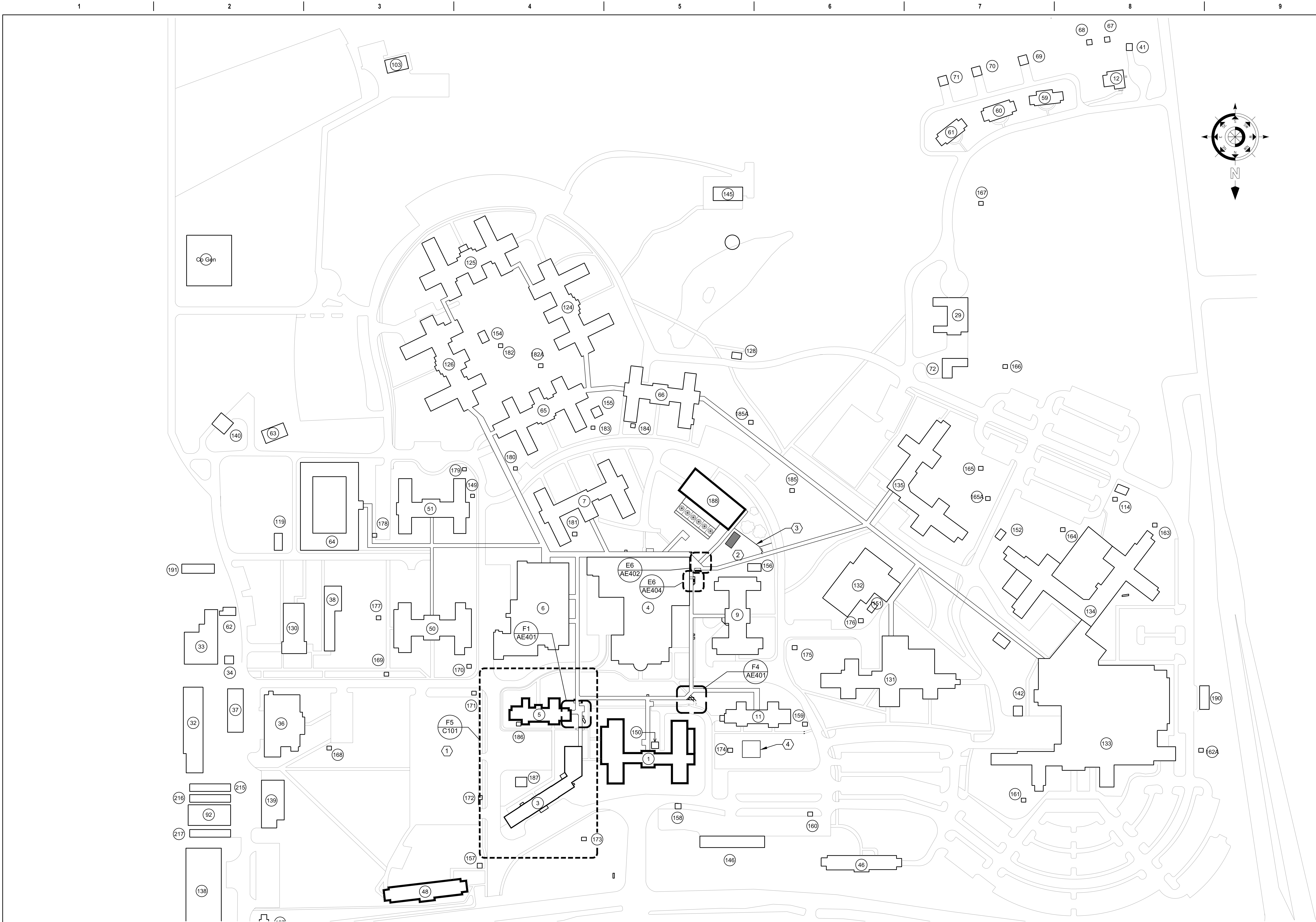
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MANAGEMENT
U.S. Department
of Veterans Affairs

DRAWING TITLE
BUILDING 3 AND 5 OVERALL CIVIL
SITE PLAN
APPROVED

PHASE
100% CONSTRUCTION
DOCUMENTS - ISSUED
FOR CONSTRUCTION
FULLY SPRINKLERED

PROJECT TITLE
CHILLED WATER FROM B188 TO B3
CONTRACT NO. - 36C25219D0022
TASK ORDER NO. - 36C25223N0287
LOCATION
CAPTAIN JAMES A. LOVELL FEDERAL HEALTH CARE
CENTER (FHCC)
ISSUE DATE
JULY 26, 2024
CHECKED
LEH
DRAWN
GG

PROJECT NUMBER
556-23-104
BUILDING NUMBER
3, 5
DRAWING NUMBER
C101



GENERAL SHEET NOTES

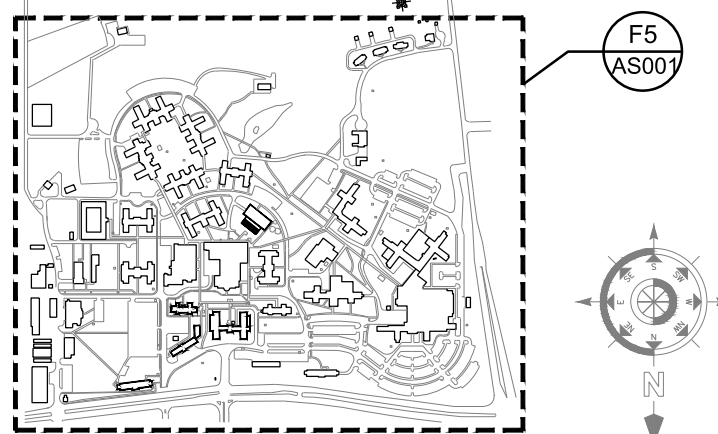
A. CONTRACTOR TO SOD AREAS AFFECTED BY CONSTRUCTION ACTIVITIES. SOD LINE IS DEMARCATED AS 4'-0" BEYOND FENCE LINES.

SHEET KEYNOTES ①

1. CONTRACTOR PARKING
2. CONTRACTOR CONSTRUCTION TRAILER AND FENCED LAYDOWN AREA, SOD AREA UPON COMPLETION.
3. CONSTRUCTION FENCE AND GATE
4. PROPOSED LAY-DOWN AREA

XXX BUILDING NUMBER

KEYPLAN



F5 OVERALL SITE PLAN
SCALE: 1" = 100'
0 50' 100' 200'

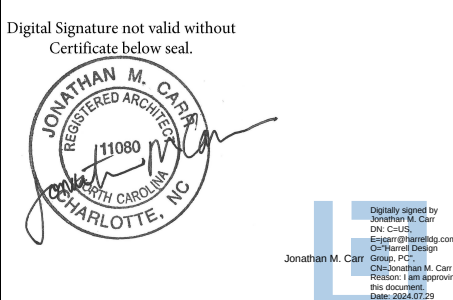
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PRESIDENT
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DRAWING TITLE

OVERALL ARCHITECTURAL
SITE PLAN

APPROVED

PHASE

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DOCUMENTS - ISSUED
FOR CONSTRUCTION

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PROJECT TITLE

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CENTER (FHCC)

ISSUE DATE
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GG

PROJECT NUMBER

556-23-104
BUILDING NUMBER

DRAWING NUMBER

AS001

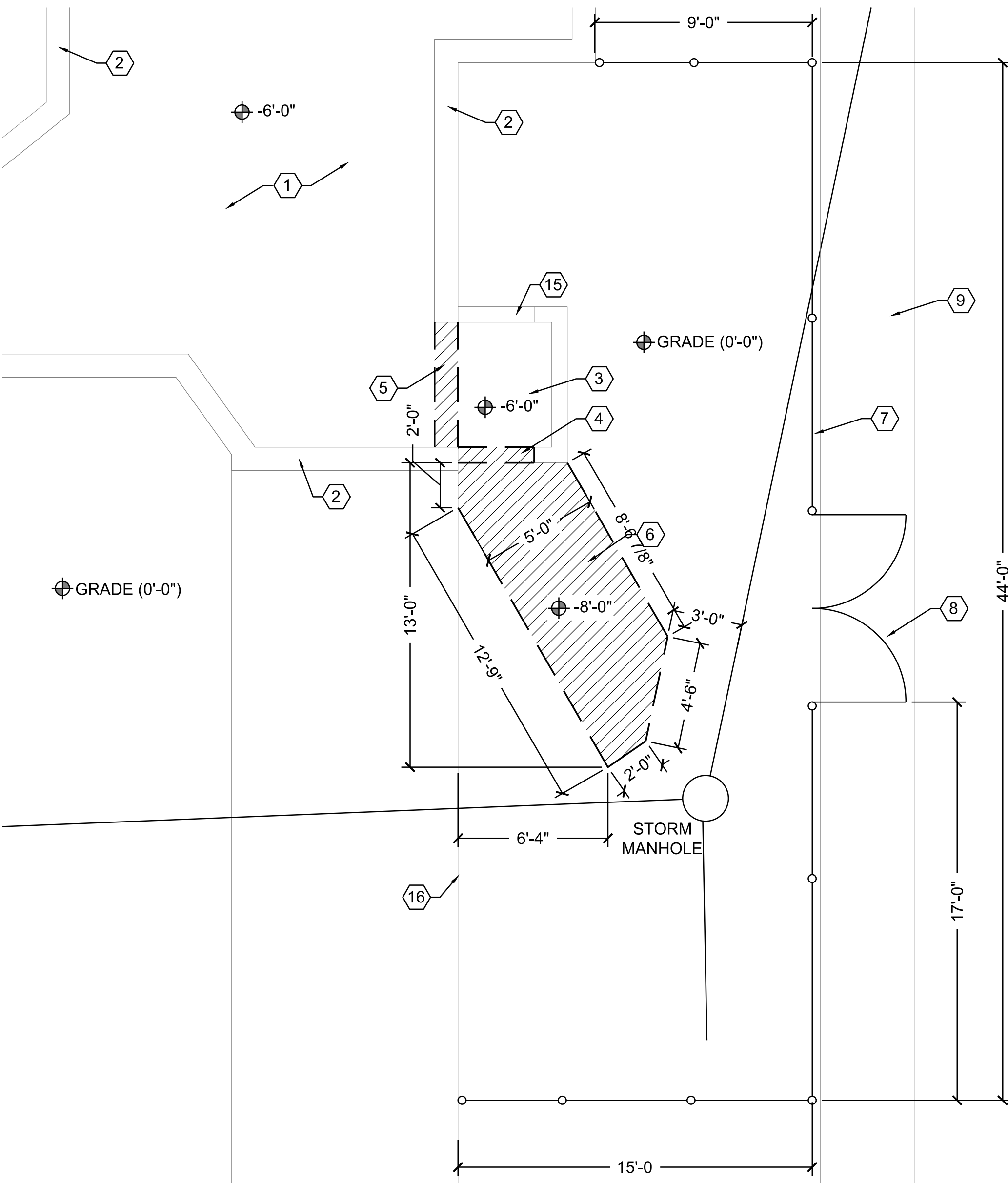
GENERAL SHEET NOTES

- A. ALL PIPE FEEDING WORK TO OCCUR OVER 5 CONSECUTIVE CALENDAR DAYS, MAXIMUM.
B. FENCE IS 4'-0" FROM EXCAVATION.
C. PROVIDE THIRD-PARTY ENGINEERED SHORING FOR INTERIOR AND EXTERIOR EXCAVATIONS (TYPICAL).
D. CONTRACTOR TO PROVIDE INFECTION CONTROL CLASS 3 FOR CORRIDOR 101.
E. FOR ALL INTERIOR EXCAVATIONS, CONTRACTOR TO HAND-DIG AND NOTE ANY UTILITY OBSTRUCTIONS.

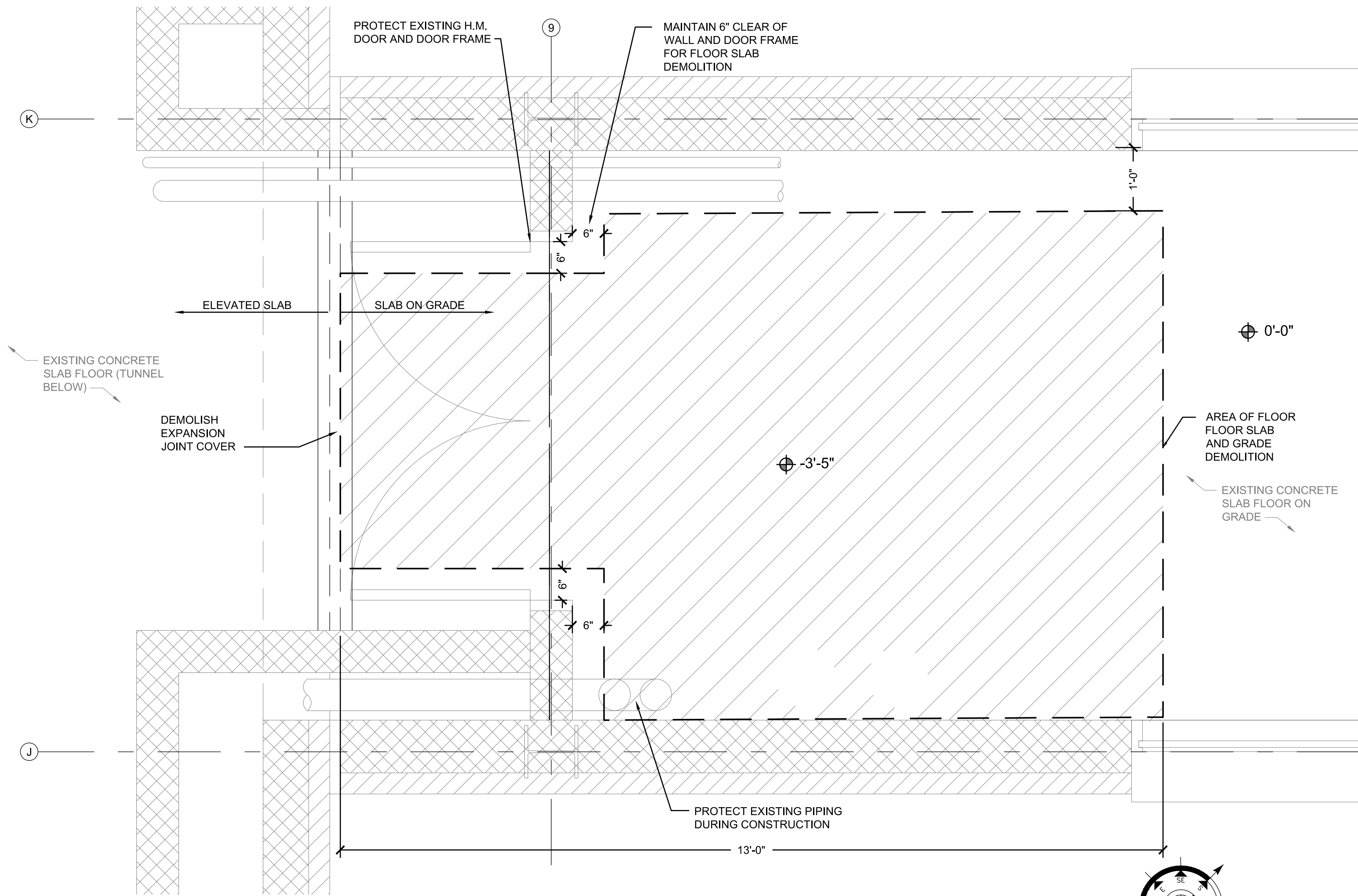
SHEET KEYNOTES ①

1. EXISTING TRAMWAY TUNNEL
2. EXISTING TRAMWAY TUNNEL WALL
3. EXISTING TUNNEL ACCESS AREAWAY
4. EXISTING BRICK KNOCKOUT PANELS IN ACCESS PIT TO BE REMOVED.
5. DEMO EXISTING ACCESS DOORS AND FRAMES IN TUNNEL WALL. REPLACE WITH NEW TREATED FRAME AND WOOD BATTEN DOORS.
6. EXCAVATE EXISTING GRADE TO BOTTOM OF TUNNEL ACCESS AREAWAY. SHORE TRENCH.
7. TEMPORARY CONSTRUCTION FENCE ON GRADE.
8. PAIR 4'-0" GATES ON GRADE
9. EXISTING SIDEWALK ON GRADE
10. EXISTING MANHOLE
11. EXISTING PAVEMENT ON GRADE
12. TRAMWAY DOORS ABOVE AT GRADE
13. DEMOLISH EXISTING SIDEWALK ON GRADE
14. EXISTING TRAMWAY ABOVE
15. EXISTING BRICK KNOCKOUT TO REMAIN
16. BELOW GRADE WATERPROOFING AT ALL EXPOSED CONCRETE AND NEW BRICK INFILL (TYPICAL)
17. EXISTING FIRE HYDRANT
18. EXISTING WATER VALVE
19. REMOVE EXISTING SHRUBS AND REPLACE AFTER CONSTRUCTION IS COMPLETE.

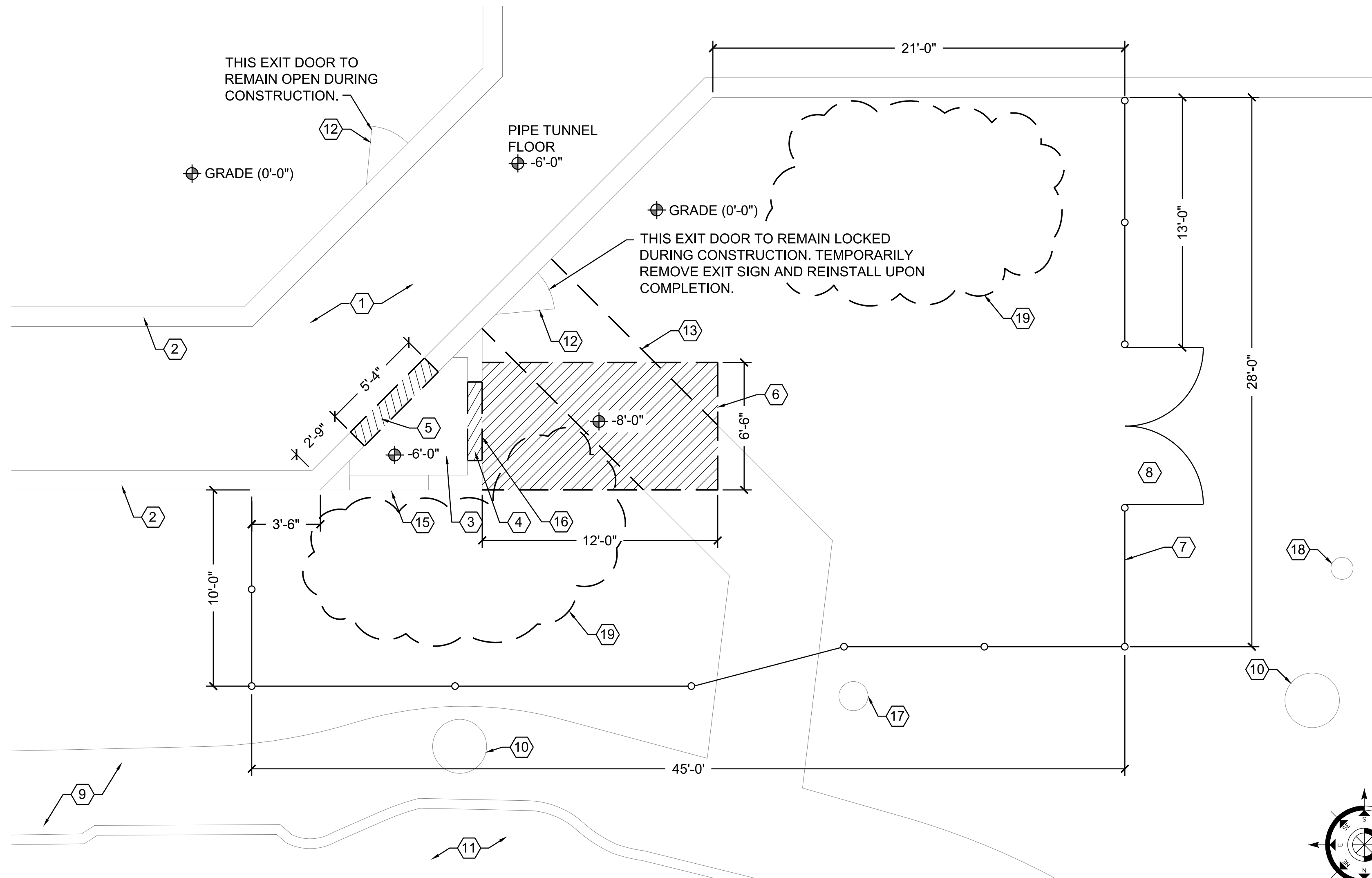
A
B
C
D
E
F



F3 ENLARGED DEMOLITION PLAN
SCALE: 1/4" = 1'-0"



C2 ENLARGED DEMOLITION PLAN
SCALE: 3/4" = 1'-0"

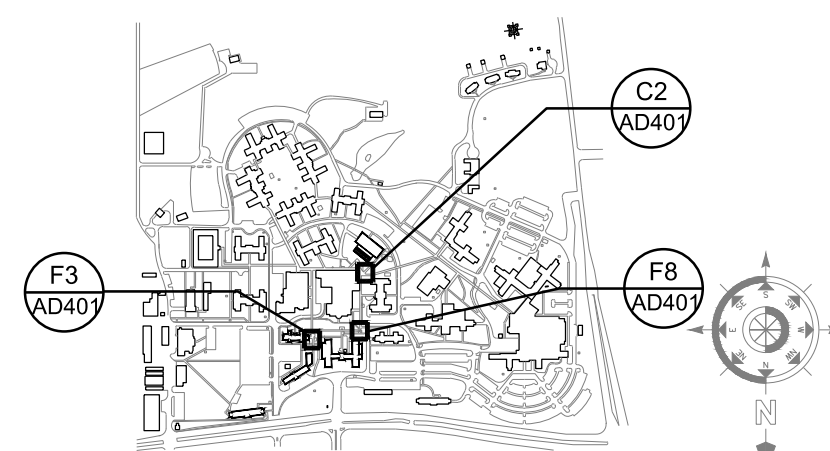


F8 ENLARGED DEMOLITION PLAN
SCALE: 1/4" = 1'-0"

WALL RATING LEGEND

--- 2-HOUR RATED PARTITION

KEYPLAN



1	75% DD	05/17/2024
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CENTER (FHCC)
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U.S. Department
of Veterans Affairs

DRAWING TITLE
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APPROVED

PHASE
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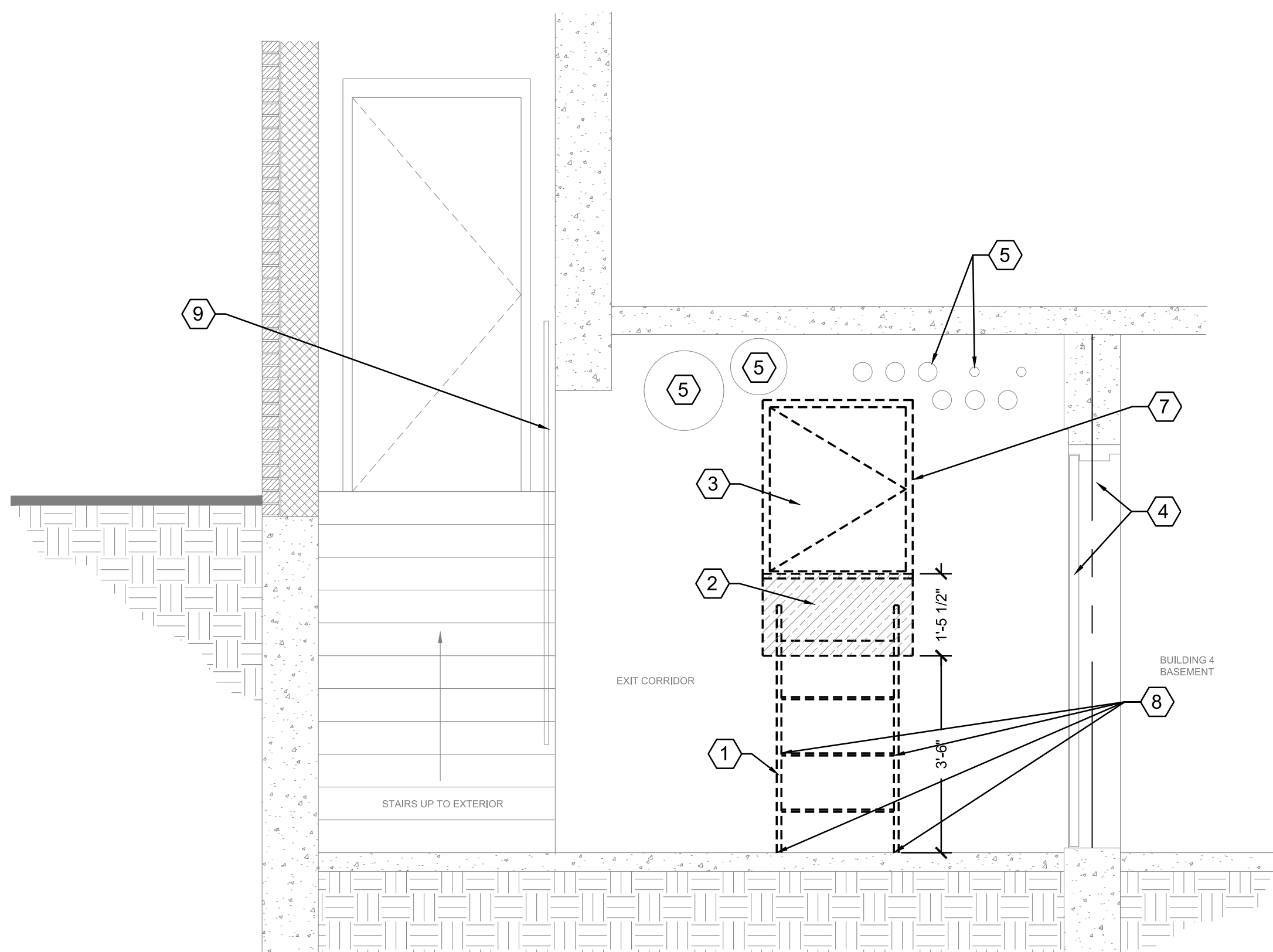
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JC
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GG

PROJECT NUMBER
556-23-104
BUILDING NUMBER
X
DRAWING NUMBER
AD401

SHEET KEYNOTES

- DEMOLISH WALL-MOUNTED LADDER
 SAW CUT AND REMOVE PORTION OF WALL FOR INSTALLATION OF
 NEW ACCESS HATCH. REPAIR DAMAGES TO CONCRETE WALL, INFILL
 GAPS, AND REPAIR WALLS.
 DEMOLISH EXISTING ACCESS HATCH AND FRAME
 PROTECT EXISTING H.M. DOOR AND FRAME
 PROTECT EXISTING INSULATED PIPE AND CONDUIT
 REMOVE EXISTING PAINT AND PREP SURFACES TO REPAINT ALL
 WALLS, DOORS, AND DOOR FRAMES IN THIS ROOM, IN COLORS TO
 MATCH EXISTING.
 PATCH AND PAINT CONCRETE UPON REMOVAL OF ANCHOR BOLTS
 PATCH AND PAINT FLOOR AND WALL UPON REMOVAL OF ANCHOR
 BOLTS.
 EXISTING HANDRAILS TO REMAIN.

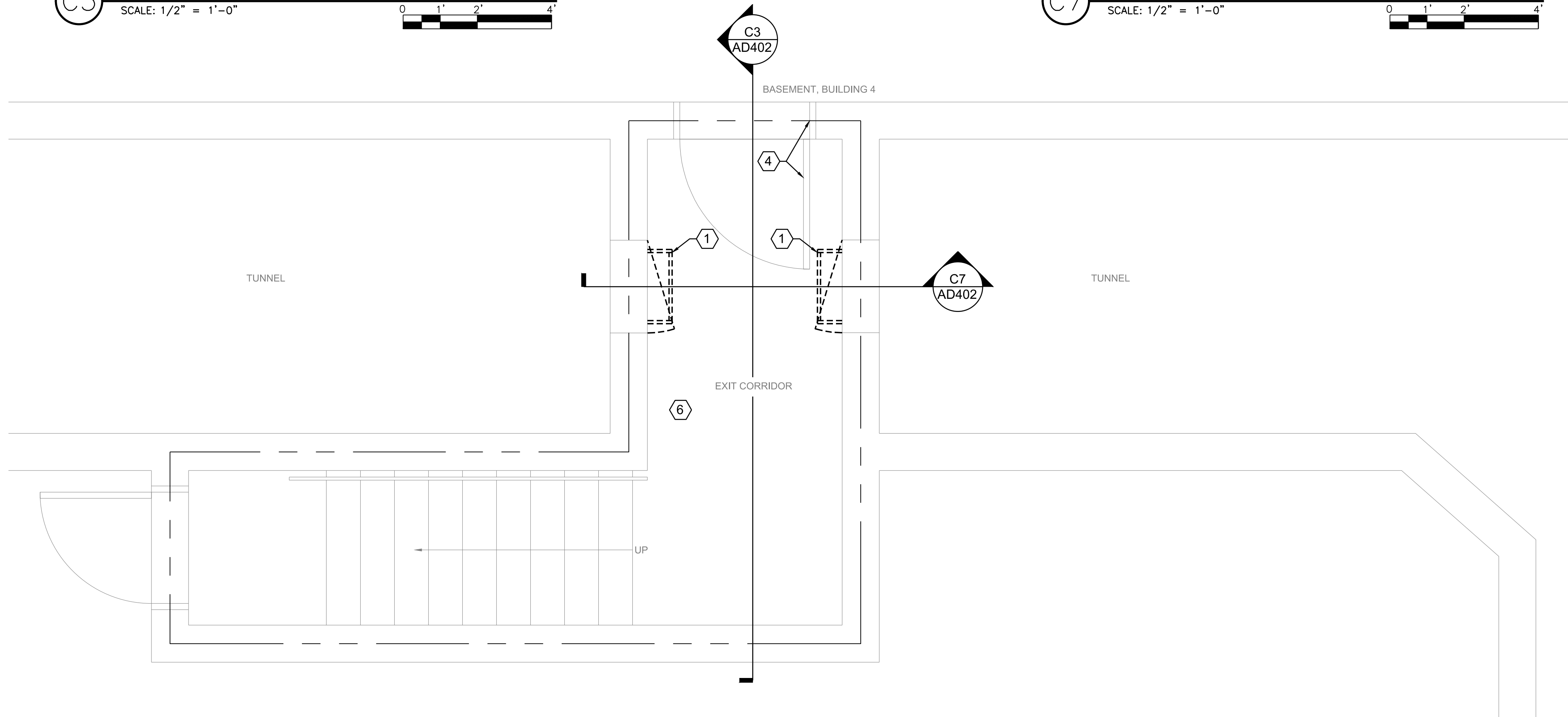
 AREA OF DEMOLITION



DEMOLITION SECTION FOR EXIT
CORRIDOR AT BUILDING 4

(C7) SCALE: 1/2" = 1'-0"

0 1' 2' 4'



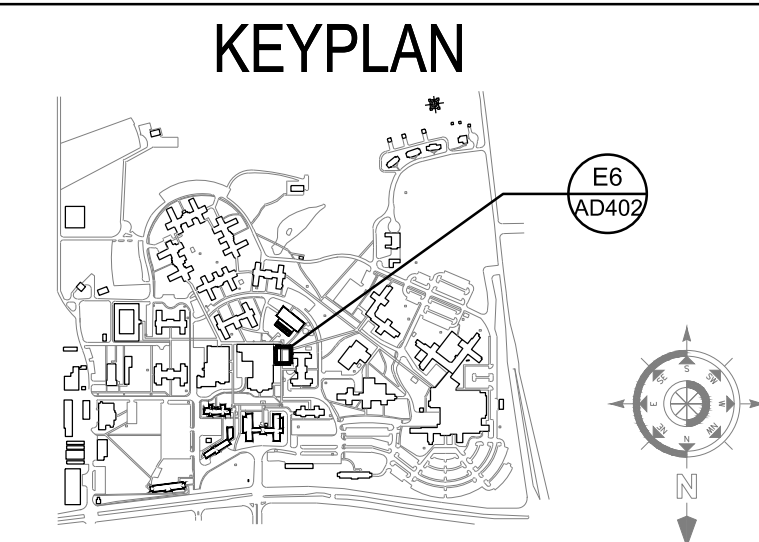
DEMOLITION PLAN FOR EXIT CORRIDOR
CORRIDOR AT BUILDING 4

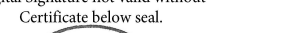
E6

SCALE: 1/2" = 1'-0"

0 1' 2' 4'

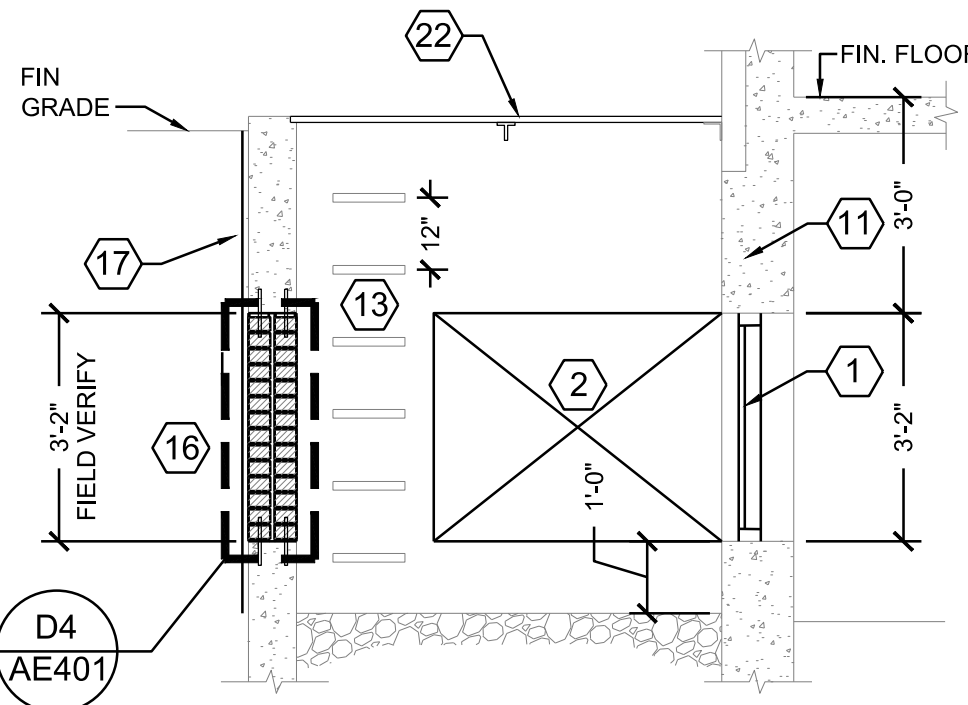
————— 2-HOUR RATED PARTITION



ARCHITECT/ENGINEER OF RECORD		STAMP		OFFICE OF CONSTRUCTION AND FACILITIES MANAGEMENT		DRAWING TITLE DEMOLITION FOR EXIT CORRIDOR AT BUILDING 4		PHASE 100% CONSTRUCTION DOCUMENTS - ISSUED FOR CONSTRUCTION		PROJECT TITLE CHILLED WATER FROM B188 TO B3 CONTRACT NO. - 36C25219D0022 TASK ORDER NO. - 36C25223N0287		PROJECT NUMBER 556-23-104	
A/E: HARRELL DESIGN GROUP PC 10720 SIKES PLACE, SUITE 100 CHARLOTTE, NC 28227 704-814-1346 POC: LEE HARRELL P.E. LEED AP, QCP PRESIDENT HDG PROJECT: 22018		 HARRELL DESIGN GROUP, P.C. DIGITAL SIGNATURE NOT VALID WITHOUT "CERTIFICATE" AND "PUBLIC KEY"  				APPROVED		FULLY SPRINKLERED		LOCATION CAPTAIN JAMES A. LOVELL FEDERAL HEALTH CARE CENTER (FHCC)		DRAWING NUMBER 4	
				 U.S. Department of Veterans Affairs						ISSUE DATE JULY 26, 2024		CHECKED JC	
										DRAWN GG		AD402	

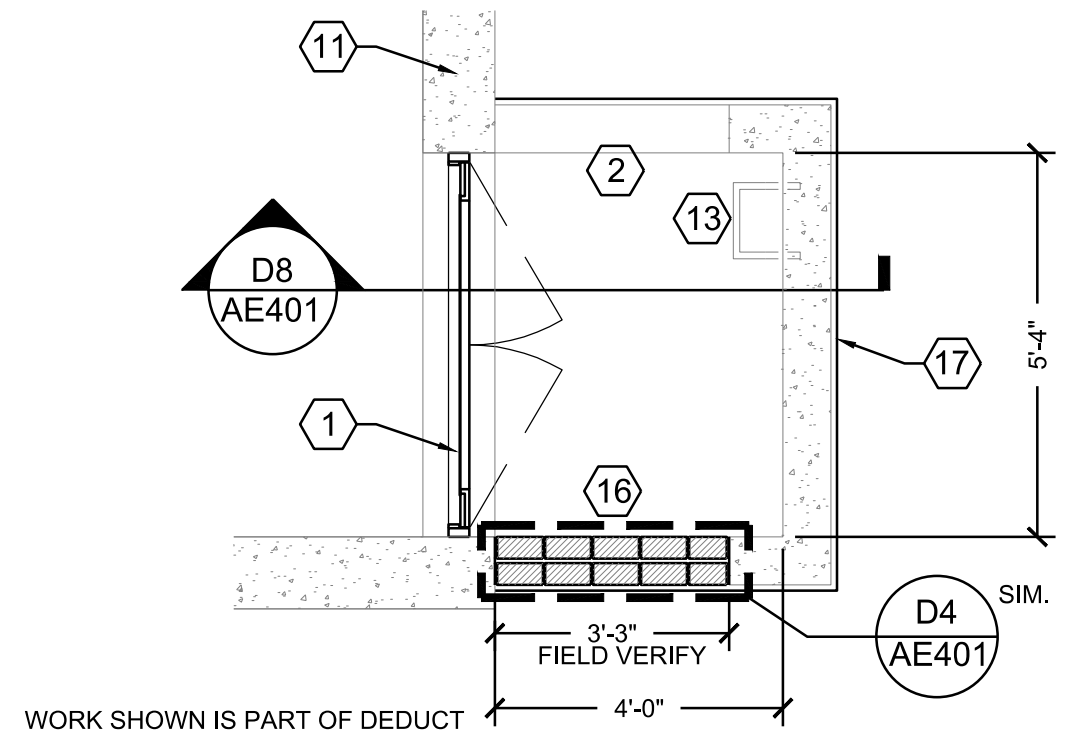
SHEET KEYNOTES

- ### KEYPLAN
-

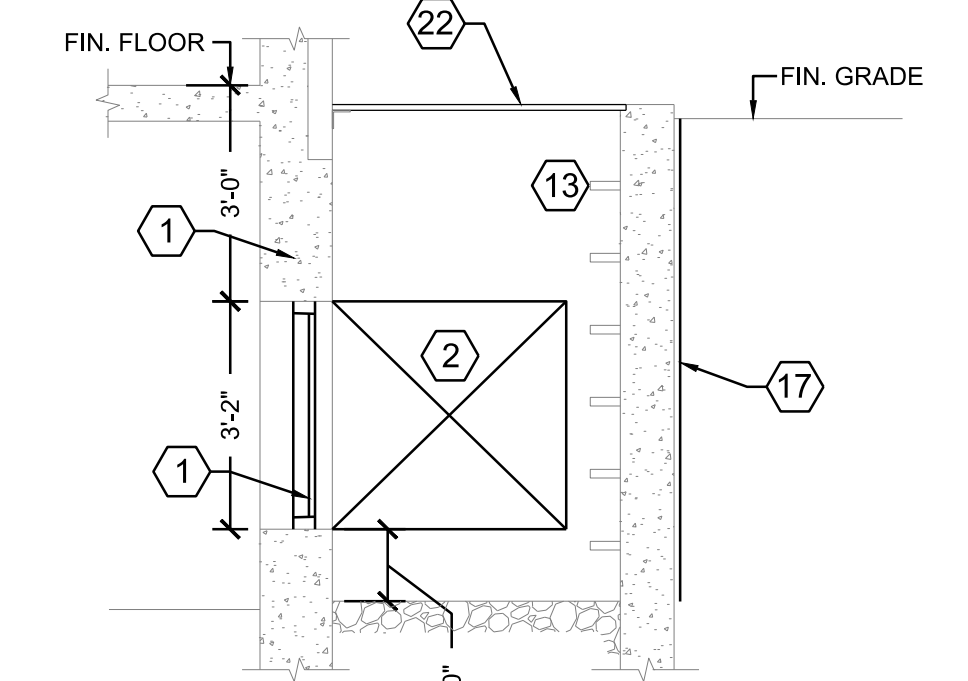


B8 SECTION

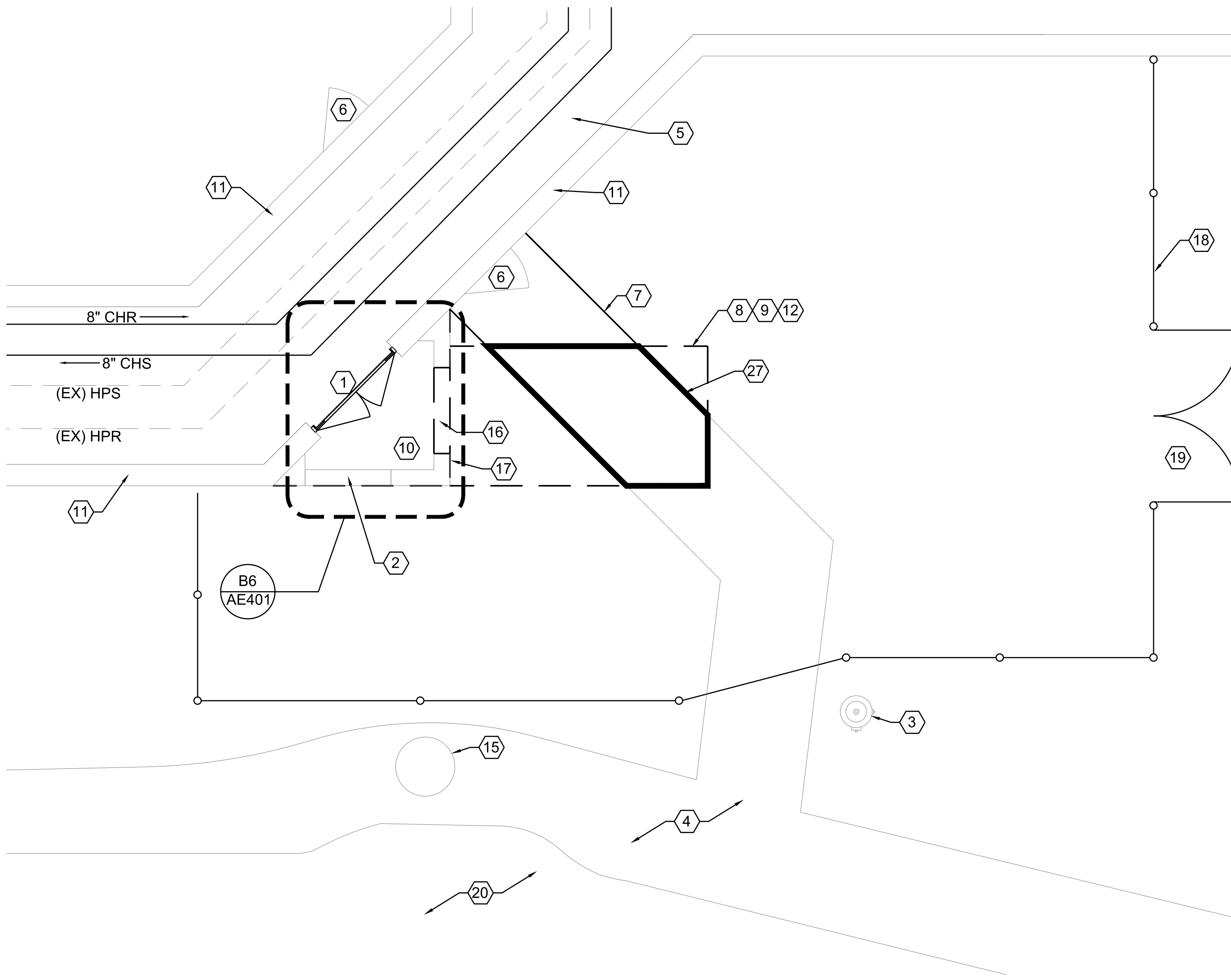
SCALE: $\frac{3}{8}" = 1'-0"$



D6 ENLARGED PLAN
SCALE: 3/8" = 1'-0" 0 2' 4' 6'



D8 SECTION
SCALE: 3/8" = 1'-0"



ENLARGED SITE PLAN
PIPE FEED LOCATION

F4

SCALE: 1/4" = 1'-0"

0 2' 4'

ARCHITECT/ENGINEER OF RECORD		STAMP	
A/E: HARRELL DESIGN GROUP PC 10720 SIKES PLACE, SUITE 100 CHARLOTTE, NC 28227 704-814-1346 POC: LEE HARRELL PE, LEED AP, QCP* PRESIDENT HDG PROJECT: 22018		 HARRELL DESIGN GROUP, P.C.  	
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 U.S. Department
of Veterans Affairs

APPROVED

FULLY SPRINKLERED

LOCATION
CAPTAIN JAMES A. LOVELL FEDERAL HEALTH CARE
CENTER (FUSC)

ISSUE DATE	CHECKED	DRAWN
JULY 26, 2024	JC	GG

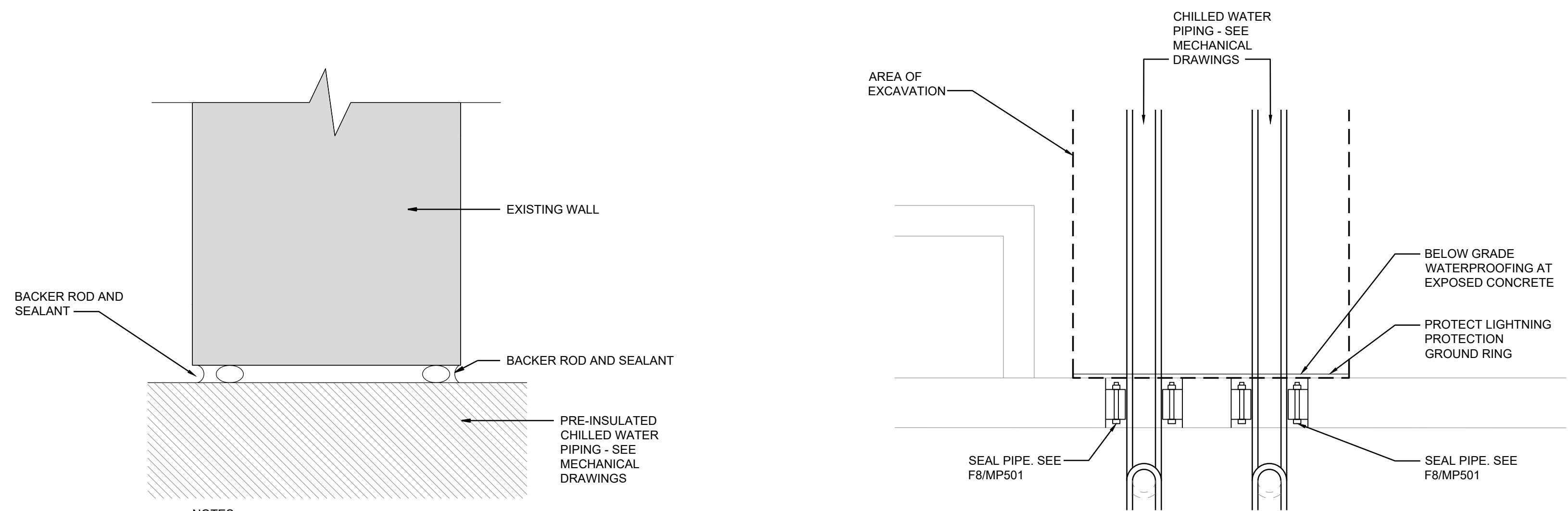
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GENERAL SHEET NOTES

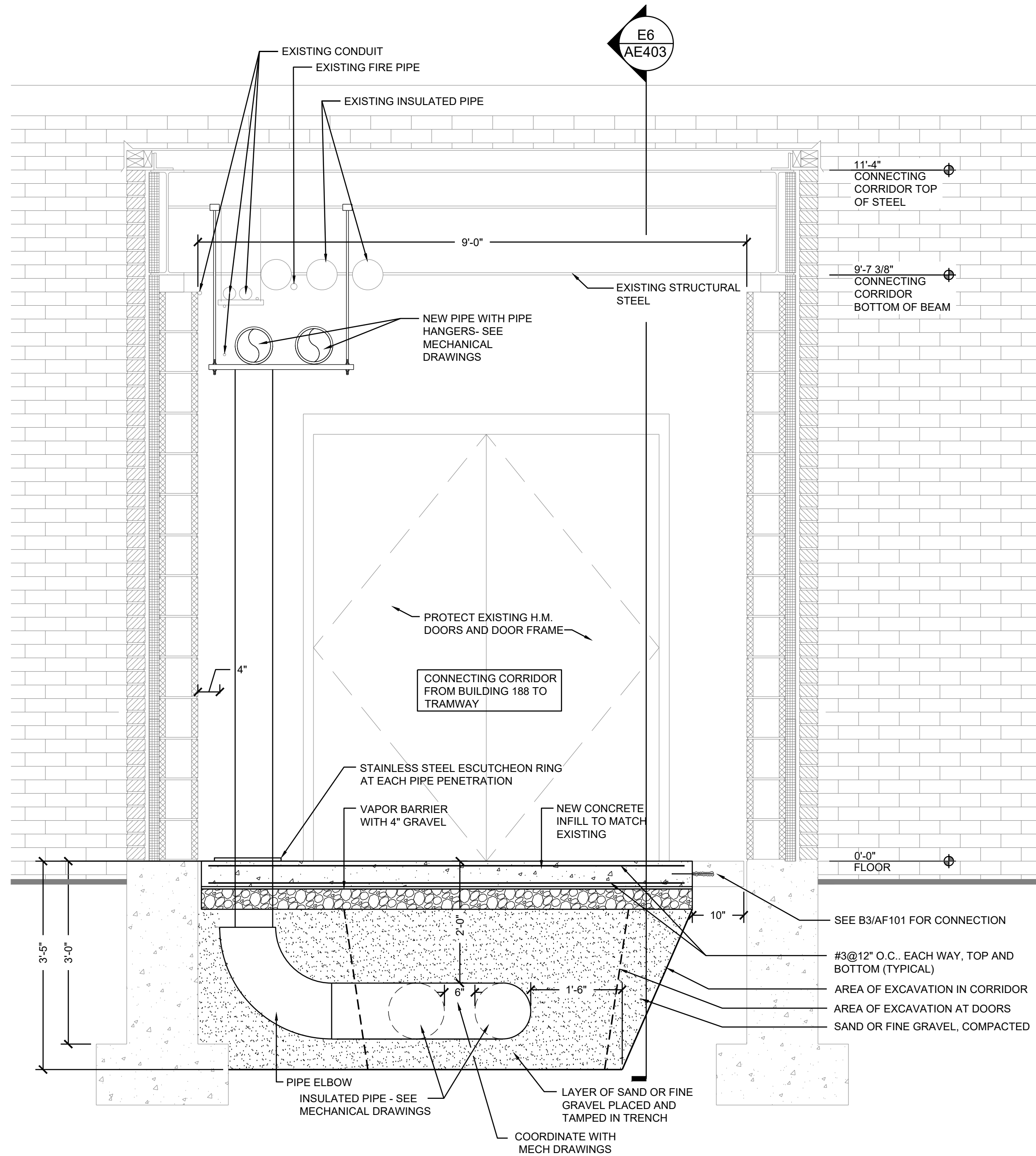
- A. FIELD VERIFY EXISTING OPENINGS PRIOR TO FABRICATION.
B. USE EXISTING DOORS AS A MODEL FOR CONSTRUCTION NEW DOORS.
C. ALL GRILLES TO BE GALVANIZED, PAINTED.
D. ALL WOOD TO BE TREATED, PAINTED.
E. CONTROL JOINT THICKNESS IS EXISTING SLAB. USE STRIPPABLE CONTROL JOINT.
F. STOP REINFORCING EACH SIDE OF JOINT.

SHEET KEYNOTES ①

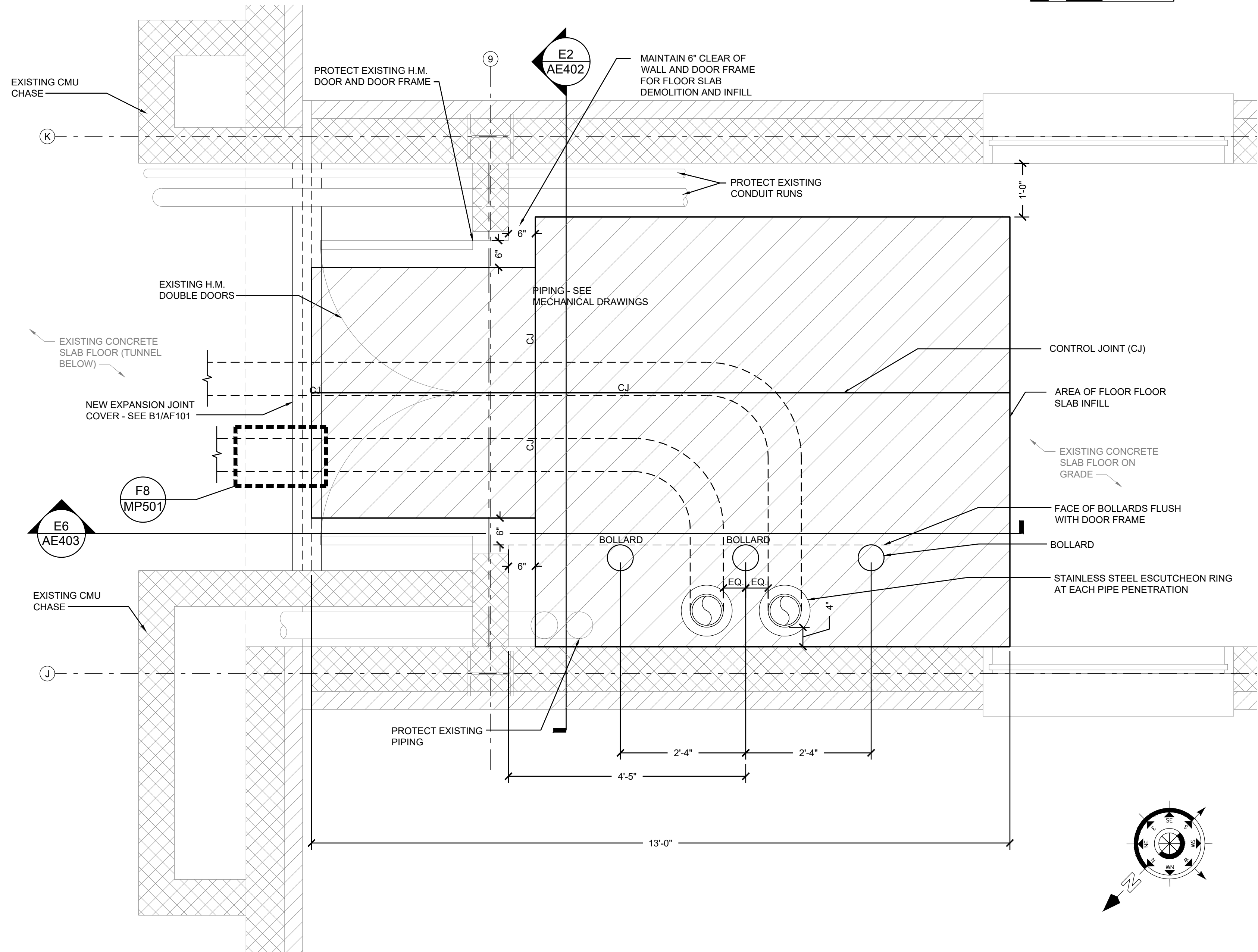


SECTION-MECHANICAL PENETRATION TRAMWAY WALL
SCALE: NONE

PARTIAL BUILDING 3 CRAWL SPACE CHILLED WATER PIPING PLAN
SCALE: 1/2" = 1'-0"



E2 CORRIDOR SECTION
SCALE: 3/4" = 1'-0"



E6 CORRIDOR PLAN
SCALE: 3/4" = 1'-0"

WALL RATING LEGEND

2-HOUR RATED PARTITION

KEYPLAN



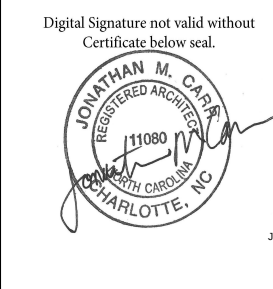
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2	75% DD	05/17/2024
3	95% DD	06/18/2024
4	100% CD	07/26/2024
Revision Number		Revisions:
		Date:

ARCHITECT/ENGINEER OF RECORD

A/E:
HARRELL DESIGN GROUP PC
10720 SIKES PLACE, SUITE 100
CHARLOTTE, NC 28227
704-814-1346
POC: LEE HARRELL PE, LEED AP, QCxP
PRESIDENT
HDG PROJECT: 22018



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OFFICE OF CONSTRUCTION AND FACILITIES MANAGEMENT



DRAWING TITLE
CORRIDOR PLAN AND SECTION

APPROVED

PHASE
100% CONSTRUCTION DOCUMENTS - ISSUED FOR CONSTRUCTION

FULLY SPRINKLERED

PROJECT TITLE
CHILLED WATER FROM B188 TO B3
CONTRACT NO. - 36C25219D0022
TASK ORDER NO. - 36C25223N0287

LOCATION
CAPTAIN JAMES A. LOVELL FEDERAL HEALTH CARE CENTER (FHCC)

ISSUE DATE
JULY 26, 2024

CHECKED
JC

DRAWN
GG

PROJECT NUMBER

556-23-104
BUILDING NUMBER
X

DRAWING NUMBER

AE402

GENERAL SHEET NOTES

- A. PROVIDE GPR WALL SCANNING FOR ALL CORES THROUGH CONCRETE WALLS. TO AVOID CORING THROUGH REBAR REINFORCEMENT (TYPICAL).
- B. INSTALL NEW FOUNDATION WATERPROOFING AT ALL EXCAVATED AREAS.
- C. FOR TYPICAL SLAB REINFORCEMENT DETAILS, SEE E2/AE402.

WALL RATING LEGEND

2-HOUR RATED PARTITION

BUILDING 3 ADDITION C-
EXTERIOR FOUNDATION WALL SECTION
F2 SCALE: 3/4" = 1'-0"

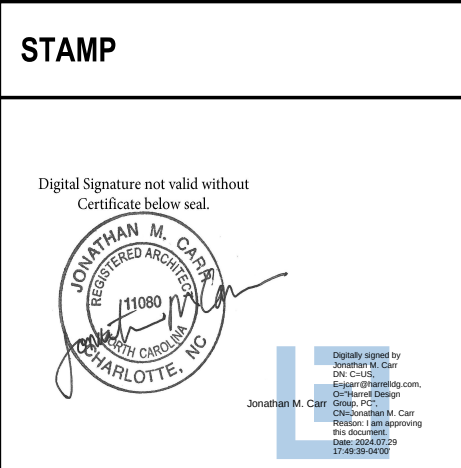
E6 CORRIDOR SECTION
SCALE: 3/4" = 1'-0"

1	95% DD	06/18/2024
2	100% CD	07/26/2024
Revision Number	Revisions:	Date:

ARCHITECT/ENGINEER OF RECORD

A/E:
HARRELL DESIGN GROUP PC
10720 SIKES PLACE, SUITE 100
CHARLOTTE, NC 28227
704-814-1346
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PRESIDENT
HDG PROJECT: 22018

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DRAWING TITLE
CORRIDOR SECTION BETWEEN
BUILDINGS 188 AND 4

APPROVED

PHASE
100% CONSTRUCTION
DOCUMENTS - ISSUED
FOR CONSTRUCTION

FULLY SPRINKLERED

PROJECT TITLE
CHILLED WATER FROM B188 TO B3
CONTRACT NO. - 36C25219D0022
TASK ORDER NO. - 36C25223N0287

LOCATION
CAPTAIN JAMES A. LOVELL FEDERAL HEALTH CARE
CENTER (FHCC)

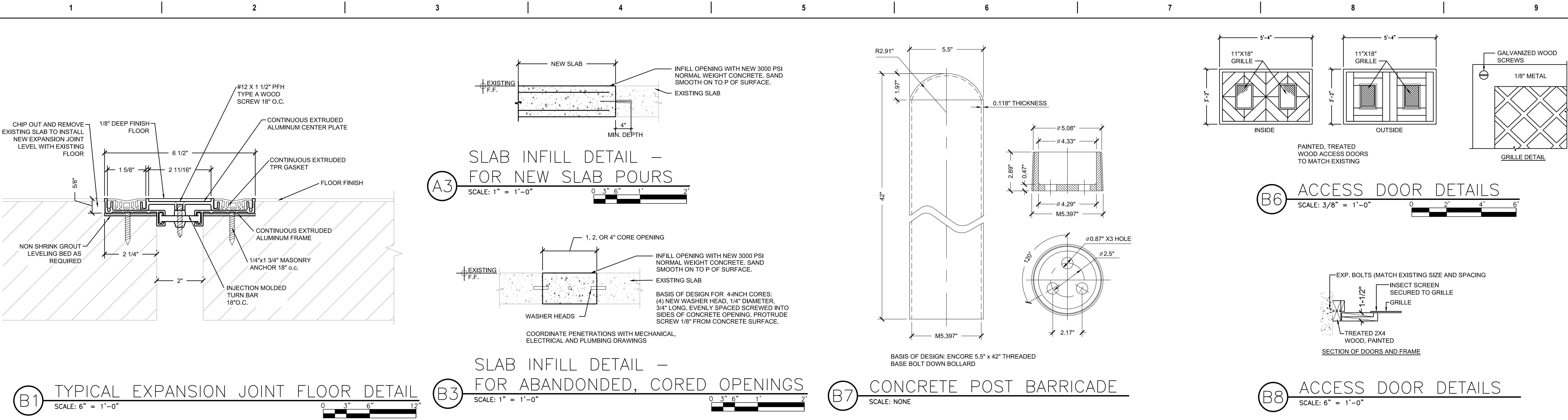
ISSUE DATE
JULY 26, 2024

CHECKED
JC

DRAWN
GG

PROJECT NUMBER
556-23-104
BUILDING NUMBER
X

DRAWING NUMBER
AE403



GENERAL SHEET NOTES

- A. INSTALL FINISHES AFTER HOURS.
B. TEST EXISTING AND NEW CONCRETE PRIOR TO INSTALLING LVT. SEE SPECS.

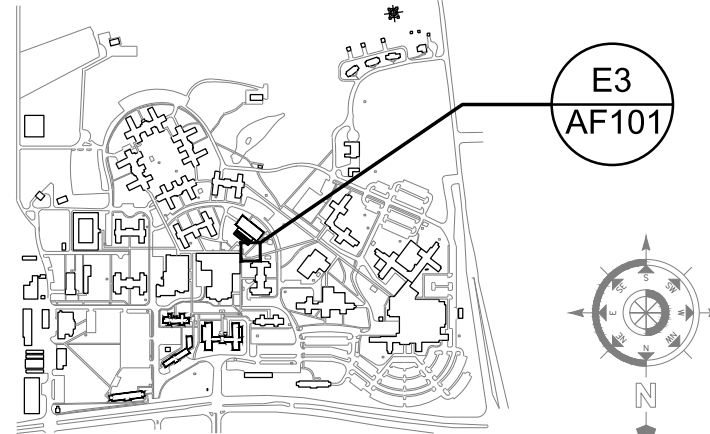
SHEET LEGEND

- COMPLETELY REMOVE EXISTING FLOORING AND INSTALL NEW LVT FLOORING IN GRAIN DIRECTION SHOWN. THIS AREA RECEIVES NEW WALL BASE.
- NEW WALL BASE

WALL RATING LEGEND

- 2-HOUR RATED PARTITION

KEYPLAN



1	95% DD	06/18/2024
2	100% CD	07/26/2024
Revision Number	Revisions:	Date:

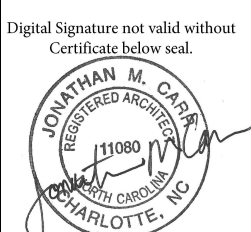
ARCHITECT/ENGINEER OF RECORD

A/E:
HARRELL DESIGN GROUP PC
10720 SIKES PLACE, SUITE 100
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POC: LEE HARRELL PE, LEED AP, QCxP
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HDG PROJECT: 22018



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MANAGEMENT



DRAWING TITLE

FINISH PLAN AND DETAILS

APPROVED

PHASE
100% CONSTRUCTION
DOCUMENTS - ISSUED
FOR CONSTRUCTION

FULLY SPRINKLERED

PROJECT TITLE

CHILLED WATER FROM B188 TO B3
CONTRACT NO. - 36C25219D0022
TASK ORDER NO. - 36C25223N0287

LOCATION

CAPTAIN JAMES A. LOVELL FEDERAL HEALTH CARE
CENTER (FHCC)

ISSUE DATE

JULY 26, 2024

CHECKED

JC

DRAWN

GG

PROJECT NUMBER

556-23-104

BUILDING NUMBER

188

DRAWING NUMBER

AF101

A

B

C

D

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F

A

B

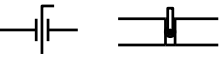
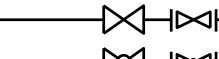
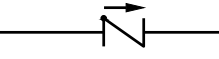
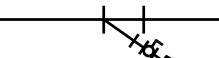
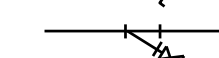
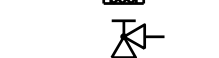
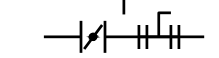
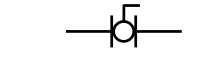
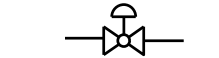
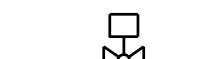
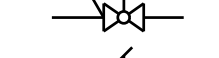
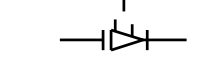
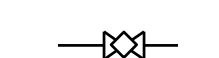
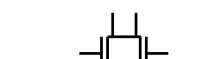
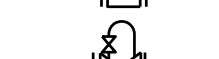
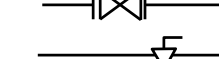
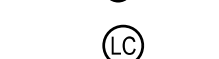
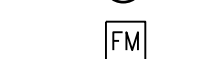
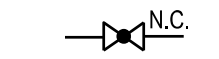
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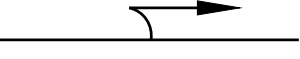
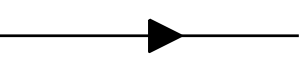
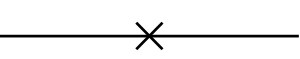
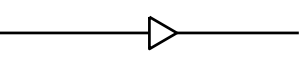
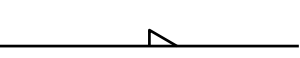
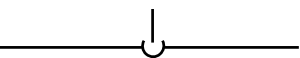
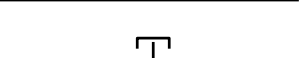
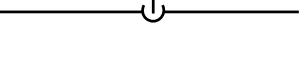
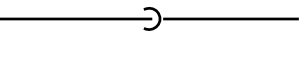
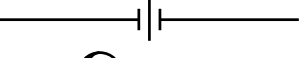
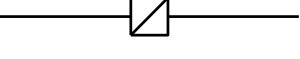
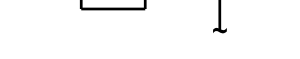
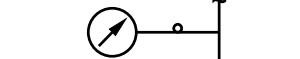
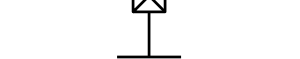
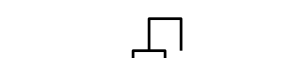
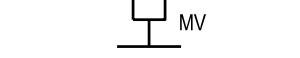
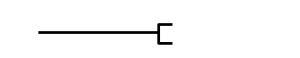
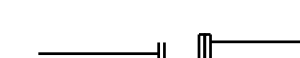
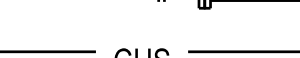
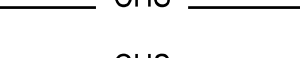
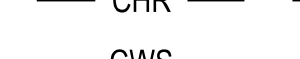
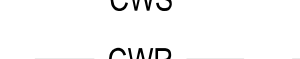
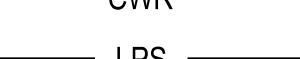
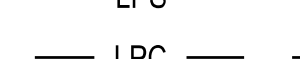
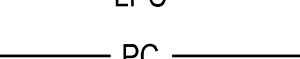
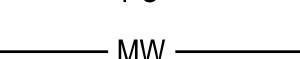
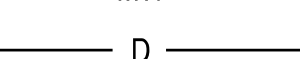
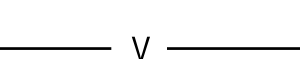
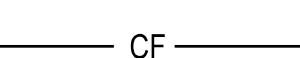
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F

ABBREVIATIONS			
AD	ACCESS DOOR	OA	OUTSIDE AIR
AFF	ABOVE FINISHED FLOOR	O.C.	ON CENTER
AHU	AIR HANDLING UNIT	OFL	OVERFLOW
AMP	AMPERAGE	OSD	OPEN SIGHT DRAIN
APD	AIR PRESSURE DROP	P	PUMP
ARI	AIR CONDITIONING AND REFRIGERATION INSTITUTE	PC	PUMPED CONDENSATE
ASME	AMERICAN SOCIETY OF MECHANICAL ENGINEERS	PD	PRESSURE DROP
ASTM	AMERICAN SOCIETY FOR TESTING AND MATERIALS	PRV	PRESSURE REGULATING VALVE
BHP	BRAKE HORSEPOWER	PSI	POUNDS PER SQUARE INCH
BIW	BACKWARD INCLINED WHEEL (FAN)	PSIG	POUNDS PER SQUARE INCH – GAGE
BLDG	BUILDING	PSV	PRESSURE SAFETY VALVE
BTU	BRITISH THERMAL UNIT	PVC	POLYVINYL CHLORIDE
BTUH	BRITISH THERMAL UNIT PER HOUR		
CFM	CUBIC FEET PER MINUTE	RA	RETURN AIR
CFP	CHEMICAL FEED PUMP	RH	RELATIVE HUMIDITY
CHP	CHILLED WATER PUMP	RPM	REVOLUTIONS PER MINUTE
CO	CARBON MONOXIDE	SA	SUPPLY AIR
COR	CONTRACTING OFFICER'S REPRESENTATIVE	SAT	SUPPLY AIR TEMPERATURE
CP	CONDENSATE PUMP	SCFM	STANDARD CUBIC FEET PER MINUTE
CV	CONSTANT VOLUME	SD	SMOKE DETECTOR
CW	COLD WATER (POTABLE)	SEN	SENSIBLE HEAT
CWP	CONDENSER WATER PUMP	SEC	SECONDARY
DB	DECIBELS	SF	SUPPLY FAN
D _b	DRY-BULB TEMPERATURE	SG	SPECIFIC GRAVITY
DDC	DIRECT DIGITAL CONTROLS	SI	SQUARE INCHES
DEG	DEGREE	SP	STATIC PRESSURE
DIA	DIAMETER	SMACNA	SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION
DP	DEW POINT TEMPERATURE	SPS	STATIC PRESSURE SENSOR
EAT	ENTERING AIR TEMPERATURE	SQ FT	SQUARE FOOT (FEET)
ECWT	ENTERING CONDENSER WATER TEMPERATURE	SS	STAINLESS STEEL
EER	ENERGY EFFICIENCY RATIO	SW	SOFTENED WATER
EF	EXHAUST FAN	TAB	TESTING, ADJUSTING, BALANCE
ESP	EXTERNAL STATIC PRESSURE	TDH	TOTAL DYNAMIC HEAD
ETR	EXISTING TO REMAIN	TSP	TOTAL STATIC PRESSURE
EWT	ENTERING WATER TEMPERATURE	TYP	TYPICAL
EX	EXISTING	UL	UNDERWRITERS LABORATORY
F	FAHRENHEIT	V	VENT
FD	FLOOR DRAIN	VA	VETERANS ADMINISTRATION
FLA	FULL LOAD AMPS	VD	VOLUME DAMPER (MANUAL OPPOSED BLADE)
GAL	GALLON	VFD	VARIABLE FREQUENCY DRIVE
GPM	GALLONS PER MINUTE	VFM	VENTURI FLOW METER
HD	HEAD	VHA	VETERANS HEALTH ADMINISTRATION
HP	HORSEPOWER	VTR	VENT THRU ROOF
HZ	HERTZ	W	WATTS
I/O	INPUT/OUTPUT	W _b	WET-BULB (TEMPERATURE)
IBC	INTERNATIONAL BUILDING COUNCIL	WG	WATER GAGE
IN	INCHES		
IN HG	INCHES OF MERCURY		
IN WG	INCH WATER GAUGE		
KH	KITCHEN HOOD		
KW	KILOWATT		
LAT	LEAVING AIR TEMPERATURE		
LBS/HR	POUNDS PER HOUR		
LEWT	LEAVING EVAPORATOR WATER TEMPERATURE		
LPDA	LOW PRESSURE DRIP ASSEMBLY		
LPR	LOW PRESSURE RETURN (STEAM CONDENSATE)		
LPS	LOW PRESSURE STEAM		
LWT	LEAVING WATER TEMPERATURE		
MAV	MANUAL AIR VENT		
MAX	MAXIMUM		
MBH	1000 BTUH		
MCA	MINIMUM BRANCH CIRCUIT AMPACITY		
MIN	MINIMUM		
MOCp	MAXIMUM OVERCURRENT PROTECTION		
MPDA	MEDIUM PRESSURE DRIP ASSEMBLY		
MSS	MANUFACTURER'S STANDARDIZATION SOCIETY		
NA	NOT APPLICABLE		
NC	NORMALLY CLOSED		
NO	NORMALLY OPEN		
NTS	NOT TO SCALE		

VALVE SYMBOLS	
	ISOLATION VALVE
	GATE VALVE - THREADED/FLANGED
	GLOBE VALVE - THREADED/FLANGED
	GATE VALVE WITH 3/4" HOSE ADAPTER
	CHECK VALVE
	WYE STRAINER (WITH BALL VALVE & HOSE CONNECTION)
	WYE STRAINER WITH VALVED DRAIN AND QUICK-COUPLE HOSE CONNECTOR
	FLEXIBLE CONNECTION
	ANGLE GLOBE VALVE
	BUTTERFLY VALVE
	BALL VALVE
	MODULATING CONTROL VALVE
	MODULATING CONTROL BUTTERFLY VALVE
	TWO POSITION CONTROL VALVE
	THREE-WAY MODULATING CONTROL VALVE
	THREE-WAY TWO POSITION CONTROL VALVE
	PRESSURE REGULATING VALVE
	PRESSURE SAFETY VALVE
	AUTOMATIC BALANCING CONTROL VALVE
	WATER BALANCE DEVICE
	CIRCUIT SETTER VALVE
	GATE VALVE WITH GLOBE-VALVED BYPASS
	PLUG VALVE
	CONTROL VALVE (CV) - FLOAT-OPERATED
	PRESSURE REDUCING VALVE (PRV)
	WATER LEVEL CONTROLLER
	FLOW METER
	NORMALLY CLOSED VALVE

MECHANICAL SYMBOLS	
	CONNECT TO EXISTING
	POINT OF DISCONNECT
	THERMOSTAT

GENERAL PIPING SYMBOLS	
	DIRECTION OF PIPE PITCH (DOWN)
	DIRECTION OF FLOW
	ANCHOR
	REDUCER OR INCREASER
	ECCENTRIC REDUCER
	TOP CONNECTION, 45° OR 90°
	BOTTOM CONNECTION, 45° OR 90°
	SIDE CONNECTION
	CAPPED OUTLET
	RISE OR DROP IN PIPE
	UNION
	PIPE UP
	PIPE DOWN
	FLOAT & THERMOSTATIC TRAP SET INCLUDING PIPING ACCESSORIES SEE DETAIL
	THERMOMETER
	PRESSURE GAGE
	TEST PLUG (PRESSURE/TEMPERATURE)
	AUTOMATIC AIR VENT
	MANUAL AIR VENT
	QUICK-COUPLE HOSE CONNECTOR
	BLIND FLANGE
	CHILLED WATER SUPPLY - SMALL SCALE DRAWINGS
	CHILLED WATER SUPPLY - LARGE SCALE DRAWINGS
	CHILLED WATER SUPPLY - SMALL SCALE DRAWINGS
	CHILLED WATER RETURN - SMALL SCALE DRAWINGS
	CHILLED WATER RETURN - LARGE SCALE DRAWINGS
	CHILLED WATER SUPPLY
	CHILLED WATER RETURN
	CONDENSER WATER SUPPLY
	CONDENSER WATER RETURN
	LOW PRESSURE STEAM (15 PSIG AND BELOW)
	LOW PRESSURE STEAM CONDENSATE RETURN
	CONDENSATE PUMP DISCHARGE
	MAKE-UP WATER
	DRAIN LINE
	VENT LINE
	CHEMICAL FEED
	OVERFLOW
	COLD WATER (CITY WATER)
	SOFTENED WATER
	EXISTING PIPE
	DEMOLITION OF EXISTING PIPE
	EXISTING PIPING CONTINUATION ON SUBSEQUENT VIEWS OR SHEETS.
	NEW PIPING CONTINUATION ON SUBSEQUENT VIEWS OR SHEETS.

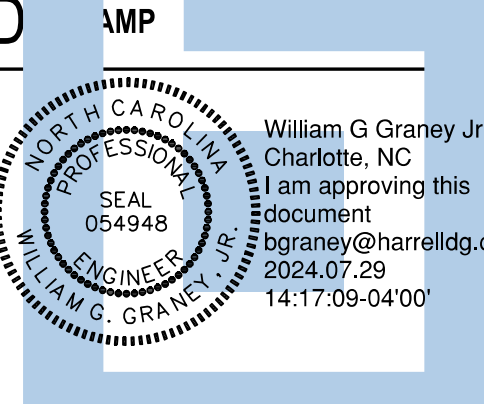
GENERAL NOTES	
1.	CLEAN AND FLUSH NEW PIPING. PIPING MUST BE CAPPED DURING CONSTRUCTION TO MAINTAIN INTEGRITY OF SYSTEM.
2.	ALL PRESSURES LISTED ARE GAUGE PRESSURE UNLESS OTHERWISE NOTED.
3.	EXISTING CONDITIONS AS SHOWN ON THE DRAWINGS ARE TAKEN FROM EXISTING DRAWINGS AND ABOVE-GROUND FIELD SURVEY. MECHANICAL CONTRACTOR SHALL FIELD VERIFY ALL EXISTING SITE CONDITIONS PRIOR TO BEGINNING WORK.
4.	WHERE A SINGLE PRODUCT IS SPECIFIED OR INDICATED ON THE DRAWINGS OR IN THE SPECIFICATIONS BY MANUFACTURER AND MODEL NUMBER, IT IS DONE SO TO ESTABLISH THE STANDARD OF QUALITY OF THE PRODUCTS INTENDED AND IS NOT INTENDED TO LIMIT SIMILAR MANUFACTURERS OR COMPETITION.
5.	COORDINATE ALL PHASING WITH THE OWNER TO MINIMIZE SYSTEM DOWNTIME. SYSTEM SHUTDOWNS SHALL OCCUR DURING NIGHTS AND WEEKENDS, AS DIRECTED BY THE CONTRACTING OFFICER'S REPRESENTATIVE.
6.	WRITTEN SPECIFICATIONS SHALL TAKE PRECEDENCE OVER ANY DRAWING OR DETAIL.
7.	ALL NEW WORK SHALL BE COORDINATED WITH PHASING REQUIREMENTS AND GENERAL CONTRACTOR SHALL ENSURE THAT ALL CONTRACTORS ARE RESPONSIBLE FOR COORDINATION ACROSS THE ENTIRE SET OF CONSTRUCTION DOCUMENTS.
8.	ALL WORK DONE IN THE UTILITY TUNNEL UNDER THE TRAMWAY SHALL BE IN ACCORDANCE WITH VA SAFETY REQUIREMENTS. SEE SPECIFICATION SECTION 01 35 26 FOR MORE INFORMATION.
9.	ONCE PROPOSED CHILLED WATER SUPPLY AND RETURN PIPING TESTING HAS BEEN FLUSHED AND HYDRAULICALLY PRESSURE TESTED, DRAIN PIPING.
10.	ALL NEW CHILLED WATER PIPING SHALL BE SEGMENTED IN LENGTHS NO GREATER THAN 10'-0".
11.	RELOCATE ANY EXISTING CONDUIT AND/OR OBSTRUCTION ENCOUNTERED DURING CHILLED WATER PIPE INSTALLATION.

1	75% DD	05/17/2024
2	95% DD	06/18/2024
3	100% CD	07/26/2024
Revision Number Revisions:		Date:

ARCHITECT/ENGINEER OF RECORD

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HDG PROJECT: 22018


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 U.S. Department
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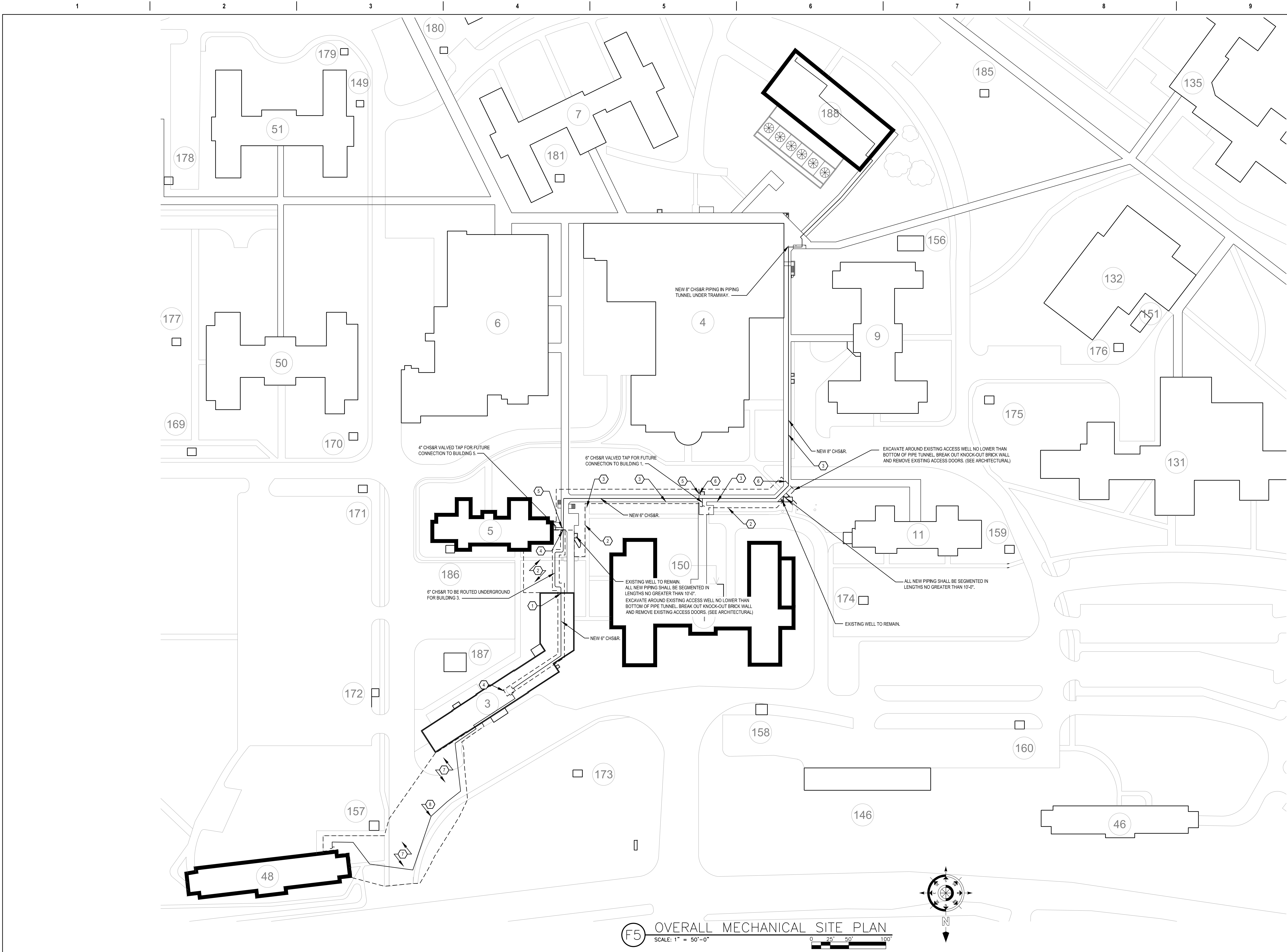
DRAWING TITLE
ABBREVIATIONS, LEGENDS, AND GENERAL NOTES
APPROVED

PHASE
100% CONSTRUCTION DOCUMENTS - ISSUED FOR CONSTRUCTION
FULLY SPRINKLERED

PROJECT TITLE
CHILLED WATER FROM B188 TO B3 CONTRACT NO. - 36C25219D0022 TASK ORDER NO. - 36C25223N0287
LOCATION CAPTAIN JAMES A. LOVELL FEDERAL HEALTH CARE CENTER (FHCC)
ISSUE DATE JULY 26, 2024
CHECKED WGG
DRAWN ANB

PROJECT NUMBER
556-23-104
BUILDING NUMBER

DRAWING NUMBER
MP001



GENERAL SHEET NOTES

- A. NEW CHILLED WATER PIPING SHALL BE ROUTED WITHIN PIPING TUNNEL, UNLESS NOTED OTHERWISE.
- B. SEE SHEET MP102 AND MP103 FOR DEDUCTIVE ALTERNATES.
- C. FOR AREAS REQUIRING GROUND PENETRATING RADAR (GPR) TESTING, SEE SPEC SECTION 31 20 00.
- D. PROVIDE GPR SURVEY RESULTS IN THIS CONTRACT FOR FUTURE UNDERGROUND PIPE WORK CONNECTING BUILDING 3 TO BUILDING 48.

SHEET KEYNOTES

- ROUTE NEW 6" CHILLED WATER PIPING INTO CRAWL SPACE OF BUILDING 3.
- ALL WORK AREAS, THIS INCLUDES ALL AREAS OUTSIDE PIPE TUNNELS, EXISTING ACCESS WELLS AND WITHIN THE PIPE TUNNELS SHALL BE SCANNED VIA GROUND PENETRATING RADAR TO LOCATE ALL UNKNOWN SUB-SURFACE OBSTRUCTIONS.
- UTILIZE THE EXISTING NATURAL VENTILATION OPENINGS TO CONNECT TEMPORARY VENTILATION FANS TO EVACUATE FUMES GENERATED DURING CONSTRUCTION.
- FOR DEDUCTIVE ALTERNATE NO. 1: INCLUDES BASE BID MINUS ALL WORK ASSOCIATED WITH ROUTING 6" PIPING TO BASEMENT OF BUILDING 3, INCLUDING TEES, ISOLATION VALVES, BLIND FLANGES, BYPASS REDUCERS, BUILDING 3 PIPE SUPPORTS, CORE DRILLING OF PIPE TUNNEL/BUILDING 3 EXTERIOR WALL AND EQUIPMENT SLAB INSTALLATION. THE BRANCH PIPING SERVING BUILDING 5 SHALL REMAIN. IF DEDUCT IS TAKEN, INSTALL SHUT-OFF VALVES, BLANK-OFF PLATES AND BYPASS VALVE ASSEMBLY ON THE MAIN LINE. PROVIDE POWER AND CONTROL POINTS FOR BYPASS VALVE AND PRESSURE SENSOR.
- FOR DEDUCTIVE ALTERNATE NO. 2: INCLUDES DEDUCTIVE ALTERNATE NO.1 AND THE REMOVAL OF THE SUPPLY AND RETURN CHILLED WATER PIPING, SHUT-OFF VALVES AND BLANK-OFF PLATES SERVING BUILDING 5 AND 3 BACK TO THE CONNECTION POINT OF BUILDING 1. THE BRANCH PIPING SERVING BUILDING 1 SHALL REMAIN. IF DEDUCT IS TAKEN, INSTALL SHUT-OFF VALVES, BLANK-OFF PLATES AND BYPASS VALVE ASSEMBLY ON THE MAIN LINE. PROVIDE POWER AND CONTROL POINTS FOR BYPASS VALVE AND PRESSURE SENSOR.
- FOR DEDUCTIVE ALTERNATE NO. 3 SHALL INCLUDE DEDUCT ALTERNATE NO.2 MINUS THE INSTALLATION OF THE CHILLED WATER PIPING SERVING BUILDING 1 BACK TO THE POINT WHERE THE PIPING HAS A CHANGE IN DIRECTION FROM PLAN NORTHSOUTH TO PLAN EASTWEST. IF DEDUCT IS TAKEN, INSTALL SHUT-OFF VALVES, BLANK-OFF PLATES AND BYPASS VALVE ASSEMBLY ON THE MAIN LINE. PROVIDE POWER AND CONTROL POINTS FOR BYPASS VALVE AND PRESSURE SENSOR.
- THE CONTRACTOR SHALL PERFORM A GROUND PENETRATING RADAR (GPR) SURVEY WITHIN OUTLINED AREA.
- PROPOSED ROUTE OF 4" CHSR FROM BUILDING 3 TO BUILDING 48.

F5 OVERALL MECHANICAL SITE PLAN
SCALE: 1" = 50'-0"

1	25% SD	05/25/2023
2	50% DD	07/07/2023
3	50% DD RESUBMISSION	08/29/2023
4	75% DD	05/17/2024
5	95% DD	06/18/2024
6	100% CD	07/26/2024
Revision Number	Revisions:	Date:

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HDG PROJECT: 22018

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U.S. Department
of Veterans Affairs

DRAWING TITLE
OVERALL MECHANICAL SITE
PLAN

APPROVED

PHASE
100% CONSTRUCTION
DOCUMENTS - ISSUED
FOR CONSTRUCTION

FULLY SPRINKLERED

PROJECT TITLE
CHILLED WATER FROM B188 TO B3
CONTRACT NO. - 36C25219D0022
TASK ORDER NO. - 36C25223N0287

LOCATION
CAPTAIN JAMES A. LOVELL FEDERAL HEALTH CARE
CENTER (FHCC)

ISSUE DATE
JULY 26, 2024

CHECKED
WGG

DRAWN
DM

PROJECT NUMBER
556-23-104

BUILDING NUMBER

DRAWING NUMBER
MS001

GENERAL SHEET NOTES

- A. NEW CHILLED WATER PIPING SHALL BE ROUTED OVERHEAD CLOSE TO EXISTING ROOF STRUCTURE WITHIN BUILDING 188 AND TRAMWAY OFFICE AREA.
B. GROUND MOUNTED PIPING SUPPORTS SHALL BE INSTALLED 10'-0" ON CENTER, UNLESS NOTED OTHERWISE. SEE DETAIL D08P501 FOR MORE INFORMATION.

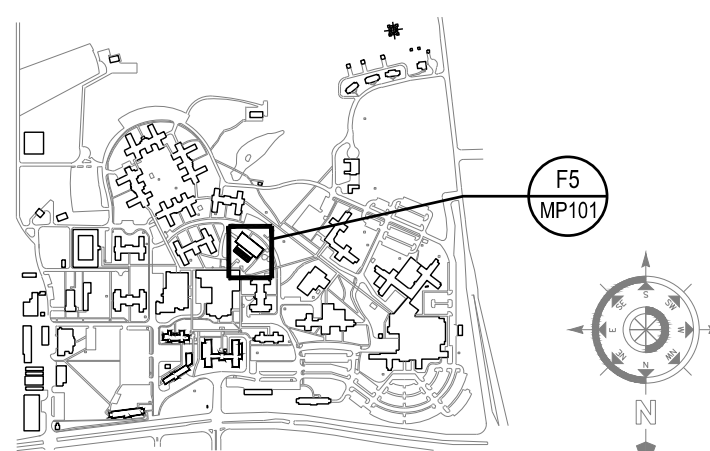
SHEET KEYNOTES ①

- ROUTE NEW 8" CH&R PIPING FROM OVERHEAD EXISTING WALL THROUGH NEW FLOOR PENETRATIONS. MOUNT PIPING TIGHT TO EXISTING WALL.
- SEE PIPING SUPPORT DETAIL FOR MORE INFORMATION. PROVIDE PIPE SUPPORT EVERY 10'-0" UNLESS NOTED OTHERWISE.
- CONNECT TO NEW CHILLED WATER PIPING TO EXISTING BLIND FLANGES WITH ISOLATION VALVE ON EXISTING CHILLED WATER SUPPLY/RETURN MAIN PIPING.
- MOUNT NEW 8" CHILLED WATER PIPING 16'-0" ABOVE FINISHED FLOOR WITHIN THE CLEAR SPACE BETWEEN EXISTING STRUCTURE AND EXISTING CONDUITS. PROVIDE PIPE SUPPORT EVERY 10'-0".
- ROUTE NEW CHILLED WATER PIPING DOWN TO 1'-6" OF THE EXISTING STEEL STRUCTURE OF THE TRAMWAY. MATCH ELEVATION OF THE EXISTING CHILLED WATER PIPING. FIELD COORDINATE EXACT HEIGHT.
- ROUTE NEW 8" CHILLED WATER PIPING 3'-0" BELOW FLOOR DOWN TO EXISTING PIPE TUNNEL. CONTRACTOR TO FIELD ADJUST PIPE ELEVATION TO SUITE SITE CONDITIONS.
- ROUTE NEW 8" CHS PIPING OVER THE EXISTING CHILLED WATER PUMP HEADER AND UNDER THE EXISTING CAMPUS CHILLED WATER SUPPLY PIPE.
- NEW CHILLED WATER PIPING, ROUTED OVER PUMP AREA, SHALL BE INSTALLED BELOW EXISTING CONDUITS AT APPROXIMATELY 13'-0" ABOVE FINISHED FLOOR. PROVIDE PIPE SUPPORT EVERY 10'-0".
- ROUTE 8" CHW PIPING UP TO MATCH PIPING ELEVATION OF THE CHILLED WATER PIPING INSTALLED WITHIN CLEAR SPACE BELOW EXISTING STRUCTURE.
- CONTRACTOR TO PROVIDE 1/16" TH. ALUMINUM PIPE JACKETING FOR PIPE PROTECTION. JACKETING SHALL EXTEND FROM TOP OF FINISHED FLOOR TO 7'-6".
- EXISTING 4" CHILLED WATER PIPING. FIELD VERIFY EXACT LOCATION. TYPICAL.
- EXISTING FIRE RISER.
- ROUTE NEW 8" CHILLED WATER PIPING OVER EXISTING PIPING. PROVIDE 2'-3" OF CLEAR SPACE BETWEEN NEW PIPING AND EXISTING STEAM PIPING. FIELD ADJUST PIPING ELEVATION.
- NEW 8" CHILLED WATER PIPING TO BE ROUTED OVER EXISTING CHILLED WATER AND STEAM PIPING AND UNDER EXISTING ELECTRICAL / COMMUNICATION PULLEX. APPROXIMATELY 4'-6" ABOVE FINISHED GRADE TO CENTER LINE OF NEW CHILLED WATER PIPE.
- CHANGE CHILLED WATER PIPE ELEVATION TO MATCH TOP OF EXISTING SLAB.
- ROUTE NEW CHILLED WATER PIPING DOWN AND TRANSITION DIRECTLY BENEATH EXISTING STEAM PIPING.
- CONTRACTOR SHALL ROUTE THE NEW CHILLED WATER PIPING IN THE 24"-33" CLEAR SPACE BENEATH THE EXISTING STEAM PIPING.
- CORE DRILL WALL OF EXISTING UTILITY TUNNEL. EACH OF THE TWO HOLES SHALL BE 12" IN DIA. GROUT AND SEAL WATER TIGHT UPON COMPLETION OF PIPE INSTALLATION. AS MEASURED FROM INSIDE OF THE PIPE TUNNEL, THE CENTER LINE OF THE HOLES SHALL BE 4'-0" ABOVE GRADE OF THE TUNNEL AND 4" ABOVE THE EXISTING STEAM PIPE.
- 8" CHW PIPING AND RUN BELOW EXISTING ELECTRICAL CONDUIT AND FIRE PROTECTION PIPING.
- REMOVE EXISTING ABANDONED STEAM PRESSURE REDUCING STATION AND ASSOCIATED EQUIPMENT. THE CONTRACTOR SHALL CONTACT THE COR AND DETERMINE IF THE VA WOULD PREFER TO RETAIN THIS EQUIPMENT. IF NOT, DISPOSE OF PROPERLY OFF-SITE.
- SEE B4/AE402 FOR SEALANT AT WALL PENETRATION. PROVIDE ESCUTCHEON PLATES ON BOTH SIDES OF THE WALL.
- REFER TO DETAIL F05P501.
- ROUTE CHILLED WATER PIPING UP WITHIN CLEAR SPACE ABOVE EXISTING PIPING AND CONDUIT.
- ROUTE PIPING DOWN FROM CLEAR SPACE TO BELOW ELECTRICAL CONDUIT AND FIRE PROTECTION PIPING.
- TRANSITION TO PRE-INSULATED PIPING FROM CONVENTIONAL STEEL PIPING WITH FIELD APPLIED INSULATION AND ALL SERVICE JACKETING AT ELBOW WITHIN THE OVERHEAD CHILLED WATER PIPING.

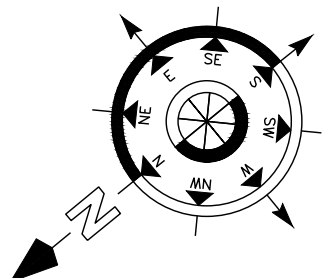
WALL RATING LEGEND

----- 2-HOUR RATED PARTITION

KEYPLAN



F5 BUILDING 188 FLOOR PLAN
SCALE: 1/16" = 1'-0"



1	25% SD	05/25/2023
2	50% DD	07/07/2023
3	50% DD RESUBMISSION	08/29/2023
4	75% DD	05/17/2024
5	95% DD	06/18/2024
6	100% CD	07/26/2024
Revision Number Revisions:		Date:

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PRESIDENT
HDG PROJECT: 22018
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DRAWING TITLE
BUILDING 188 FLOOR PLAN

APPROVED

PHASE
100% CONSTRUCTION
DOCUMENTS - ISSUED
FOR CONSTRUCTION

FULLY SPRINKLERED

PROJECT TITLE
CHILLED WATER FROM B188 TO B3
CONTRACT NO. - 36C25219D0022
TASK ORDER NO. - 36C25223N0287

LOCATION
CAPTAIN JAMES A. LOVELL FEDERAL HEALTH CARE
CENTER (FHCC)

ISSUE DATE
JULY 26, 2024

CHECKED
WGG

DRAWN
DM

PROJECT NUMBER

556-23-104

BUILDING NUMBER

188

DRAWING NUMBER

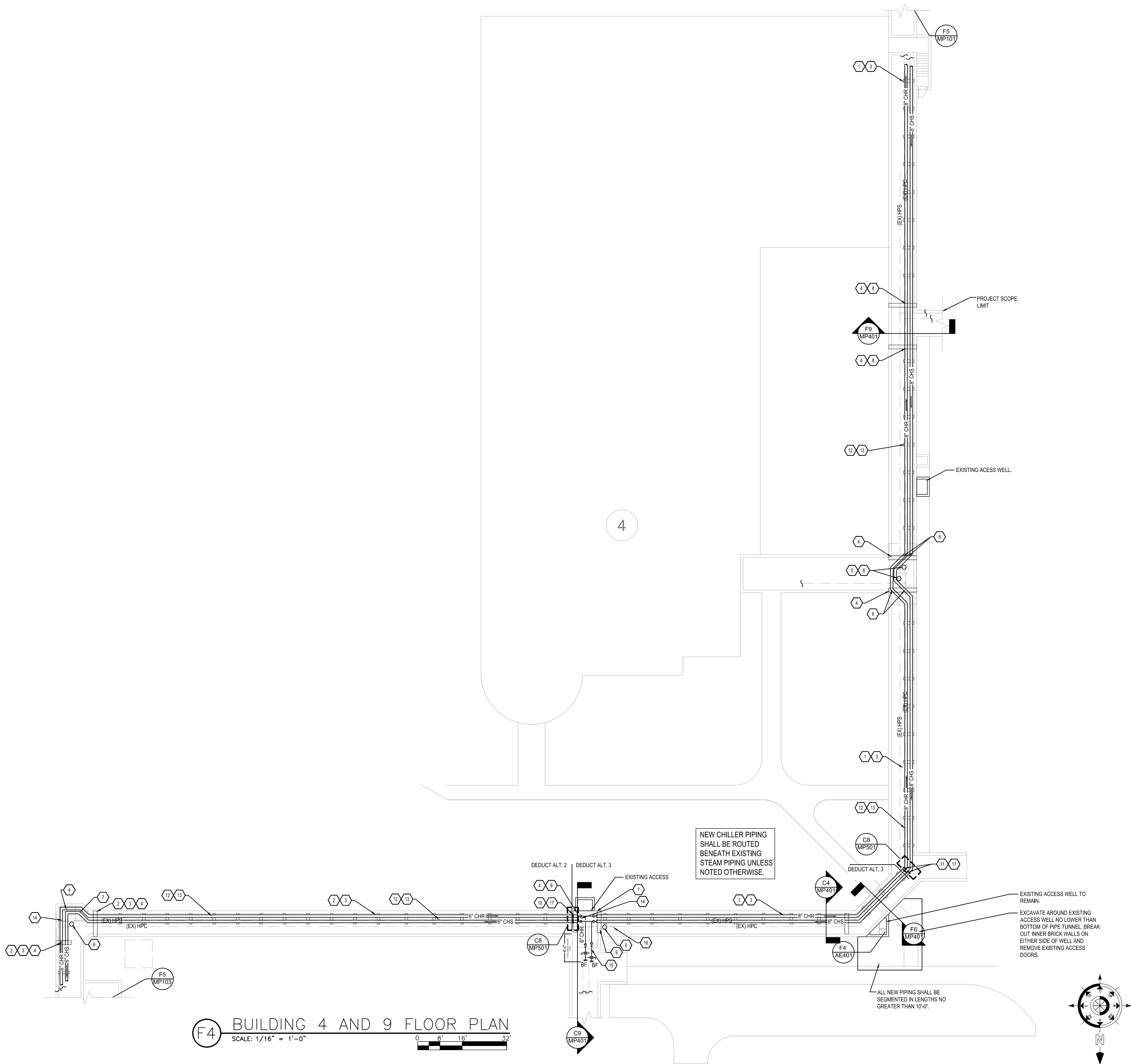
MP101

GENERAL SHEET NOTES

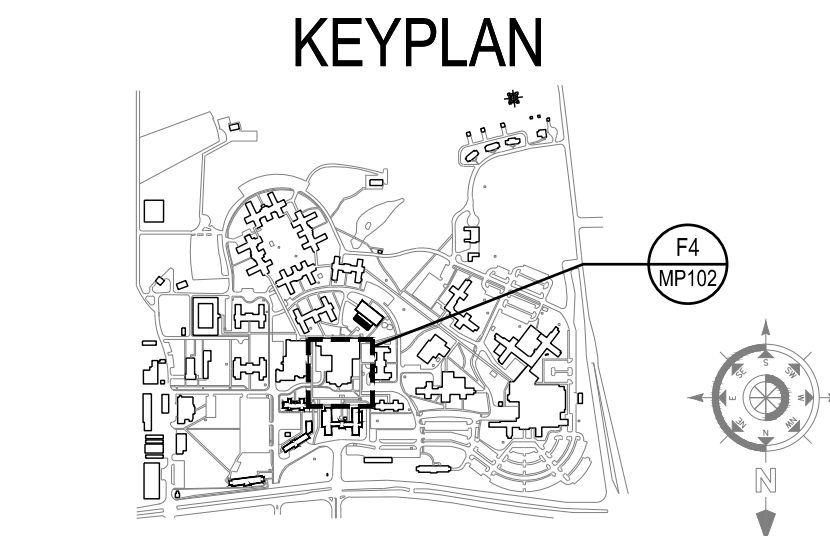
- A. NEW CHILLED WATER PIPING SHALL BE ROUTED WITHIN PIPING TUNNEL, UNLESS NOTED OTHERWISE.
- B. ROUTE NEW CHILLED WATER PIPING IN THE VERTICAL CLEAR SPACE BENEATH EXISTING STEAM AND CONDENSATE PIPING, UNLESS NOTED OTHERWISE.
- C. GROUND MOUNTED PIPING SUPPORTS SHALL BE INSTALLED 10'-0" ON CENTER, UNLESS NOTED OTHERWISE. SEE DETAIL D8MP501 FOR MORE INFORMATION.

SHEET KEYNOTES 1

1. UNDERGROUND RATED 8" CHSAR PRE-INSULATED PIPING WITH FITTINGS.
2. UNDERGROUND RATED 6" CHSAR PRE-INSULATED PIPING WITH FITTINGS.
3. CHILLED WATER PIPING TO BE ROUTED ABOVE GROUND, TYPICAL.
4. ROUTE CHILLED WATER PIPING OVER GRADE BEAM WHICH IS 4" ABOVE GRADE. INSTALL PIPE ROLLER SUPPORTS DIRECTLY ON EXISTING CONCRETE FOOTER/SLAB.
5. ROUTE CHILLED WATER PIPING TO AVOID CONFLICT BETWEEN NEW CHILLED WATER PIPING AND EXISTING STEAM TRAP/STREAM EQUIPMENT.
6. EXISTING STEAM TRAP/STREAM EQUIPMENT SHALL REMAIN FULLY FUNCTIONAL DURING CONSTRUCTION.
7. HAND EXCAVATE IN CONFINED AREA.
8. ROUTE PIPE ABOVE EXISTING GRADE BEAM.
9. CONTRACTOR SHALL NOT ROUTE UNDER GROUND BELOW FOOTER OF EXISTING PIPE SUPPORTS. INSTALL PIPE ROLLER SUPPORTS DIRECTLY ON EXISTING CONCRETE.
10. FOR DEDUCTIVE ALTERNATE NO. 2, INCLUDES DEDUCTIVE ALTERNATE NO. 1 AND THE REMOVAL OF THE SUPPLY AND RETURN CHILLED WATER PIPING, SHUT-OFF VALVES AND BLANK-OFF PLATES SERVING BUILDING 5 AND 3 BACK TO THE CONNECTION POINT OF BUILDING 1. THE BRANCH PIPING SERVING BUILDING 1 SHALL REMAIN. IF DEDUCT IS TAKEN, INSTALL SHUT-OFF VALVES, BLANK-OFF PLATES AND BYPASS VALVE ASSEMBLY ON THE MAIN LINE. PROVIDE POWER AND CONTROL POINTS FOR BYPASS VALVE AND PRESSURE SENSOR.
11. FOR DEDUCTIVE ALTERNATE NO. 3 SHALL INCLUDE DEDUCT ALTERNATE NO. 2 MINUS THE INSTALLATION OF THE CHILLED WATER PIPING SERVING BUILDING 1 BACK TO THE POINT WHERE THE PIPING HAS A CHANGE IN DIRECTION FROM PLAN NORTH/SOUTH TO PLAN EAST/WEST. IF DEDUCT IS TAKEN, INSTALL SHUT-OFF VALVES, BLANK-OFF PLATES AND BYPASS VALVE ASSEMBLY ON THE MAIN LINE. PROVIDE POWER AND CONTROL POINTS FOR BYPASS VALVE AND PRESSURE SENSOR.
12. PROVIDE PIPE SUPPORTS EVERY 10'-0".
13. NEW CHILLED WATER PIPING TO BE ROUTED BELOW EXISTING STEAM PIPING, TYPICAL.
14. ROUTE CHILLED WATER PIPING 2'-0" BELOW GRADE TO AVOID CONFLICT BETWEEN NEW CHILLED WATER PIPING AND EXISTING STEAM TRAP/STREAM EQUIPMENT.
15. IF NEW BUTTERFLY VALVE, BLIND FLANGE AND BLANK-OFF PLATE FOR FUTURE CONNECTION TO BUILDING 1.
16. ON DISCHARGE SIDE OF VALVE AT DRIP LEGS INSTALL NIPPLE AND ELBOW. PIPING INLET AND OUTLET FOR CONDENSATE TRAP SHALL RUN PARALLEL TO EXISTING STEAM PIPING. DISCHARGE OF CONDENSATE TRAP SHALL RISE BETWEEN EXISTING STEAM AND CONDENSATE PIPING AND BE TIED BACK INTO CONDENSATE PIPING RUNNING PARALLEL TO UTILITY TUNNEL SIDEWALL AT CEILING. MATCH EXISTING PIPE SIZE AND RE-INSTALL CONDENSATE TRAP. REPLACE PIPING, UNIONS AND ISOLATION VALVES. RE-INSULATE PIPING WITH FIBERGLASS INSULATION TO MATCH EXISTING CONDITIONS. RELABEL PIPING. TIE BACK INTO TO EXISTING PIPING PRIOR TO EXISTING PIPE HANGER AND INSTALL NEW PIPE HANGER AS CONDENSATE PIPING RISES.
17. REFER TO DETAIL C8MP501 FOR BYPASS VALVE ASSEMBLY DESIGNATED FOR USE IF DEDUCT ALTERNATES ARE TAKEN.



F4 BUILDING 4 AND 9 FLOOR PLAN
SCALE: 1/16" = 1'-0"

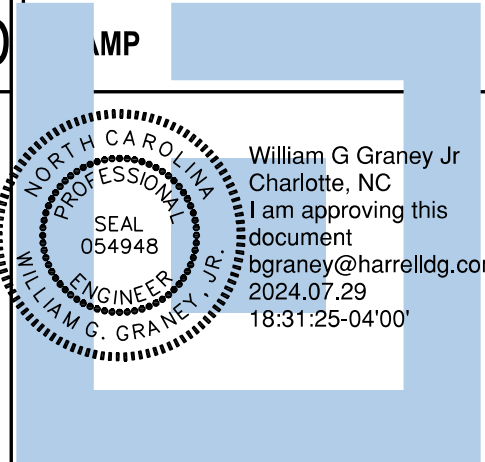


1	25% SD	05/25/2023
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DRAWING TITLE
BUILDING 4 AND 9 FLOOR PLAN

APPROVED

PHASE
100% CONSTRUCTION
DOCUMENTS - ISSUED
FOR CONSTRUCTION

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PROJECT TITLE
CHILLED WATER FROM B188 TO B3
CONTRACT NO. - 36C25219D0022
TASK ORDER NO. - 36C25223N0287

PROJECT NUMBER
556-23-104
BUILDING NUMBER
4, 9
DRAWING NUMBER
MP102

LOCATION
CAPTAIN JAMES A. LOVELL FEDERAL HEALTH CARE
CENTER (FHCC)

ISSUE DATE
JULY 26, 2024

CHECKED
WGG

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DM

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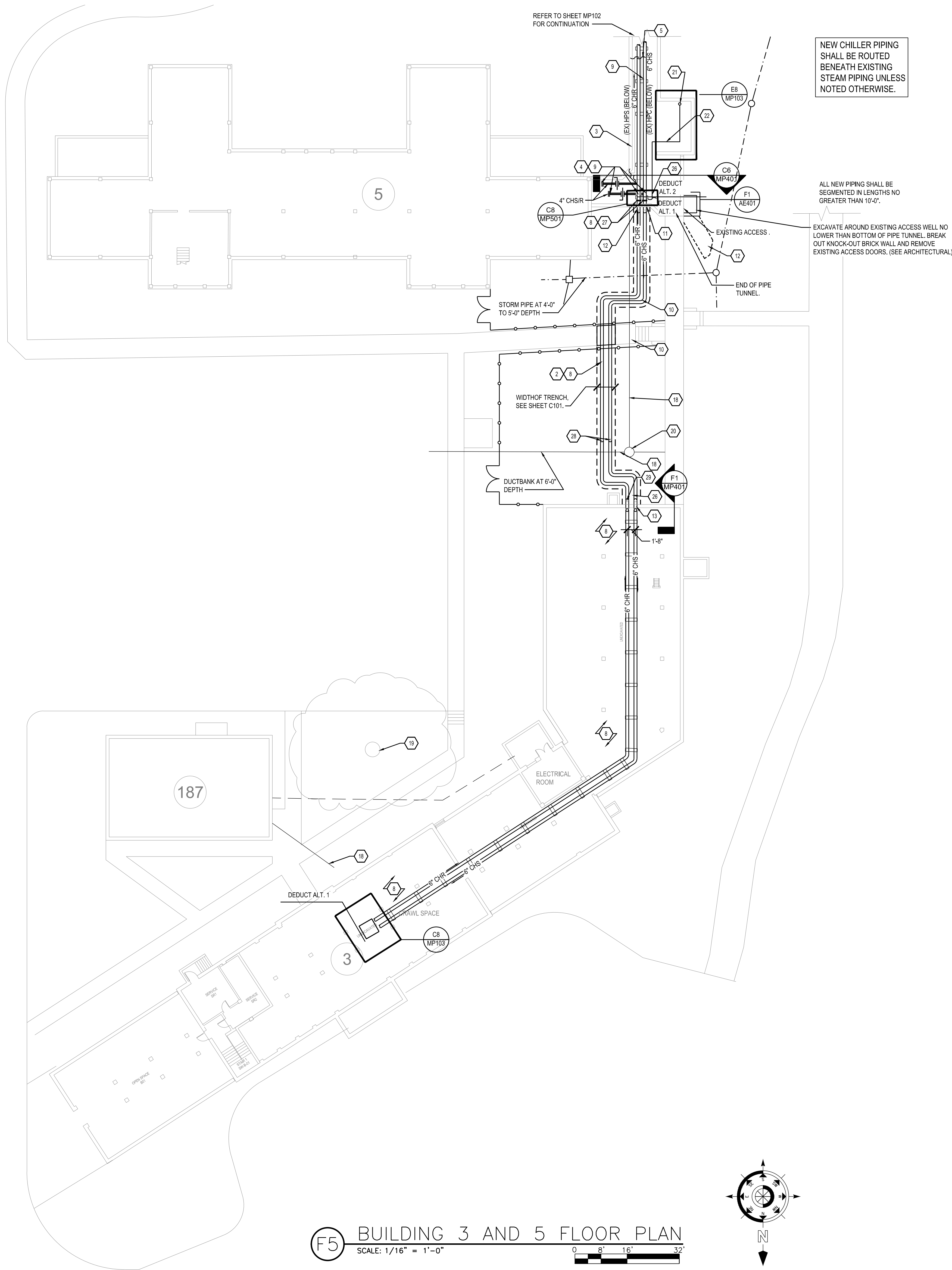
B

C

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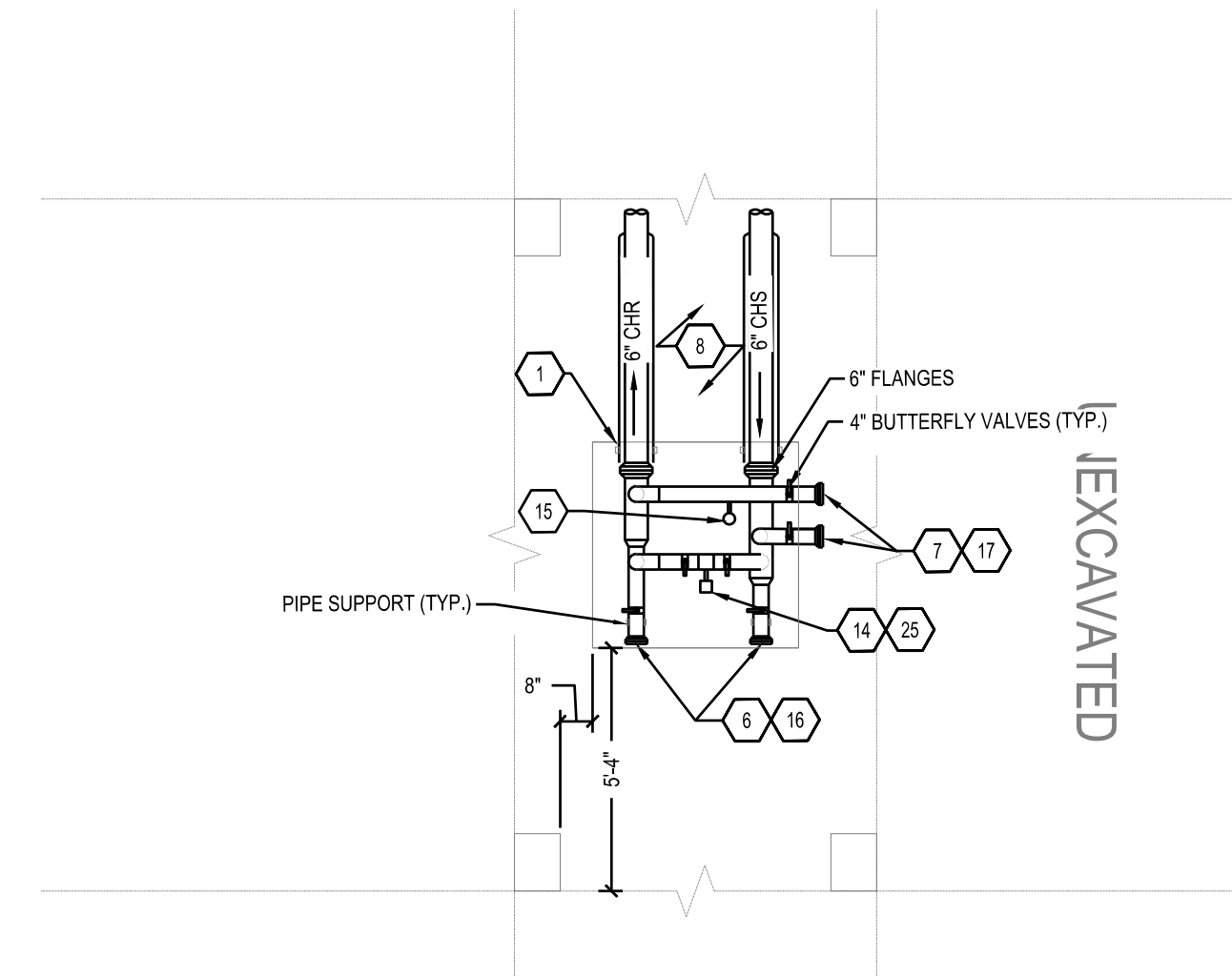
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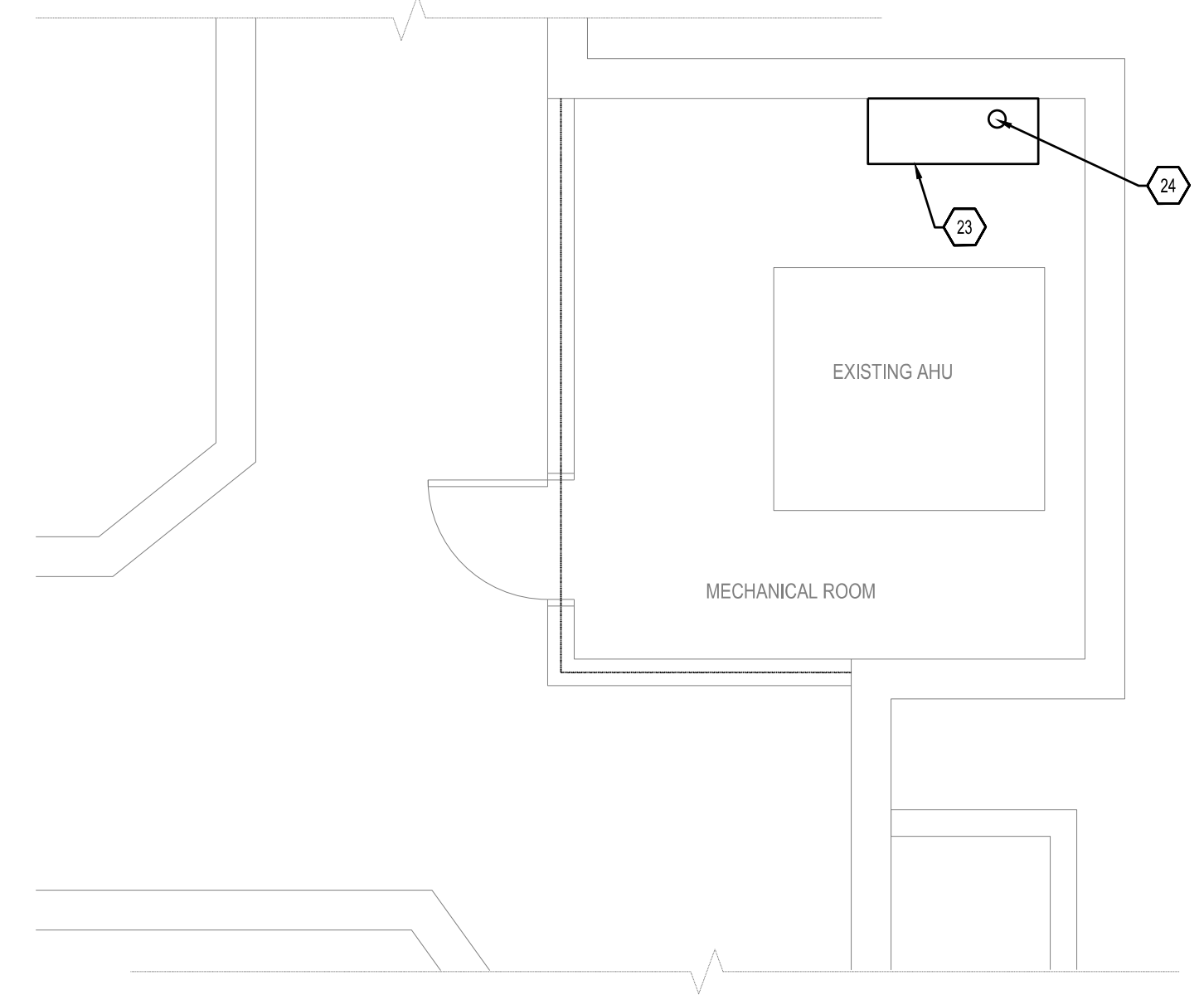
NEW CHILLER PIPING
SHALL BE ROUTED
BENEATH EXISTING
STEAM PIPING UNLESS
NOTED OTHERWISE.

ALL NEW PIPING SHALL BE
SEGMENTED IN LENGTHS NO
GREATER THAN 10'-0".

EXCAVATE AROUND EXISTING ACCESS WELL NO
LOWER THAN BOTTOM OF PIPE TUNNEL. BREAK
OUT KNOCK-OUT BRICK WALL AND REMOVE
EXISTING ACCESS DOORS. (SEE ARCHITECTURAL)



PARTIAL BUILDING 3 CRAWL SPACE
CHILLED WATER PIPING PLAN
SCALE: 1/2" = 1'-0"



TRAMWAY MECHANICAL ROOM
AT BUILDING 5 CONNECTOR PLAN
SCALE: 1/2" = 1'-0"

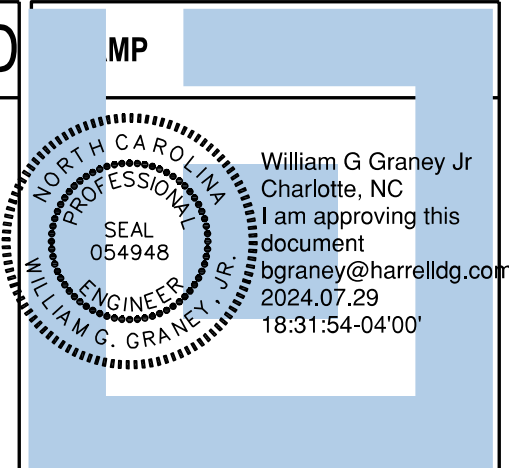
F5 BUILDING 3 AND 5 FLOOR PLAN
SCALE: 1/16" = 1'-0"

1	25% SD	05/25/2023
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DRAWING TITLE BUILDING 3 AND 5 FLOOR PLAN	PHASE 100% CONSTRUCTION DOCUMENTS - ISSUED FOR CONSTRUCTION	PROJECT TITLE CHILLED WATER FROM B188 TO B3 CONTRACT NO. - 36C25219D0022 TASK ORDER NO. - 36C25223N0287	PROJECT NUMBER 556-23-104 BUILDING NUMBER 3, 5
APPROVED	FULLY SPRINKLERED	LOCATION CAPTAIN JAMES A. LOVELL FEDERAL HEALTH CARE CENTER (FHCC)	DRAWING NUMBER MP103
		ISSUE DATE JULY 26, 2024	CHECKED WGG DRAWN DM

GENERAL SHEET NOTES

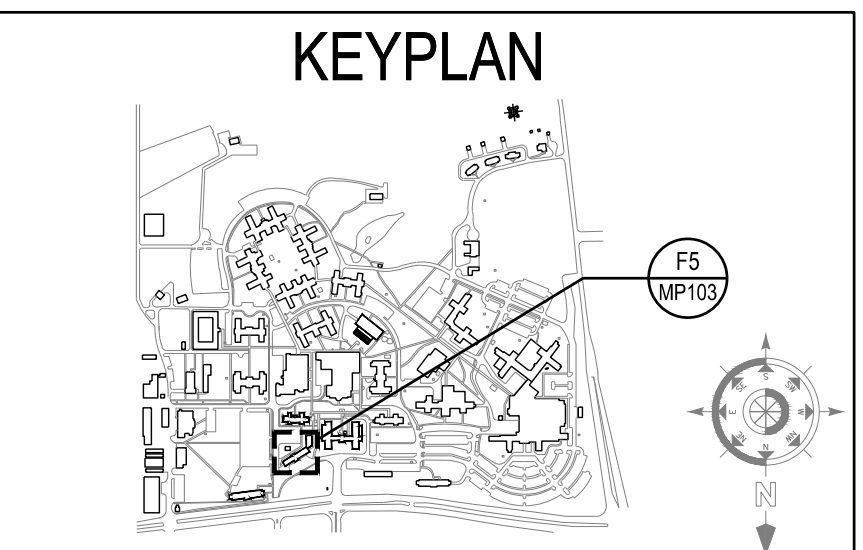
- NEW CHILLED WATER PIPING SHALL BE ROUTED WITHIN PIPING TUNNEL, UNLESS NOTED OTHERWISE.
- GROUND MOUNTED PIPING SUPPORTS SHALL BE INSTALLED 10'-0" ON CENTER, UNLESS NOTED OTHERWISE. SEE DETAIL D1AE404 FOR MORE INFORMATION.
- SEE SHEET C101 FOR ADDITIONAL INFORMATION.

SHEET KEYNOTES 1

- PROVIDE 12" THICK, 4'-6" x 4'-6" CONCRETE HOUSE KEEPING PAD LOCATED WITHIN THE CRAWL SPACE OF BUILDING 3. REFER TO DETAIL D1AE404 FOR MORE INFORMATION. SEE DETAIL D1AE404.
- UNDERGROUND RATED 6" CHSR PRE-INSULATED PIPING, TYPICAL OF ALL.
- INSULATED 6" CHILLED WATER PIPING ROUTED WITHIN EXISTING PIPING TUNNEL.
- INSTALL 4" VALVE AND BLIND FLANGE FOR FUTURE CHILLED WATER TAP SERVING BUILDING 5.
- REFER TO K501 FOR MORE INFORMATION.
- 4" NEW BUTTERFLY VALVE, BLIND FLANGE AND BLANK-OFF PLATE FOR FUTURE CONNECTION TO BUILDING 3.
- 4" NEW BUTTERFLY VALVE, BLIND FLANGE AND BLANK-OFF PLATE FOR FUTURE CONNECTION TO BUILDING 4B.
- FOR DEDUCTIVE ALTERNATE NO. 1, INCLUDES BASE BID MINUS ALL WORK ASSOCIATED WITH ROUTING 6" PIPING TO BASEMENT OF BUILDING 3, INCLUDING TEES, ISOLATION VALVES, BLIND FLANGES, BYPASS REDUCERS, BUILDING 3 PIPE SUPPORTS, CORE DRILLING OF PIPE TUNNEL BUILDING 3 EXTERIOR WALL AND EQUIPMENT SLAB INSTALLATION, THE BRANCH PIPING SERVING BUILDING 5 SHALL REMAIN, IF DEDUCT IS TAKEN, INSTALL SHUT-OFF VALVES, BLANK-OFF PLATES AND BYPASS VALVE ASSEMBLY ON THE MAIN LINE, PROVIDE POWER AND CONTROL POINTS FOR BYPASS VALVE AND PRESSURE SENSOR.
- FOR DEDUCTIVE ALTERNATE NO. 2, INCLUDES DEDUCTIVE ALTERNATE NO. 1 AND THE REMOVAL OF THE SUPPLY AND RETURN CHILLED WATER PIPING, SHUT-OFF VALVES AND BLANK-OFF PLATES SERVING BUILDING 5 AND 3 BACK TO THE CONNECTION POINT OF BUILDING 1, THE BRANCH PIPING SERVING BUILDING 1 SHALL REMAIN, IF DEDUCT IS TAKEN, INSTALL SHUT-OFF VALVES, BLANK-OFF PLATES AND BYPASS VALVE ASSEMBLY ON THE MAIN LINE, PROVIDE POWER AND CONTROL POINTS FOR BYPASS VALVE AND PRESSURE SENSOR.
- ALL SOIL REMOVAL, WITHIN THE OUTLINED AREA SHALL BE HAND EXCAVATED.
- NEW CHSR PIPING SHALL TURN DOWN TO 7'-0" BELOW EXISTING GRADE, AS MEASURED FROM BOTTOM OF TRENCH. ROUTE NEW CHILLED WATER PIPING BENEATH EXISTING STORM PIPING. ADJUST PIPE ELEVATION TO SUIT FIELD CONDITIONS.
- CORE DRILL PIPING AT APPROXIMATELY 5'-0" BE LOW FINISHED GRADE. CONTRACTOR SHALL FIELD DETERMINE EXACT HEIGHT OF PIPE PENETRATIONS.
- CHILLED WATER PIPING SHALL PENETRATE CONCRETE SUB-SURFACE WALL ABOVE EXISTING WALL FOUNDATION. PIPING SHALL BE INSTALLED NO LESS THAN 3'-0" BELOW GRADE. PRIOR TO THE START OF CONSTRUCTION, G.P.R. EXISTING WALL TO AVOID EXISTING REBAR FIELD ADJUST PIPE ELEVATION TO AVOID CONFLICT WITH REBAR. REFER TO F1MP401 FOR MORE INFORMATION.
- MOTOR OPERATED VALVE FOR BYPASS.
- PORT FOR PRESSURE SENSING.
- PIPE BLANKS FOR BUILDING 3.
- PIPE PLANKS FOR BUILDING 4B.
- EXISTING EHRM UNDERGROUND CONDUIT.
- EXISTING TREE.
- EXISTING EHRM MANHOLE.
- CONDUIT UP TO LEVEL ABOVE.
- ROUTE CONDUIT TIGHT TO BOTTOM OF CEILING STRUCTURE. CONTINUE CONTROLS CONDUIT ROUTING TO FOLLOW 6" CHILLED WATER PIPING TO BUILDING 3 CRAWL SPACE.
- BACNET AUTOMATED LOGIC PANEL TIE IN FOR CONTROL OF BYPASS VALVE.
- PROVIDE NEW CONDUIT DOWN TO PIPE TUNNEL BELOW. SEE DETAIL F7EE501 FOR FIRE STOPPING THE B.A.S. CONDUIT.
- SEE D1MP501 FOR CONTROL POINTS AND LOGIC FOR BYPASS VALVE AND PRESSURE TRANSMITTER.
- ROUTE B.A.S. UNDERGROUND RATED CONDUIT IN PIPING TRENCH TO BYPASS IN CRAWL SPACE OF BUILDING 3.
- REFER TO DETAIL C8MP501 FOR BYPASS VALVE ASSEMBLY DESIGNATED FOR USE IF DEDUCT ALTERNATES ARE TAKEN.
- TRANSITION UNDERGROUND PIPING ELEVATION FROM 7'-0" TO 8'-0" TO AVOID CONFLICT WITH EXISTING DUCTBANK BY ROUTING PIPE BENEATH EXISTING UTILITY.
- ROUTE NEW CHILLED WATER PIPING UP FROM 8'-0" TO 3'-0" AS MEASURED FROM BOTTOM OF TRENCH.

WALL RATING LEGEND

----- 2-HOUR RATED PARTITION



A

B

C

D

E

F

A

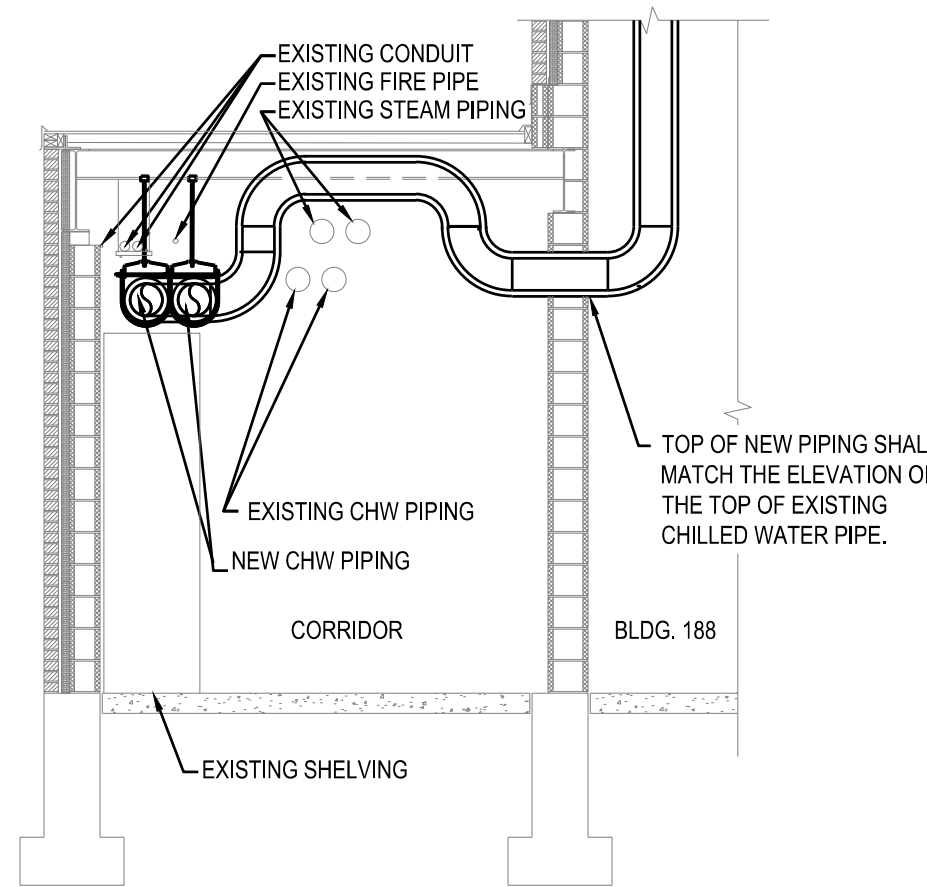
B

C

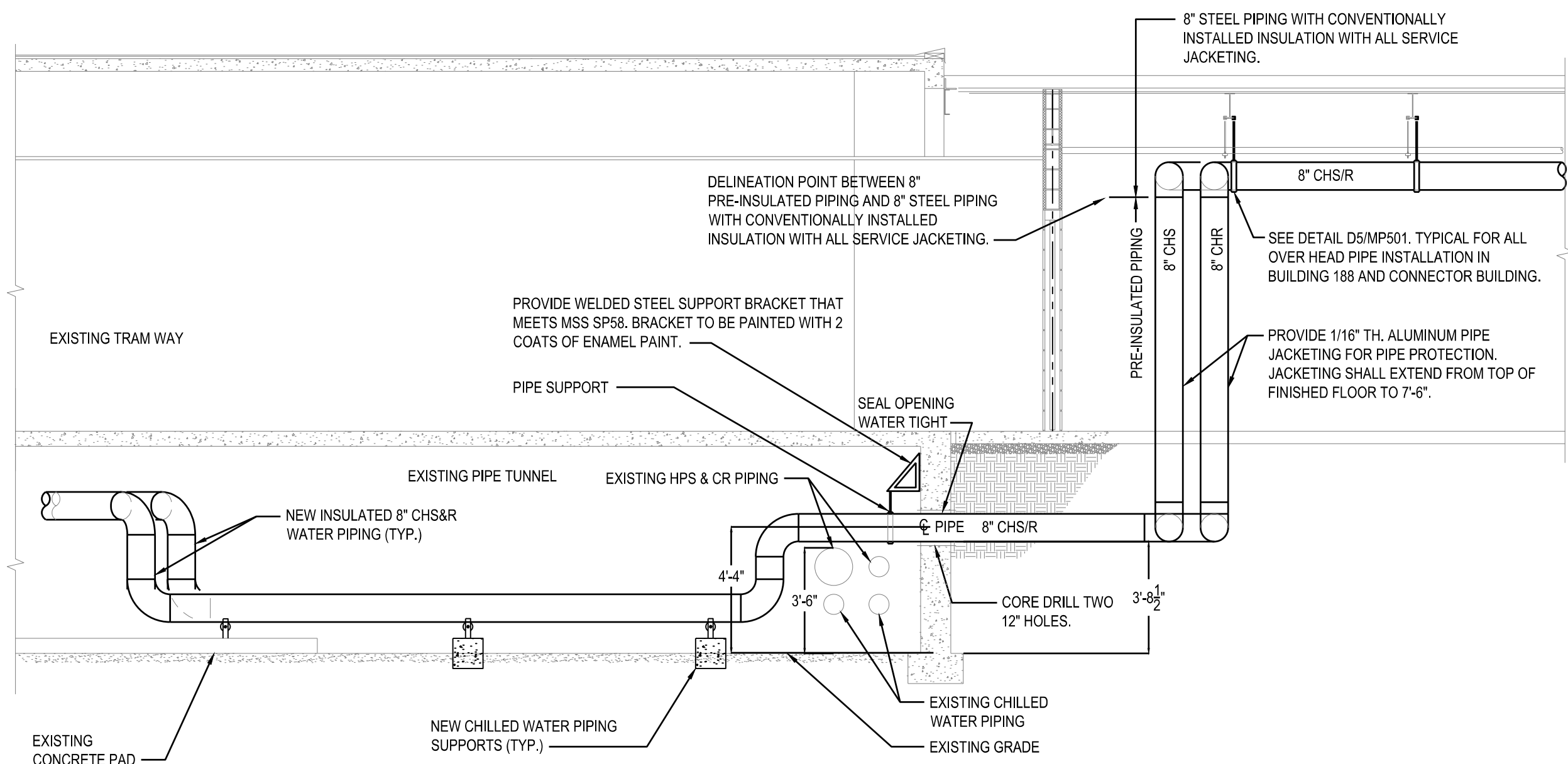
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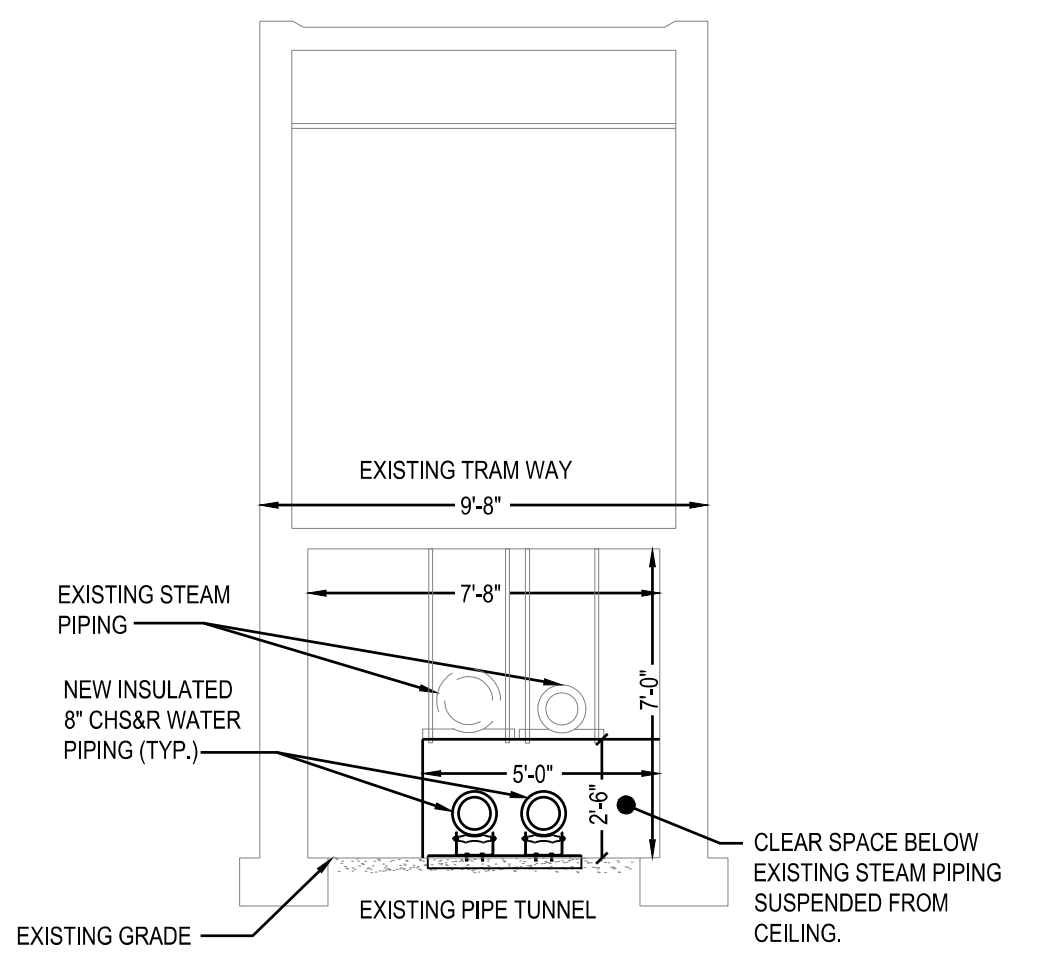
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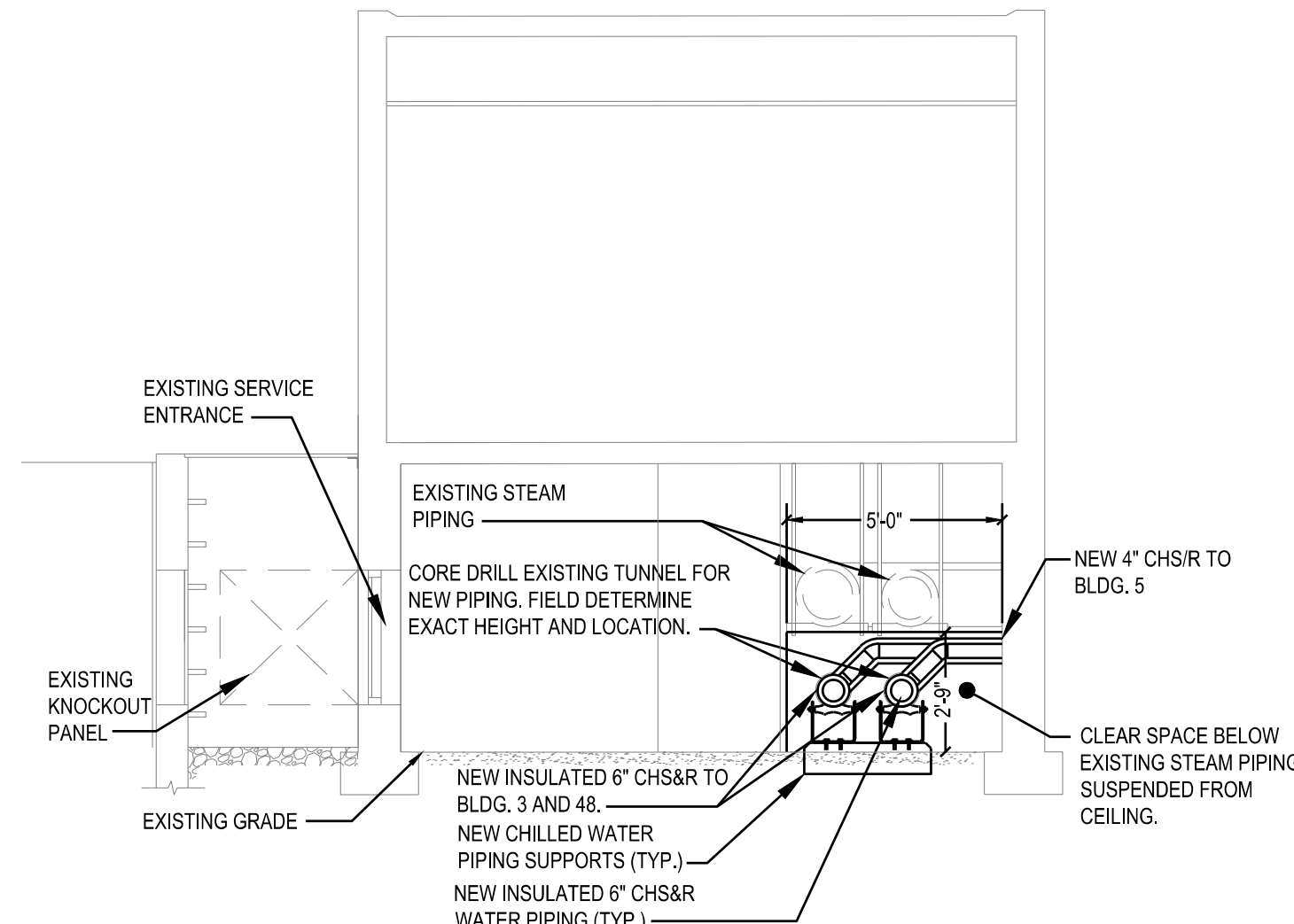
B9 CORRIDOR SECTION
SCALE: 1/4" = 1'-0"



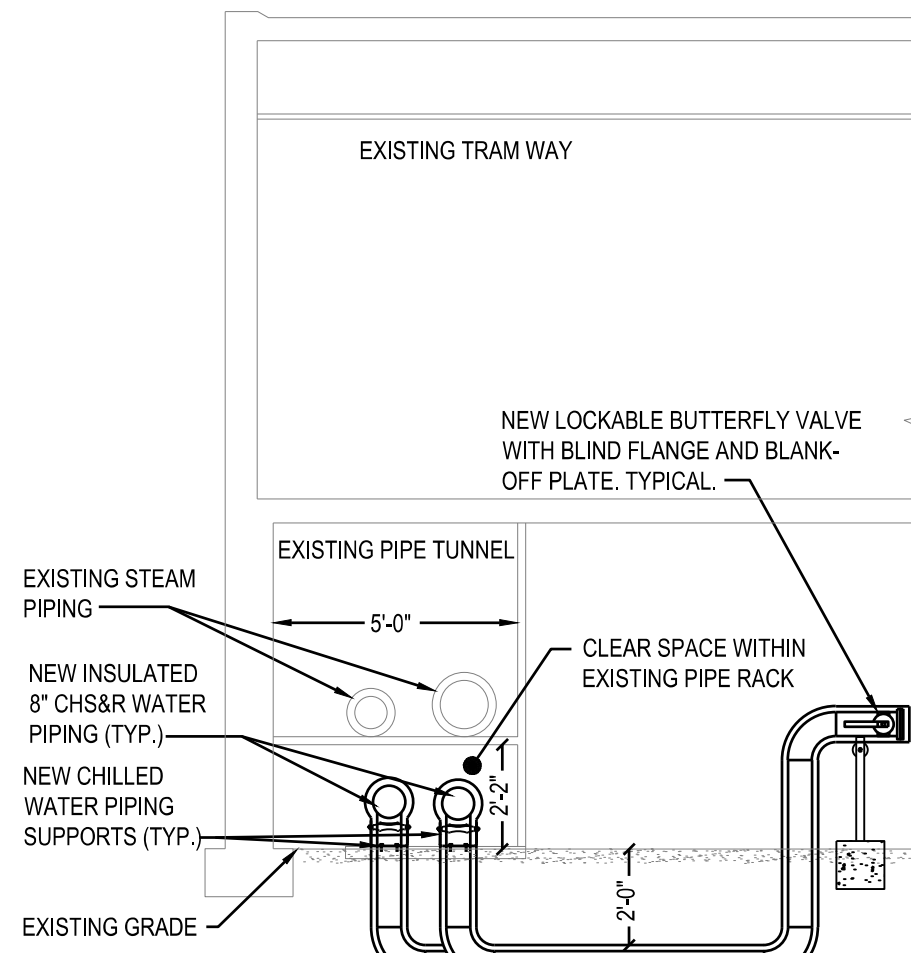
B2 TUNNEL SECTION
SCALE: 1/4" = 1'-0"



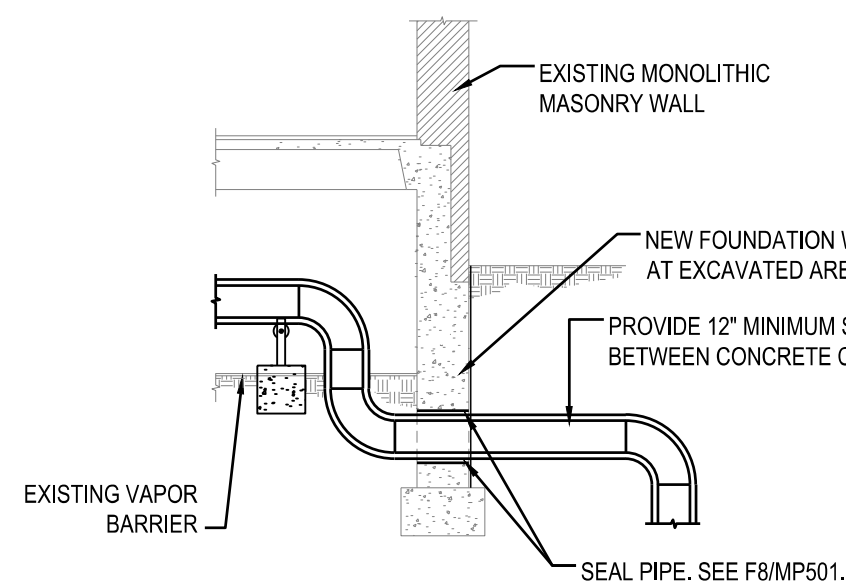
C4 TUNNEL SECTION
SCALE: 1/4" = 1'-0"



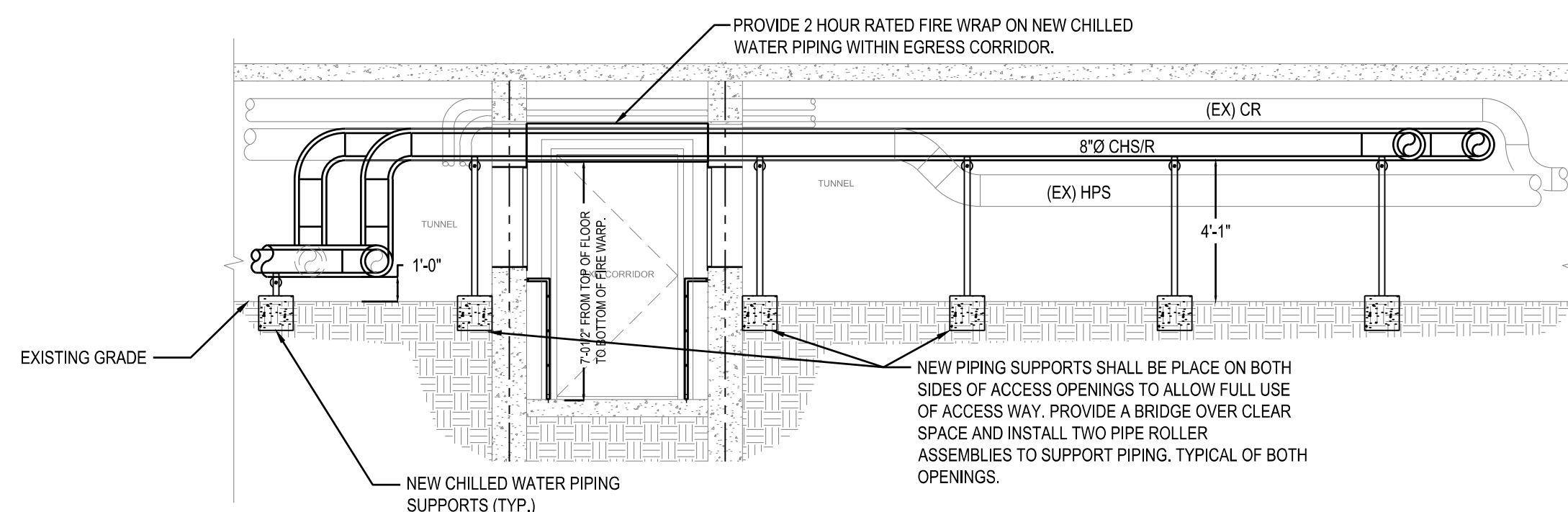
C6 TUNNEL SECTION
SCALE: 1/4" = 1'-0"



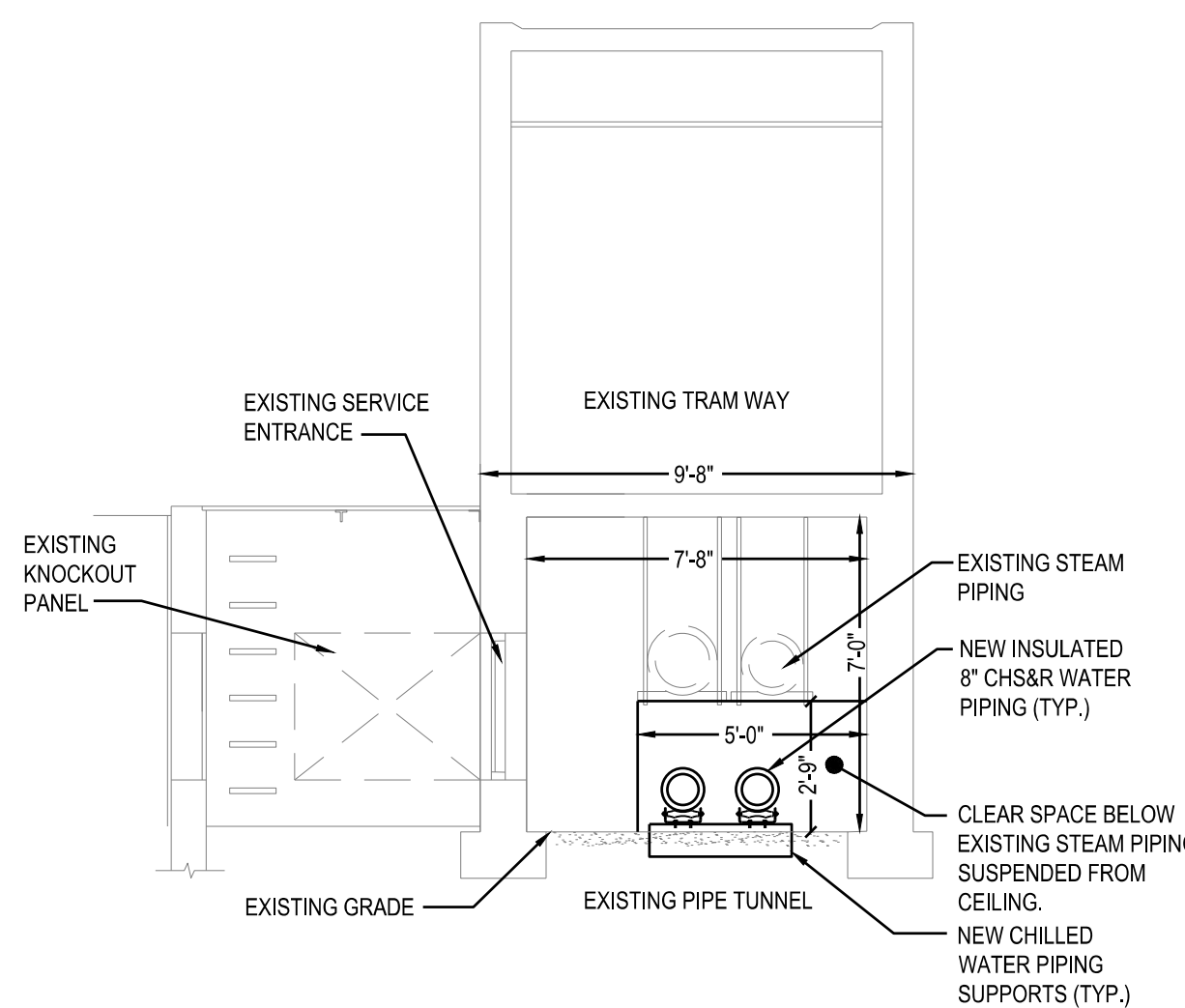
C9 TUNNEL SECTION
SCALE: 1/4" = 1'-0"



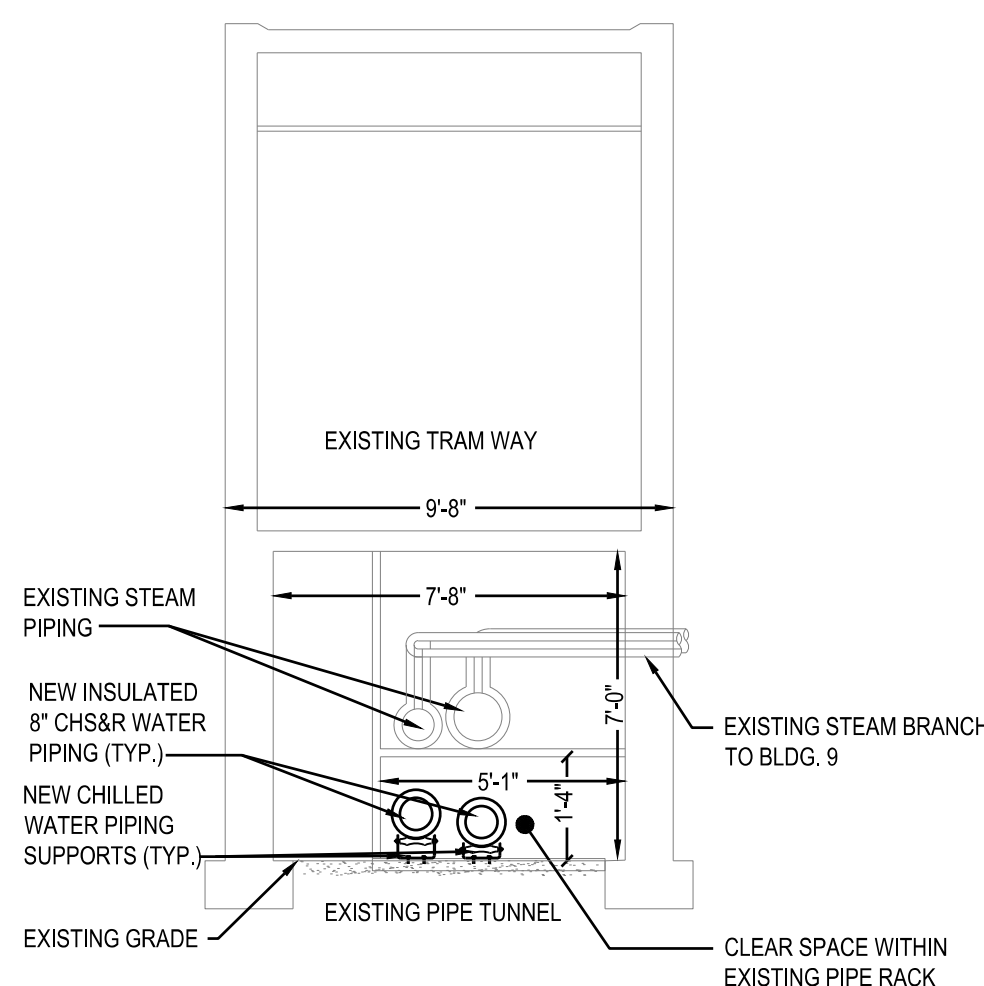
F1 BUILDING 3 EXTERIOR WALL PENETRATION
SCALE: 1/4" = 1'-0"



F3 TUNNEL SECTION
SCALE: 1/4" = 1'-0"



F6 TUNNEL SECTION
SCALE: 1/4" = 1'-0"



F9 TUNNEL SECTION
SCALE: 1/4" = 1'-0"

WALL RATING LEGEND

----- 2-HOUR RATED PARTITION

1	75% DD	05/17/2024
2	95% DD	06/18/2024
3	100% CD	07/26/2024
Revisions:		
Revision Number	Revisions:	Date:

ARCHITECT/ENGINEER OF RECORD
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PRESIDENT
HDG PROJECT: 22018
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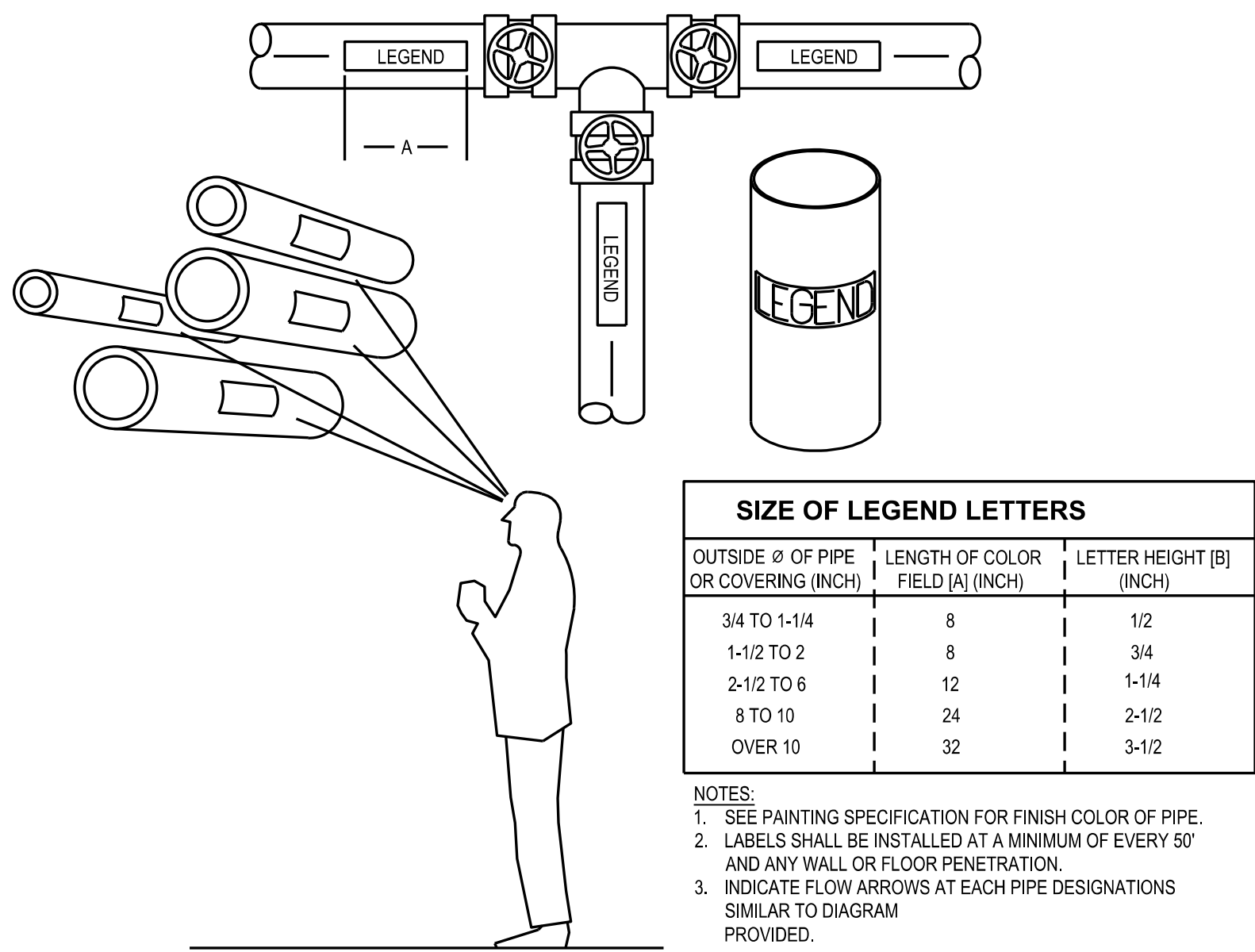
DRAWING TITLE
MECHANICAL SECTIONS AND
SCHEDULES
APPROVED

PHASE
100% CONSTRUCTION
DOCUMENTS - ISSUED
FOR CONSTRUCTION
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PROJECT TITLE
CHILLED WATER FROM B188 TO B3
CONTRACT NO. - 36C25219D0022
TASK ORDER NO. - 36C25223N0287
LOCATION
CAPTAIN JAMES A. LOVELL FEDERAL HEALTH CARE
CENTER (FHCC)
ISSUE DATE
JULY 26, 2024
CHECKED
WGG
DRAWN
ANB

PROJECT NUMBER
556-23-104
BUILDING NUMBER

DRAWING NUMBER
MP401

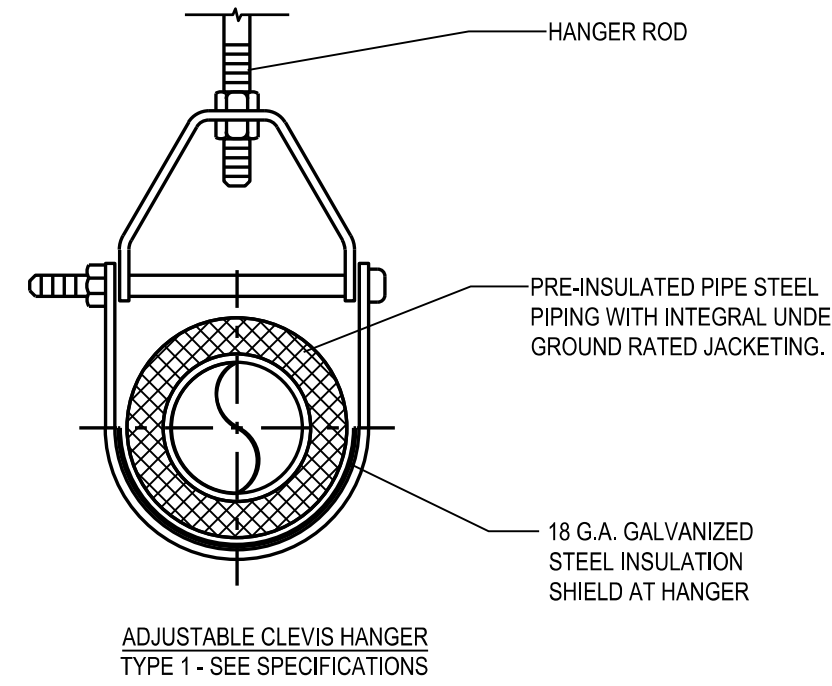


SIZE OF LEGEND LETTERS		
OUTSIDE Ø OF PIPE OR COVERING (INCH)	LENGTH OF COLOR FIELD (INCH)	LETTER HEIGHT (B) (INCH)
3/4 TO 1-1/4	8	1/2
1-1/2 TO 2	8	3/4
2-1/2 TO 6	12	1-1/4
8 TO 10	24	2-1/2
OVER 10	32	3-1/2

NOTES:
1. SEE PAINTING SPECIFICATION FOR FINISH COLOR OF PIPE.
2. LABELS SHALL BE INSTALLED AT A MINIMUM OF EVERY 50' AND ANY WALL OR FLOOR PENETRATION.
3. INDICATE FLOW ARROWS AT EACH PIPE DESIGNATIONS SIMILAR TO DIAGRAM PROVIDED.

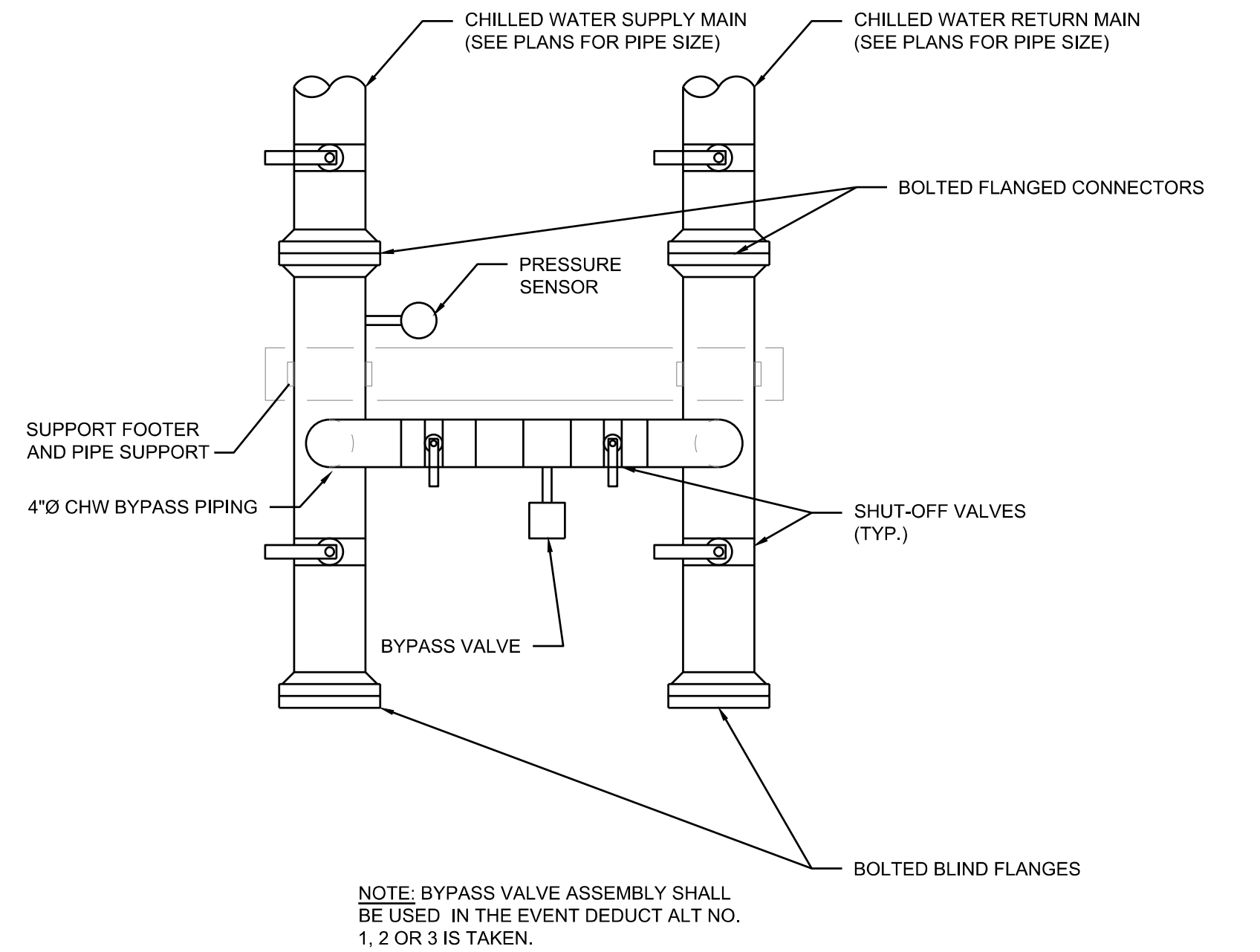
C3 PIPING SYSTEM IDENTIFICATION

SCALE: NONE



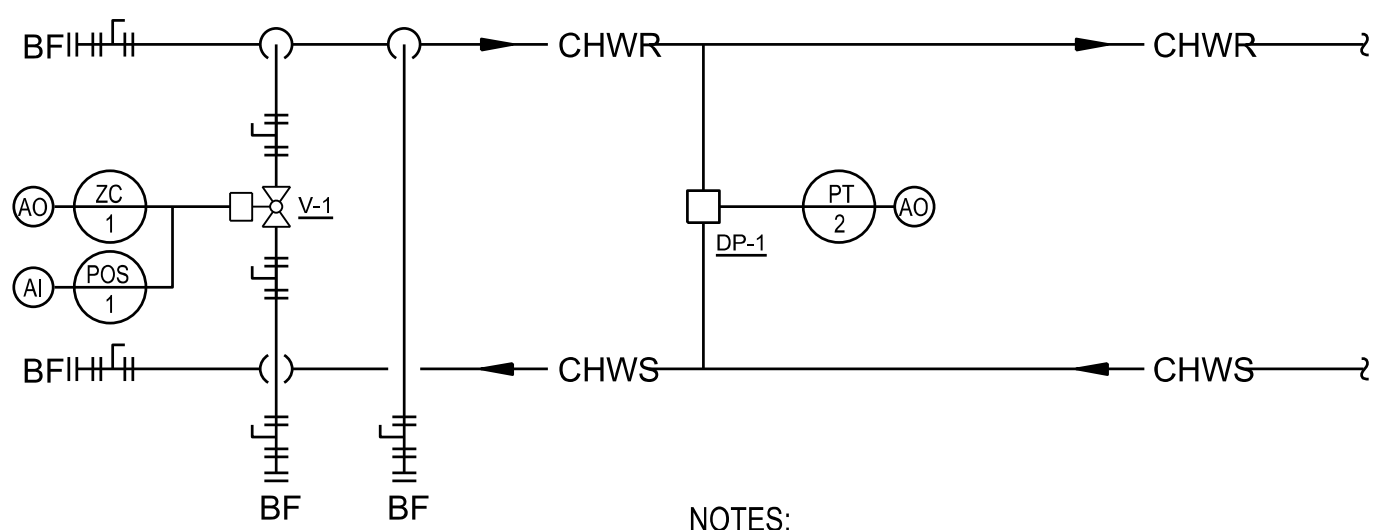
C5 PIPE HANGERS

SCALE: NONE

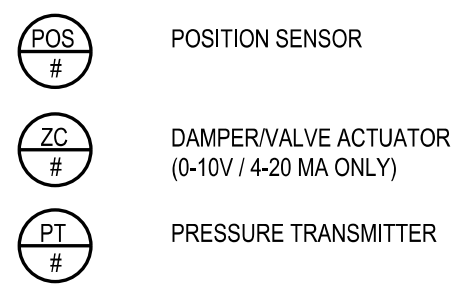


C8 CHILLED WATER PIPING BYPASS

SCALE: NONE



CONTROLS SYMBOLS

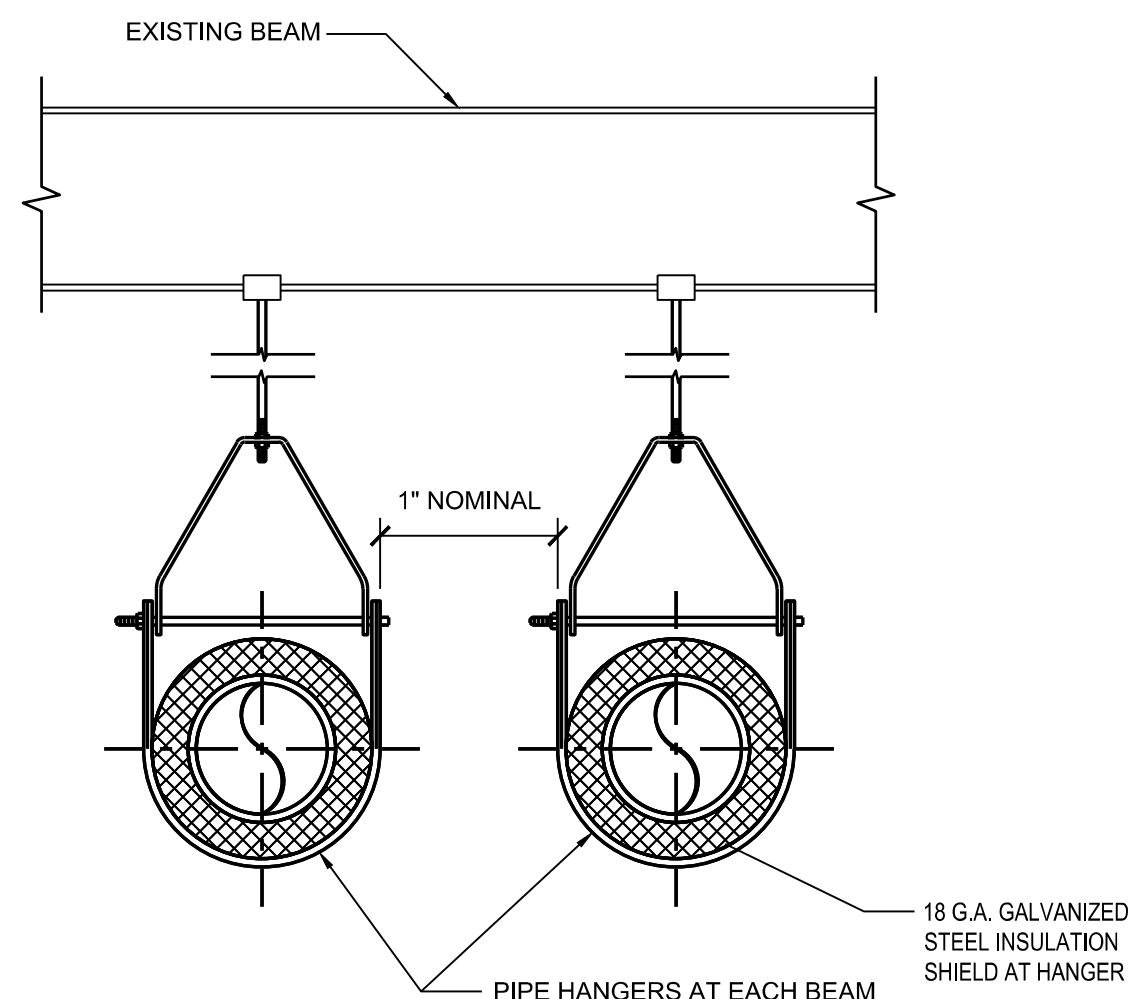


- NOTES:
1. INTEGRATE NEW BYPASS VALVE (V-1) AND DIFFERENTIAL PRESSURE SENSOR (DP-1) INTO EXISTING B.A.S. BYPASS VALVE SHALL REMAIN CLOSED UNDER NORMAL CONDITIONS.
 2. BYPASS VALVE SHALL PROVIDE CIRCULATION DURING LOW AMBIENT CONDITIONS TO ASSIST IN FREEZE PROTECTION.
 3. PRESSURE MONITORING IS TO PROVIDE SIGNAL TO CHILLED WATER PUMP VFDS TO MAINTAIN ADEQUATE CHW SUPPLY PRESSURE FOR BUILDINGS 3 & 48.
 4. BYPASS VALVE SHALL BE FLANGED IRON GLOBE VALVE WITH BRONZE SEATS AND 120 VOLT ACTUATOR WITH POSITION SENSOR BY BELIMO OR EQUAL.
 5. CONTROL ELEMENTS SHALL BE INSTALLED AND TESTED WHILE PIPING IS FLOODED. ONCE PIPING IS DRAINED, THE CONTROL ALGORITHM FOR THE BYPASS VALVE SHALL DISABLE. WHEN THE PRESSURE SENSOR DETERMINES PIPING IS FILLED WITH WATER AGAIN, AN ALARM SHALL BE PROVIDED TO ENABLE BYPASS VALVE.

SYSTEM: CHILLED WATER SUPPLY AND RETURN	POINT ID	ABBREVIATION	PROPERTY/ADJUTANT/TERMINAL ID	START/STOP	SPEED COMMAND	VALVE POSITION	PRESSURE	ALARM	PRESSURE	PRESSURE	POSITION	LOCAL ALARM PANEL	REMARKS
SYSTEM COMPONENT:													
BYPASS VALVE V-1	AO-1	V-1											1,2,4,5
BYPASS VALVE V-1	AI-1	V-1											1,2,4,5
DIFFERENTIAL PRESSURE SENSOR DP-1	AO-2	DP-1											1,3,5

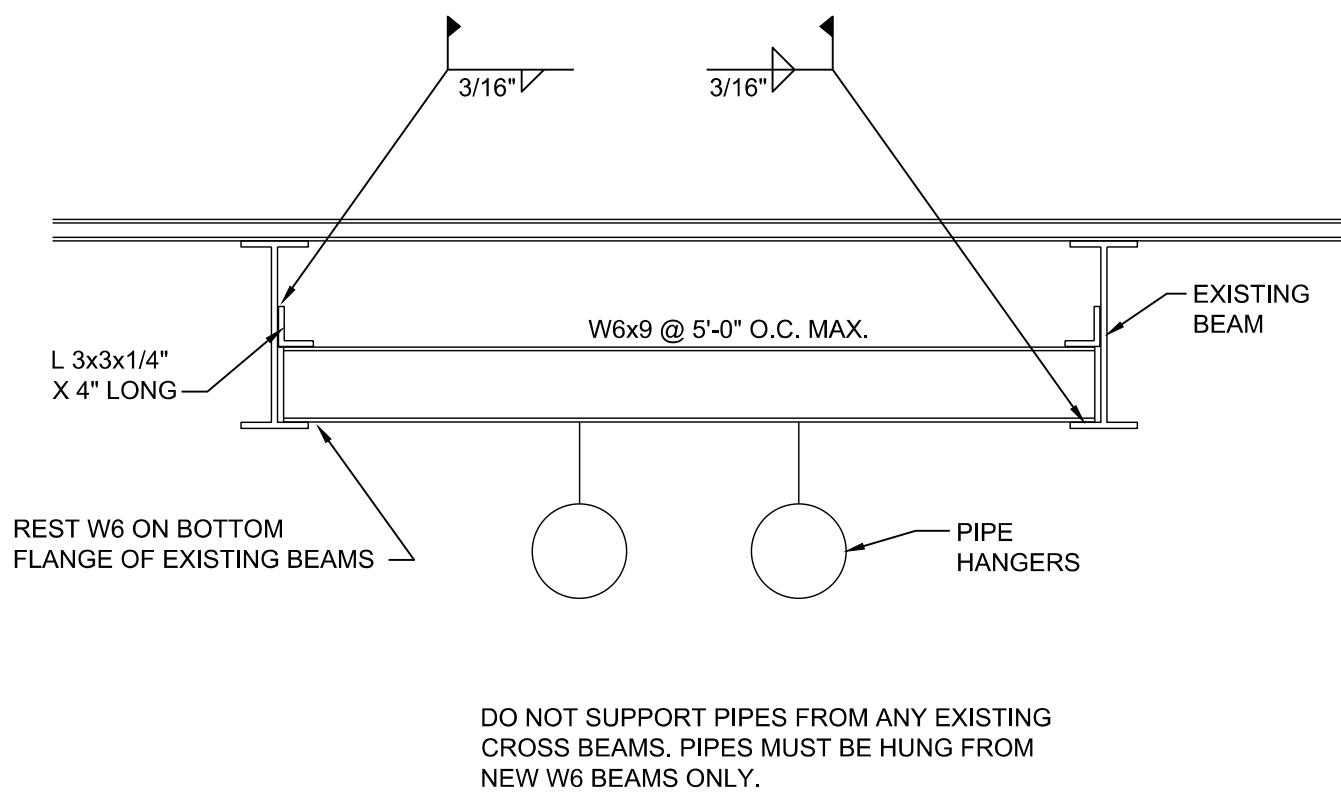
D1 PIPING CONTROLS AND POINTS LIST AT END OF PIPE RUN

SCALE: NONE



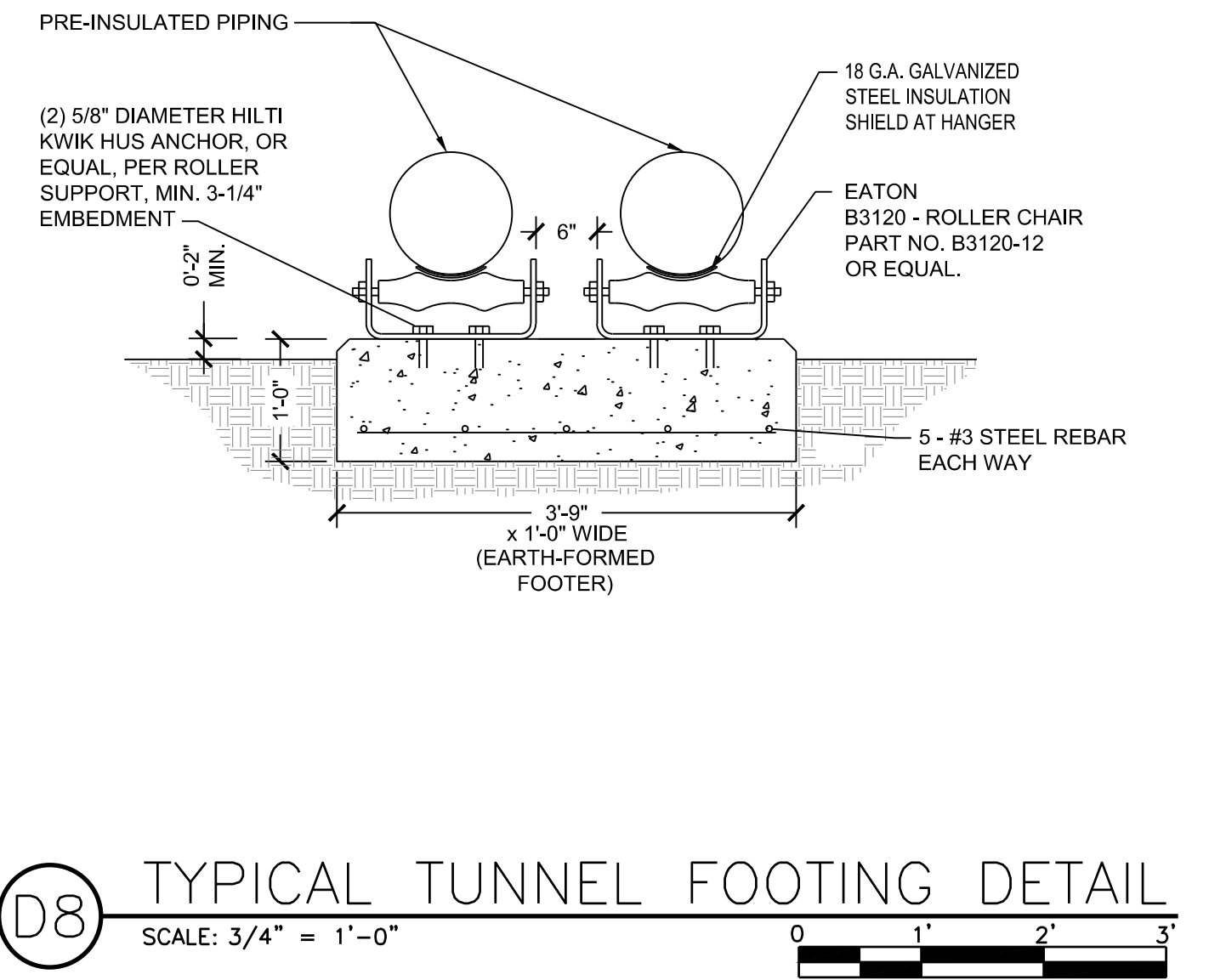
D5 PIPE RUNNING PERPENDICULAR TO EXISTING ROOF BEAMS AT CORRIDOR-BLDG 188

SCALE: NONE



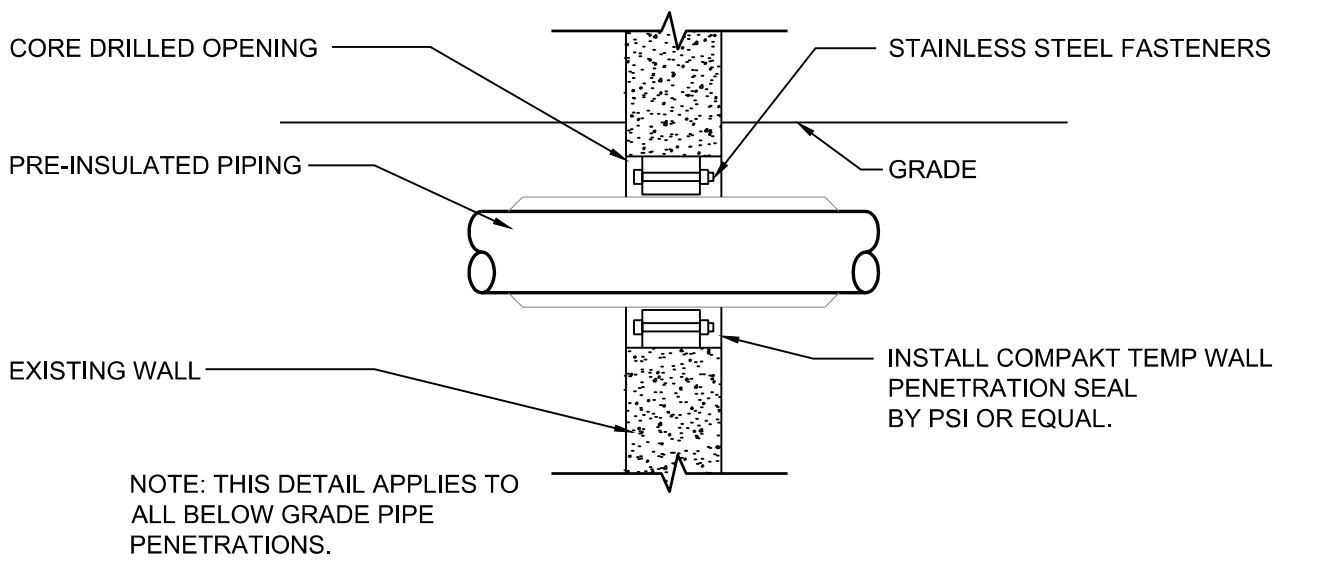
F5 PIPE SUPPORT AT ROOF BEAMS - BLDG 188

SCALE: NONE



D8 TYPICAL TUNNEL FOOTING DETAIL

SCALE: 3/4" = 1'-0"



F8 BELOW GRADE PIPING WALL PENETRATION DETAIL

SCALE: NONE

1	75% DD	05/17/2024
2	95% DD	06/18/2024
3	100% CD	07/26/2024
Revision Number	Revisions:	Date:

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DRAWING TITLE
MECHANICAL
DETAILS

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PHASE
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DOCUMENTS - ISSUED
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PROJECT TITLE
CHILLED WATER FROM B188 TO B3
CONTRACT NO. - 36C25219D0022
TASK ORDER NO. - 36C25223N0287

LOCATION
CAPTAIN JAMES A. LOVELL FEDERAL HEALTH CARE
CENTER (FHCC)

ISSUE DATE
JULY 26, 2024

CHECKED
WGG

DRAWN
ANB

PROJECT NUMBER
556-23-104
BUILDING NUMBER

DRAWING NUMBER
MP501

ELECTRICAL SYMBOL LEGEND

	SOLID LINE DENOTES CONDUIT RUN CONCEALED IN WALL, ABOVE CEILINGS, EXPOSED IN UNFINISHED AREAS, RUN PARALLEL OR PERPENDICULAR TO STRUCTURE OR WALL. DASHED LINE DENOTES CONDUIT RUN BELOW GRADE OR BELOW FINISHED FLOOR.
	HOMERUN TO PANELBOARD. QUANTITY OF ARROWS INDICATES NUMBER OF CIRCUITS. EACH UNGROUNDED CONDUCTOR SHALL BE PAIRED WITH A SEPARATE NEUTRAL (IF REQ'D). SHARING OF NEUTRAL CONDUCTORS BETWEEN BRANCH CIRCUITS IS NOT PERMITTED.
	SURFACE OR RECESSED MOUNTED LIGHTING FIXTURE CONNECTED TO EQUIPMENT BRANCH CIRCUIT. LETTER INDICATES TYPE. SEE LIGHTING FIXTURE SCHEDULE FOR EXACT REQUIREMENTS.
	SINGLE-POLE, 20 AMP, 120/277 VOLT, MOTOR RATED TOGGLE SWITCH WITH INTEGRAL OVERLOAD PROTECTION
	EXISTING WALL MOUNTED OSCILLATING FAN
	208Y/120 OR 120/240 VOLT PANELBOARD, FLUSH AND SURFACE MOUNTED RESPECTIVELY. DESIGNATION AS INDICATED. SEE PANELBOARD SCHEDULE FOR EXACT REQUIREMENTS.

ABBREVIATIONS & ACRONYMS

THIS SCHEDULE IS REPRESENTATIVE OF THOSE TYPICALLY FOUND ON THE PLANS BUT NOT ALL MAY BE UTILIZED

A	AMPERE, AMMETER	KW	KILOWATTS
AF	AMPERE FRAME	LC	LIGHTING CONTACTOR
AFF	ABOVE FINISHED FLOOR	LTG	LIGHTING
AIC	AMPERES INTERRUPTING CAPACITY		PROVIDE FURNISH, INSTALL & CONNECT
AT	AMPERE TRIP	RECPT	RECEPTACLE
ATS	AUTOMATIC TRANSFER SWITCH	RL	EXISTING ITEM TO BE RELOCATED
C	CONDUIT	RM	EXISTING ITEM TO REMAIN
CL	CENTER LINE	RP	EXISTING ITEM TO BE REPLACED
CR	CARD READER ACCESS	SC	SECURITY CAMERA OUTLET
CU	COPPER	SN	SOLID NEUTRAL
EC	EMPTY CONDUIT	SW	SWITCH
EMER	EMERGENCY	SWBD	SWITCHBOARD
EWC	ELECTRIC WATER COOLER	TV	COMBINATION TV & RECEPT OUTLET
EX	EXISTING	TYP	TYPICAL
G, GND	GROUND	UG	UNDER GROUND
GF	GROUND FAULT INTERRUPTER	UNO	UNLESS NOTED OTHERWISE
HH	HANDHOLE	UOI	UNLESS OTHERWISE INDICATED
HP	HORSEPOWER	V	VOLT
HZ	HERTZ	LC	LIGHTING CONTACTOR
IG	ISOLATED GROUND	WP	WEATHERTIGHT GFI
JB	JUNCTION BOX	WAP	WIRELESS ACCESS POINT
KVA	KILOVOLT-AMPERES	XFMR	TRANSFORMER

GENERAL NOTES

ELECTRICAL GENERAL NOTES

7. THE SITE SHALL BE INSPECTED PRIOR TO SUBMITTING PRICING. THE DRAWINGS ARE INTENDED TO COVER THE REQUIRED ELECTRICAL SYSTEMS, YET MAY NOT SHOW THE COMPLETE OR ACCURATE DETAIL OF THE BUILDING OR SYSTEM IN EVERY RESPECT. OBTAIN ANY ADDITIONAL INFORMATION AS REQUIRED.
2. COMPLY WITH THE LATEST EDITION OF THE NEC AND ALL APPLICABLE CODES.
3. FURNISH AND INSTALL ALL MATERIALS IN A NEAT AND WORKMANLIKE METHOD. ALL MATERIALS SHALL BE NEW, OF FIRST QUALITY, AND SHALL BEAR THE U.L. LABEL.
4. ALL DIMENSIONS AND CLEARANCES SHALL BE VERIFIED PRIOR TO INSTALLATION OF EQUIPMENT AND RACEWAYS. COORDINATE WORK WITH ALL OTHER TRADES AS REQUIRED.
5. CONDUIT SHALL BE ELECTRICAL METALLIC TUBING (EMT) FOR BRANCH CIRCUIT WIRING AS PERMITTED BY THE NEC. FITTINGS SHALL BE INSULATED THROAT STEEL COMPRESSION TYPE. SET SCREW OR CRIMP TYPE FITTINGS ARE NOT ACCEPTABLE. WHERE EXPOSED TO PHYSICAL DAMAGE, CONDUITS SHALL BE RIGID GALVANIZED STEEL. MINIMUM CONDUIT SIZE SHALL BE 3/4".
6. ALL CONDUCTORS SHALL BE TYPE THHN/TWN COPPER BUILDING WIRE. MINIMUM SIZE SHALL BE #12 COPPER UNLESS NOTED OTHERWISE. ALL WIRING SHALL BE CONDUCTORS IN CONDUIT. MANUFACTURED WIRING SYSTEMS (E.G. TYPE "MC" CABLE) SHALL NOT BE PERMITTED. UNDERGROUND CONDUITS SHALL BE PVC SCHEDULE 40, WITH TRANSITION TO RIGID GALVANIZED, AND WITH CORROSION PREVENTION TAPE BELOW GRADE FOR ALL EXPOSED CONDUITS.
7. PROVIDE GROUNDING FOR ALL EQUIPMENT IN ACCORDANCE WITH ARTICLE 250 OF THE NATIONAL ELECTRICAL CODE.
8. ALL ENCLOSURES SHALL BE OF THE NEMA TYPE APPLICABLE TO THE INTENDED INSTALLATION.
9. ALL SAFETY SWITCHES (FUSIBLE AND NON-FUSIBLE) SHALL BE RATED HEAVY DUTY WITH QUICK MAKE, QUICK BREAK OPERATING MECHANISMS.
10. SEAL ALL CONDUIT PENETRATIONS TO MATCH THE RATING OF THE WALL OR SURFACE THAT HAS BEEN OR IS BEING PENETRATED. APPLY FIRESTOPPING TO ELECTRICAL PENETRATIONS OF FIRE-RATED FLOOR AND WALL ASSEMBLIES TO RESTORE ORIGINAL FIRE-RESISTANCE RATING OF ASSEMBLY. COORDINATE WITH THE OWNER AND FACILITY FIRESTOPPING MATERIALS AND METHODS CONSISTENT WITH EXISTING PRACTICES IN THE OWNER'S FACILITIES. USE ONLY APPROVED ASSEMBLIES THAT ARE EQUAL TO OR EQUIVALENT WITH THE FOLLOWING:

CONDUIT PENETRATIONS OF 1 & 2 HOUR GYPSOBOARD WALLS - U.L.	#W-L-1001
CONDUIT PENETRATIONS OF 1 & 2 HOUR CONCRETE OR BLOCK WALLS - U.L.	#C-AJ-1001
CONDUIT PENETRATIONS OF 1 & 2 HOUR CONCRETE FLOORS - U.L.	#C-AJ-1001
11. ALL WORK SHALL HAVE PROPER LABELING AND NAMEPLATES. ALL CIRCUITS SHALL BE LABELED AT PANELBOARDS AND BOXES AS INDICATED. ALL DISTRIBUTION EQUIPMENT (PANELBOARDS, SAFETY SWITCHES, ETC.) SHALL BE PERMANENTLY MARKED IDENTIFYING THE EQUIPMENT SERVED. PROVIDE ALL PANELBOARDS WITH ACCURATE TYPEWRITTEN SCHEDULES.
12. ALL CIRCUIT BREAKERS ON CIRCUITS SUPPLYING HVAC EQUIPMENT SHALL BE HACR RATED.
13. THE CONTRACTOR SHALL THOROUGHLY CLEAN ALL EQUIPMENT AND SYSTEMS PRIOR TO OPERATION. RESTORE FINISHED SURFACES WHERE DAMAGED AND DELIVER THE ENTIRE INSTALLATION IN AN APPROVED CONDITION.
14. THE OWNER'S PERSONNEL OR REPRESENTATIVE SHALL BE INSTRUCTED IN THE PROPER OPERATION AND MAINTENANCE OF THE SYSTEMS. FURNISH TO THE OWNER THREE SETS OF OPERATION AND MAINTENANCE MANUALS FOR EACH SYSTEM.
15. THE WORK INSTALLED UNDER THIS CONTRACT SHALL BE GUARANTEED FOR A PERIOD OF ONE YEAR MINIMUM AFTER THE DATE OF FINAL ACCEPTANCE. DEFECTS THAT APPEAR AS A RESULT OF NORMAL USAGE SHALL BE REMEDIED BY THE CONTRACTOR TO THE COMPLETE SATISFACTION OF AND WITHOUT ADDITIONAL COST TO THE OWNER. ALL MANUFACTURER'S WARRANTY SHALL MATCH ALL SPECIFICATION SECTIONS.
16. A CURRENT SET OF CONTRACT DOCUMENTS SHALL BE MAINTAINED, INCLUDING APPROVED SUBMITTALS. FOR THE DURATION OF THE PROJECT, ALL CHANGES TO WORK SHALL BE NEATLY AND ACCURATELY MARKED IN RED AND SHALL BE TURNED OVER TO THE OWNER AT THE COMPLETION OF THE PROJECT.
17. PROVIDE COMPOSITE COORDINATION PLANS TO ENGINEERING INDICATING ALL TRADES WORK AND AREAS OF CONFLICT WITH RESOLUTIONS.
18. ALL 20 AMP BRANCH CIRCUIT WIRE SIZING SHALL BE IN ACCORD WITH THE FOLLOWING TABLE: UNLESS LISTED AS LARGER ON SCOPE OF WORK, DRAWINGS, OR SPECIFICATIONS.

VOLTAGE	DISTANCE	HOMERUN	REMAINDER OF CIRCUIT
120/208	0 - 50'	#12	#12
	51 - 100'	#10	#12
	101 - 150'	# 8	#10
277/480	0' - 125'	#12	#12
	125' - 220'	#10	#12
	220' - 330'	# 8	#10

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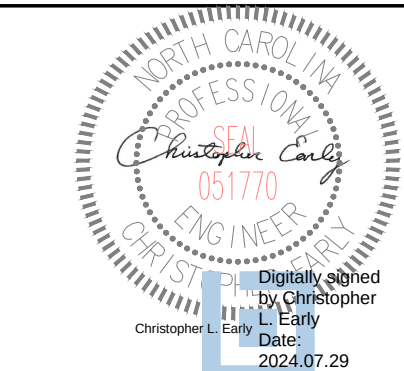
ARCHITECT/ENGINEER OF RECORD

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PRESIDENT
HDG PROJECT: 22018



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DRAWING TITLE

ABBREVIATIONS, LEGENDS, AND GENERAL NOTES

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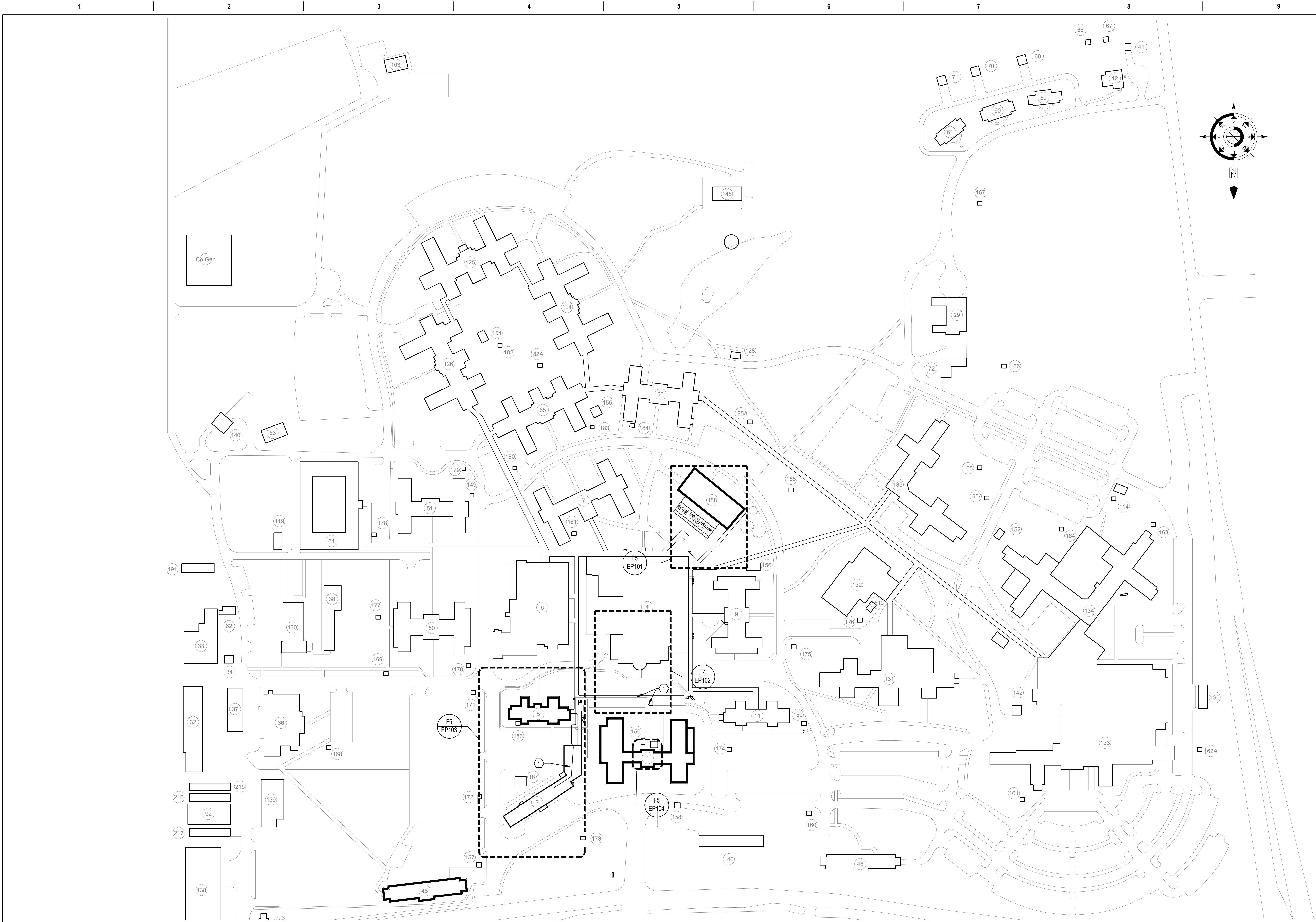
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PROJECT NUMBER

556-23-104
BUILDING NUMBER

DRAWING NUMBER

EG001



F5 OVERALL SITE PLAN
SCALE: 1" = 100'

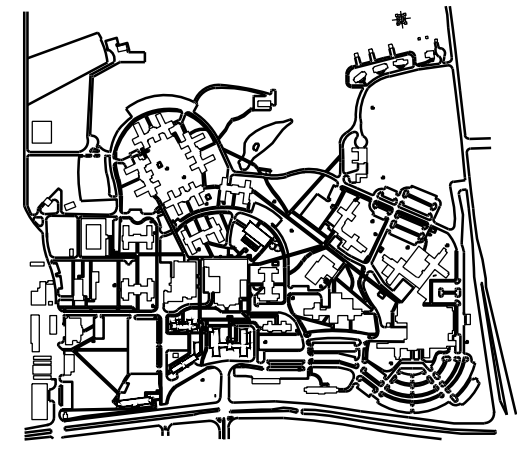
GENERAL SHEET NOTES

- A. VOLTAGE DROP SHALL NOT EXCEED VA DESIGN MANUAL GUIDELINES, 3% MAXIMUM ON BRANCH CIRCUITS AND 2% MAXIMUM ON FEEDERS.
- B. COORDINATE EXACT CONDUIT ROUTING IN THE FIELD PRIOR TO ROUGH-IN AND INSTALLATION. PATCH ANY HOLES CREATED IN HORIZONTAL AND VERTICAL SURFACES TO MAINTAIN ANY EXISTING RATINGS.
- C. PROTECT AND MAINTAIN EXISTING BUILDING GROUNDING SYSTEMS (I.E. GROUND RINGS, RODS, ETC.) OF ALL BUILDINGS IN SCOPE WHEN ENCOUNTERED.
- D. COORDINATE ALL UNDERGROUND CONSTRUCTION WITH VA CAMPUS SECURITY.
- E. DAMAGE TO ANY EXISTING UTILITIES SHALL BE REPAIRED IN ACCORDANCE WITH THE VA STANDARD CONSTRUCTION SPECIFICATIONS AT NO ADDITIONAL COST.

SHEET KEYNOTES

- 1. ELECTRICAL CONTRACTOR TO ROUTE BYPASS VALVE ELECTRICAL CONDUIT TIGHT TO BOTTOM OF CEILING STRUCTURE TO CHILLED WATER TUNNEL. CONTINUE ELECTRICAL CONDUIT ROUTING TO FOLLOW 6" CHILLED WATER PIPING TO BUILDING 3 CRAWL SPACE.

KEYPLAN



1	75% DD	05/17/2024
2	95% DD	06/18/2024
3	100% CD	07/26/2024
Revision Number	Revisions:	Date:

ARCHITECT/ENGINEER OF RECORD

A/E:
HARRELL DESIGN GROUP PC
10720 SIKES PLACE, SUITE 100
CHARLOTTE, NC 28227
704-814-1346
POC: LEE HARRELL PE, LEED AP, QCCP
PRESIDENT
HDG PROJECT: 22018

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U.S. Department
of Veterans Affairs

DRAWING TITLE

OVERALL ELECTRICAL SITE
PLAN

APPROVED

PHASE		PROJECT TITLE		PROJECT NUMBER	
100% CONSTRUCTION DOCUMENTS - ISSUED FOR CONSTRUCTION		CHILLED WATER FROM B188 TO B3 CONTRACT NO. - 36C25219D0022 TASK ORDER NO. - 36C25223N0287		556-23-104	
FULLY SPRINKLERED		LOCATION CAPTAIN JAMES A. LOVELL FEDERAL HEALTH CARE CENTER (FHCC)		BUILDING NUMBER ---	
		ISSUE DATE JULY 26, 2024	CHECKED LEH	DRAWN CLE	DRAWING NUMBER ES001

A

B

C

D

E

F

A

B

C

D

E

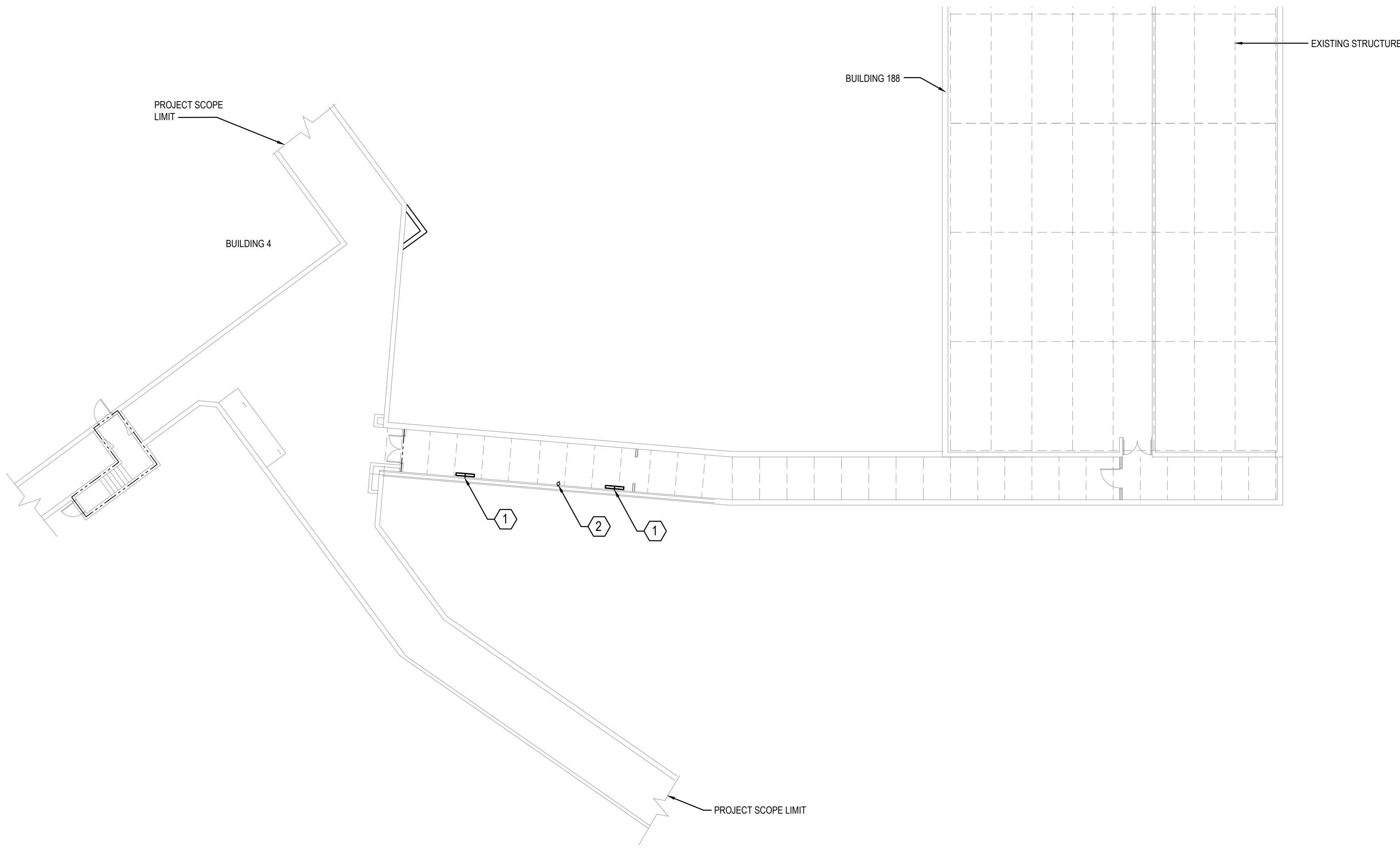
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GENERAL SHEET NOTES

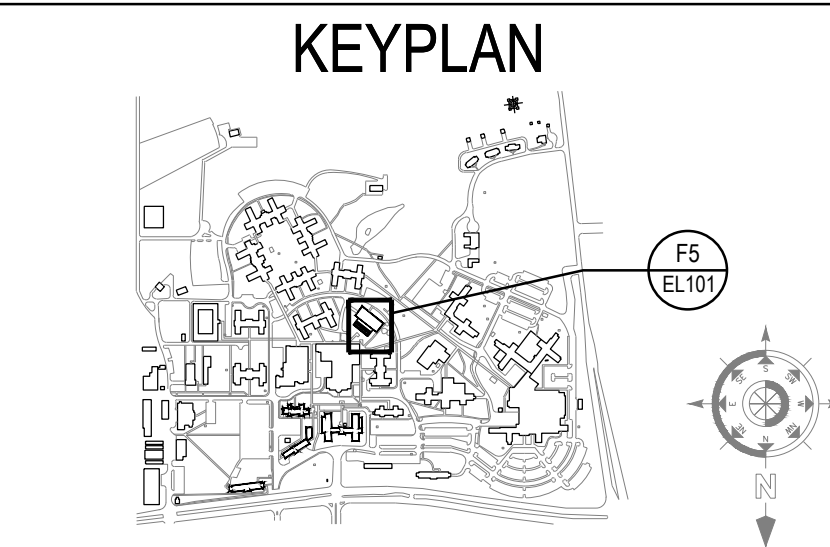
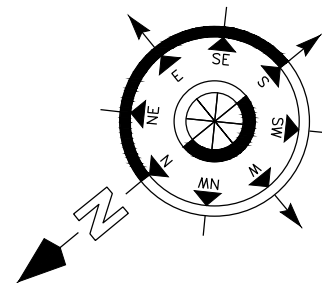
A. VOLTAGE DROP SHALL NOT EXCEED VA DESIGN MANUAL GUIDELINES, 3% MAXIMUM ON BRANCH CIRCUITS AND 2% MAXIMUM ON FEEDERS.

SHEET KEYNOTES

1. ELECTRICAL CONTRACTOR TO DISCONNECT AND REMOVE ALL LIGHTING FIXTURES WITHIN THE AREA OF WORK. COORDINATE SHUTDOWN OF EXISTING LIGHT FIXTURE CIRCUIT WITH VA PRIOR TO DEMOLITION. COORDINATE FINAL MOUNTING HEIGHT OF LIGHTING FIXTURE, DURING REINSTALLATION TO EXISTING LIGHTING CIRCUIT SERVING THE AREA.
2. ELECTRICAL CONTRACTOR TO REMOVE WALL MOUNTED OSCILLATING FAN. COORDINATE FINAL LOCATION AND MOUNTING HEIGHT OF OSCILLATING FAN DURING REINSTALLATION.



(F5) BUILDING 188 FLOOR PLAN
SCALE: 1/16" = 1'-0"



1	75% DD	05/17/2024
2	95% DD	06/18/2024
3	100% CD	07/26/2024
Revision Number	Revisions:	Date:

ARCHITECT/ENGINEER OF RECORD

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POC: LEE HARRELL PE, LEED AP, QCxP
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HDG PROJECT: 22018

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of Veterans Affairs

DRAWING TITLE

BUILDING 188 FLOOR PLAN

APPROVED

PHASE

100% CONSTRUCTION
DOCUMENTS - ISSUED
FOR CONSTRUCTION

FULLY SPRINKLERED

PROJECT TITLE

CHILLED WATER FROM B188 TO B3
CONTRACT NO. - 36C25219D0022
TASK ORDER NO. - 36C25223N0287

LOCATION
CAPTAIN JAMES A. LOVELL FEDERAL HEALTH CARE
CENTER (FHCC)

ISSUE DATE
JULY 26, 2024

CHECKED
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CLE

PROJECT NUMBER

556-23-104

BUILDING NUMBER

188

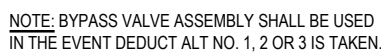
DRAWING NUMBER

EP101

4. NEW CHILLED WATER PIPING SHALL BE ROUTED WITHIN PIPING TUNNEL, UNLESS NOTED OTHERWISE.
5. VOLTAGE DROP SHALL NOT EXCEED VAS DESIGN MANUAL GUIDELINES; 3% MAXIMUM ON BRANCH CIRCUITS AND 2% MAXIMUM ON FEEDERS.
6. COORDINATE EXACT ELECTRICAL CONDUIT ROUTING IN THE FIELD PRIOR TO ROUGH-IN AND INSTALLATION. PATCH ANY HOLES CREATED IN HORIZONTAL AND VERTICAL SURFACES TO MAINTAIN ANY EXISTING RATINGS.

SHEET KEYNOTES

ELECTRICAL CONTRACTOR TO ROUTE BYPASS VALVE
ELECTRICAL CONDUIT TIGHT TO BOTTOM OF CEILING

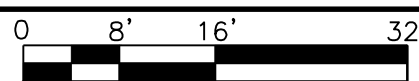


B8

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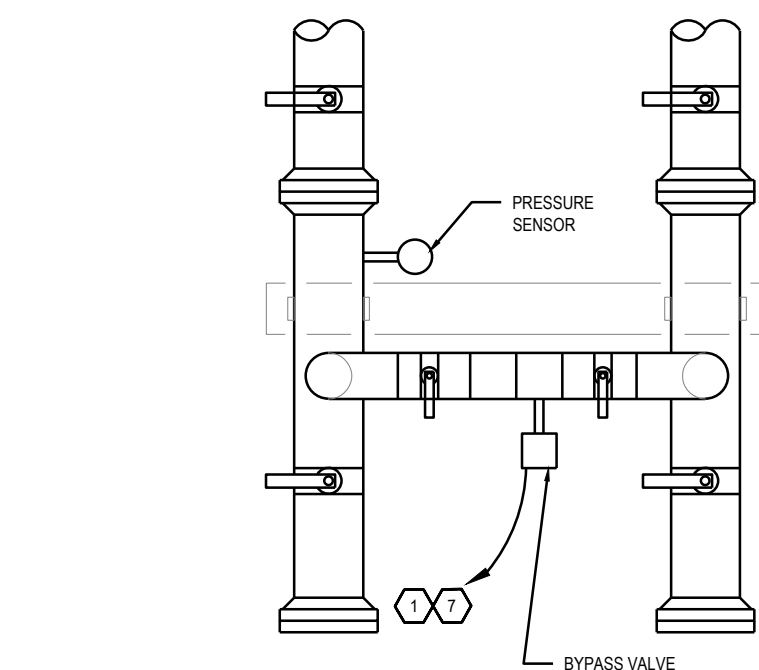


SCALE: 1/16" = 1'-0"

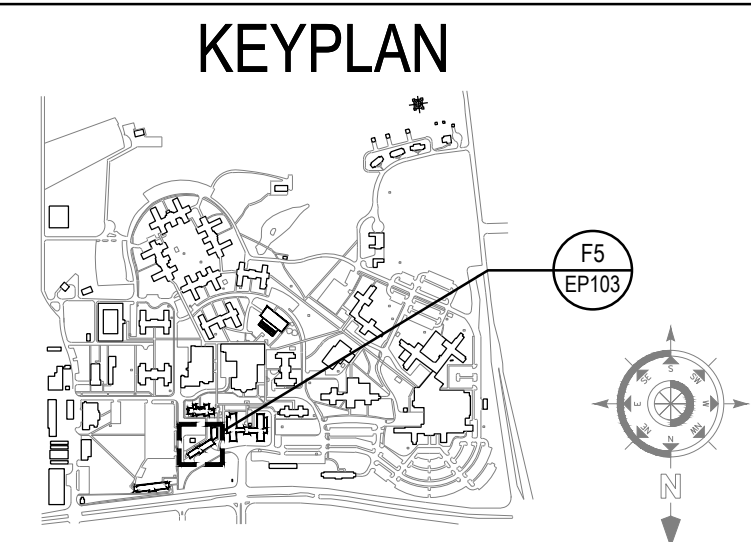
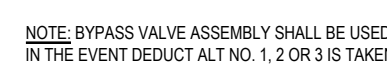


1 | 2

	5	6	7	8	9	10
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DEDUCT ALTERNATE NO. 1:
E8 CHILLED WATER PIPING BYPASS
SCALE: NONE



ARCHITECT/ENGINEER OF RECORD		STAMP	OFFICE OF CONSTRUCTION AND FACILITIES MANAGEMENT	DRAWING TITLE BUILDING 3 AND 5 FLOOR PLAN		PROJECT TITLE 100% CONSTRUCTION DOCUMENTS - ISSUED FOR CONSTRUCTION		PROJECT TITLE CHILLED WATER FROM B188 to B3 CONTRACT NO. - 36C25219D0022 TASK ORDER NO. - 36C25223N0287		PROJECT NUMBER 556-23-104	
A/E: HARRELL DESIGN GROUP PC 10720 SIKES PLACE, SUITE 100 CHARLOTTE, NC 28227 704-814-1346 POC: LEE HARRELL, PE, LEED AP, QCxP PRESIDENT HDG PROJECT: 22018			 U.S. Department of Veterans Affairs	APPROVED					LOCATION CAPTAIN JAMES A. LOVELL FEDERAL HEALTH CARE CENTER (FHCIC)		DRAWING NUMBER EP103
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[illegible]

CONDUIT COLOR CHART & NOTES

<u>SYSTEM (APPLICATION)</u>		<u>COLOR</u>
1.	FIRE ALARM	RED
2.	HVAC, CONTROL, BAS	BLUE
3.	FIBER OPTIC COMMUNICATIONS	ORANGE
4.	NORMAL POWER	GREEN
5.	EMERGENCY POWER, CRITICAL POWER	YELLOW
6.	SECURITY	BLACK
7.	MEDICAL SYSTEMS, NURSE CALL, MED GAS ALARMS	PURPLE

ALL CONDUITS ARE TO BE PRE-PAINTED AT THE FACTORY ACCORDING TO THIS LIST.

ALL CONDUITS SHALL BE UL APPROVED, FACTORY-POWDER COAT PAINTED, COLORED RIGID METAL CONDUIT (RMC) AND UL APPROVED, FACTORY POLYMER PAINTED, COLORED ELECTRICAL METAL TUBING (EMT)

GENERAL SHEET NOTES

A. VOLTAGE DROP SHALL NOT EXCEED VA DESIGN MANUAL GUIDELINES; 3% MAXIMUM ON BRANCH CIRCUITS AND 2% MAXIMUM ON FEEDERS.

B. COORDINATE EXACT CONDUIT ROUTING IN THE FIELD PRIOR TO ROUGH-IN AND INSTALLATION. CONDUIT TYPES AND INSTALLATION METHODS PER VA DESIGN STANDARDS.

C. PATCH ANY HOLES CREATED IN HORIZONTAL AND VERTICAL SURFACES TO MAINTAIN ANY EXISTING RATINGS.

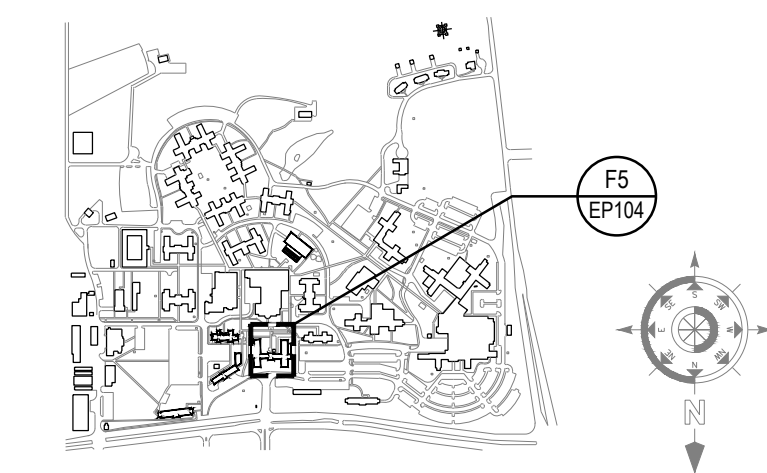
SHEET KEYNOTES

1. ELECTRICAL CONTRACTOR TO PROVIDE AND INSTALL 20A, 120V, SINGLE PHASE CIRCUIT BREAKER FOR A BYPASS CONTROL VALVE TO BUILDING 1 PANEL "VA-FREEZER 1" PANEL, SEE "ELECTRICAL" FOR FURTHER DETAILS.
2. ELECTRICAL CONTRACTOR TO ROUTE NEW ELECTRICAL CONDUIT ALONG THE CEILING, PRESERVE ALL EXISTING MOUNTED LIGHTING, AND FOLLOW ALL SYSTEM DETAILS DURING CONSTRUCTION. REINSTATE ALL WORK TO COORDINATE POWER SHUTDOWN WITH VA PRIOR TO DEMOLITION. COORDINATE EXACT ROUTING IN THE FIELD PRIOR TO ROUT-UP-IN.
3. ELECTRICAL CONTRACTOR TO ROUTE NEW ELECTRICAL CONDUIT ALONG CEILING STRUCTURE SIMILAR TO OTHER EXISTING CONDUIT IN THE FIELD PRIOR TO COORDINATE EXACT ROUTING IN THE FIELD PRIOR TO ROUT-UP-IN AND INSTALLATION.
4. RIGID, 1/2" WALL FOR A MINIMUM 1" CONDUIT. ROUTING ALONG WALL AS SHOWN.
5. ELECTRICAL CONTRACTOR TO COORDINATE, PRIOR TO ROUT-UP-IN AND INSTALLATION, ROUTING OF NEW ELECTRICAL CONDUIT TO BUILDING 3 MECHANICAL ROOM, IN MECHANICAL ROOM, PRIOR TO ROUTING DOWN TO LEVEL BELOW. CONTINUE ELECTRICAL CONDUIT ROUTING TO BUILDING 3 CHILLED WATER PIPING TO BUILDING 3 CRAWL SPACE.

WALL RATING LEGEND

————— — — — 2-HOUR RATED PARTITION

KEYPLAN



BUILDING 1 FLOOR PLAN

SCALE: 1/16" = 1'-0"

[illegible]

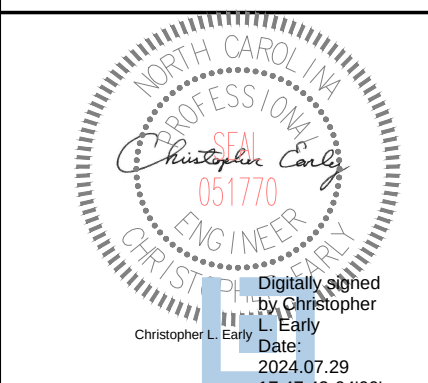
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PRESIDENT
HDG PROJECT: 22018



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	DRAWING TITLE
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BUILDING 1 FLOOR PLAN

APPROVED

PHASE	100% CONSTRUCTION DOCUMENTS - ISSUED FOR CONSTRUCTION
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FULLY SPRINKLERED

PROJECT TITLE	CHILLED WATER FROM B188 TO B3
CONTRACT NO. -	36C25219D0022
TASK ORDER NO. -	36C25223N0287

LOCATION	CAPTAIN JAMES A. LOVELL FEDERAL HEALTH CARE CENTER (FHCC)
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ISSUE DATE
JULY 26, 2024

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PROJECT NUMBER

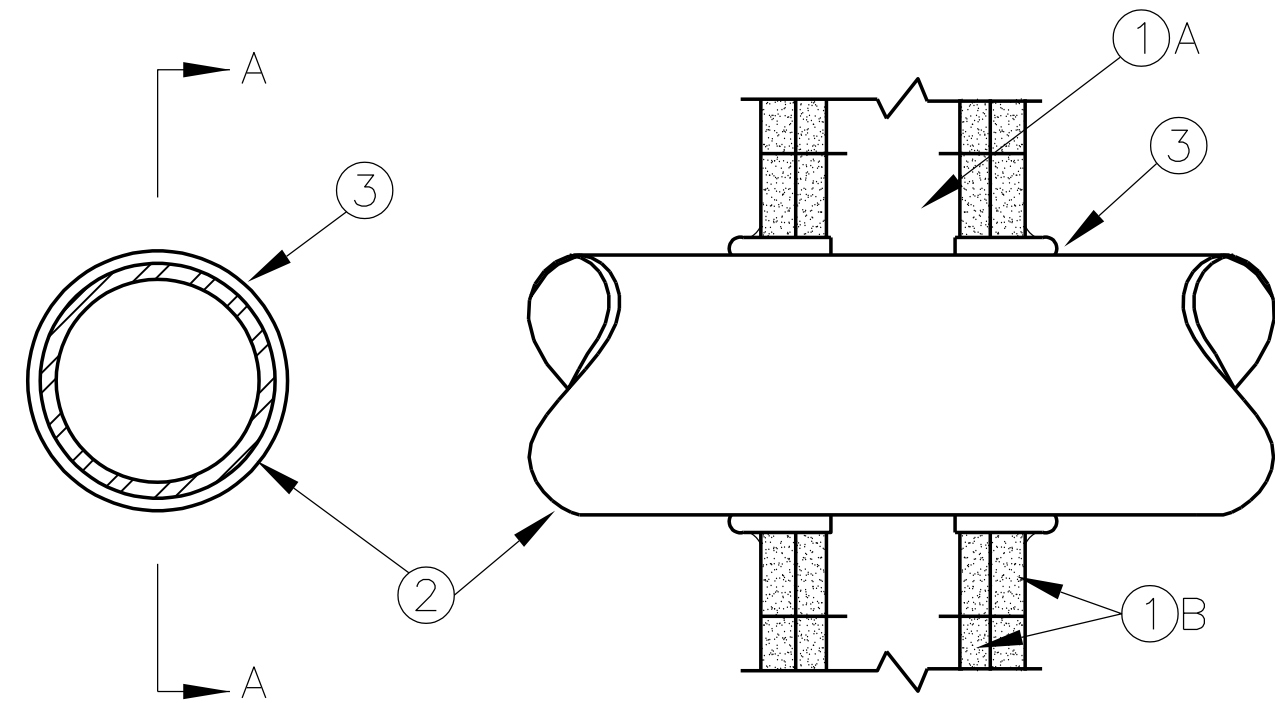
556-23-104
BUILDING NUMBER
188

	DRAWING NUMBER

EP104

SYSTEM NO. W-L-1001
(FORMERLY SYSTEM NO. 147)

F RATINGS - 1, 2, 3 AND 4 HR (SEE ITEMS 2 AND 3)
T RATINGS - 0, 1, 2, 3 AND 4 HR (SEE ITEM 3)
L RATING AT AMBIENT - LESS THAN 1 CFM PER SQ. FT.
L RATING AT 400°F - LESS THAN 1 CFM PER SQ. FT.



SECTION A-A

1. WALL ASSEMBLY – THE 1, 2, 3 OR 4 HR FIRE-RATED GYPSUM WALLBOARD/STUD WALL ASSEMBLY SHALL BE CONSTRUCTED OF THE MATERIALS AND IN THE MANNER DESCRIBED IN THE INDIVIDUAL U300 OR U400 SERIES WALL OR PARTITION DESIGNS IN THE UL FIRE RESISTANCE DIRECTORY AND SHALL INCLUDE THE FOLLOWING CONSTRUCTION FEATURES:

- A. STUDS - WALL FRAMING MAY CONSIST OF EITHER WOOD STUDS (MAX 2 HR FIRE RATED ASSEMBLIES) OR STEEL CHANNEL STUDS. WOOD STUDS TO CONSIST OF NOM 2 BY 4 IN. LUMBER SPACE 16 IN. OC WITH NOM 2 BY 4 IN. LUMBER END PLATES AND CROSS BRACES. STEEL STUDS TO BE MIN 3-5/8 IN. WIDE BY 1-3/8 IN. DEEP CHANNELS SPACED MAX 24 IN. OC.

- B. WALLBOARD, GYPSUM* - NOM 1/2 OR 5/8 IN. THICK, 4 FT. WIDE WITH SQUARE OR TAPERED EDGES. THE GYPSUM WALLBOARD TYPE, THICKNESS, NUMBER OF LAYERS, FASTENER TYPE AND SHEET ORIENTATION SHALL BE AS SPECIFIED IN THE INDIVIDUAL U300 OR U400 SERIES DESIGN IN THE UL FIRE RESISTANCE DIRECTORY. MAX DIAM OF OPENING IS 13-1/2 IN.

2. PIPE OR CONDUIT - NOM 12 IN. DIAM (OR SMALLER)
SCHEDULE 10 (OR HEAVIER) STEEL PIPE, NOM 12 IN. DIAM (OR
SMALLER) SERVICE WEIGHT (OR HEAVIER) CAST IRON SOIL
PIPE, NOM 12 IN. DIAM (OR SMALLER) CLASS 50 (OR HEAVIER)
DUCTILE IRON PRESSURE PIPE, NOM 6 IN. DIAM (OR SMALLER)

STEEL CONDUIT, NOM 4 IN. DIAM (OR SMALLER) STEEL ELECTRICAL METALLIC TUBING, NOM 6 IN. DIAM (OR SMALLER) TYPE L (OR HEAVIER) COPPER TUBING OR NOM 1 IN. DIAM (OR SMALLER) FLEXIBLE STEEL CONDUIT. WHEN COPPER PIPE IS USED, MAX F RATING OF FIRESTOP SYSTEM (ITEM 3) IS 2 H. STEEL PIPES OR CONDUITS LARGER THAN NOM 4 IN. DIAM MAY BE USED IF THEY ARE INSTALLED USING STEEL CHANNEL STUDS. A MAX OF ONE PIPE OR CONDUIT IS PERMITTED IN THE FIRESTOP SYSTEM. PIPE OR CONDUIT TO BE INSTALLED NEAR CENTER OF STUD CAVITY WIDTH AND TO BE RIGIDLY SUPPORTED ON BOTH SIDES OF WALL ASSEMBLY.

3. FILL, VOID OR CAVITY MATERIAL* - CAULK - CAULK FILL MATERIAL INSTALLED TO COMPLETELY FILL ANNULAR SPACE BETWEEN PIPE OR CONDUIT AND GYPSUM WALLBOARD AND WITH A MIN 1/4 IN. DIAM BEAD OF CAULK APPLIED TO PERIMETER OF PIPE OR CONDUIT AT ITS JOINTS FROM THE WALL ASSEMBLY. THE CAULK MUST BE INSTALLED SYMMETRICALLY ON BOTH SIDES OF WALL ASSEMBLY. THE HOURLY F RATING OF THE FIRESTOP SYSTEM IS DEPENDENT UPON THE HOURLY FIRE RATING OF THE WALL ASSEMBLY IN WHICH IT IS INSTALLED., AS SHOWN IN THE FOLLOWING TABLE. THE HOURLY F RATING OF THE FIRESTOP SYSTEM IS DEPENDENT UPON THE SIZE OF THE PIPE OR CONDUIT AND THE HOURLY FIRE RATING OF THE WALL ASSEMBLY IN WHICH IT IS INSTALLED, AS TABULATED BELOW:

MAX PIPE OR CONDUIT DIAM IN	ANNULAR SPACE IN.	F RATING, HR.	T RATING, HR
1	0 TO 3/16	1 OR 2	0+, 1 OR 2
1	1/4 TO 1/2	3 OR 4	3 OR 4
4	0 TO 1/4	1 OR 2	0
4	0 TO 1-1/2#	1 OR 2	0
6	1/4 TO 1/2	3 OR 4	0
12	3/16 TO 3/8	1 OR 2	0

+ WHEN COPPER PIPE IS USED, T RATING IS 0 HR.

0 TO 1-1/2 IN. ANNULAR SPACE APPLIES ONLY WHEN TYPE CP-25WB+ CAULK IS USED, AND ONLY WHEN THE MIN THICKNESS OF THE GYPSUM WALLBOARD IS 5/8 IN. FOR 1 HR RATED WALLS AND 1-1/4 IN. FOR 2 HR RATED WALLS.

MINNESOTA MINING & MFG. CO.—TYPECP-25 WB+

* BEARING THE UL CLASSIFICATION MARKING.

WL1001

(WALL PENETRATION DETAIL)

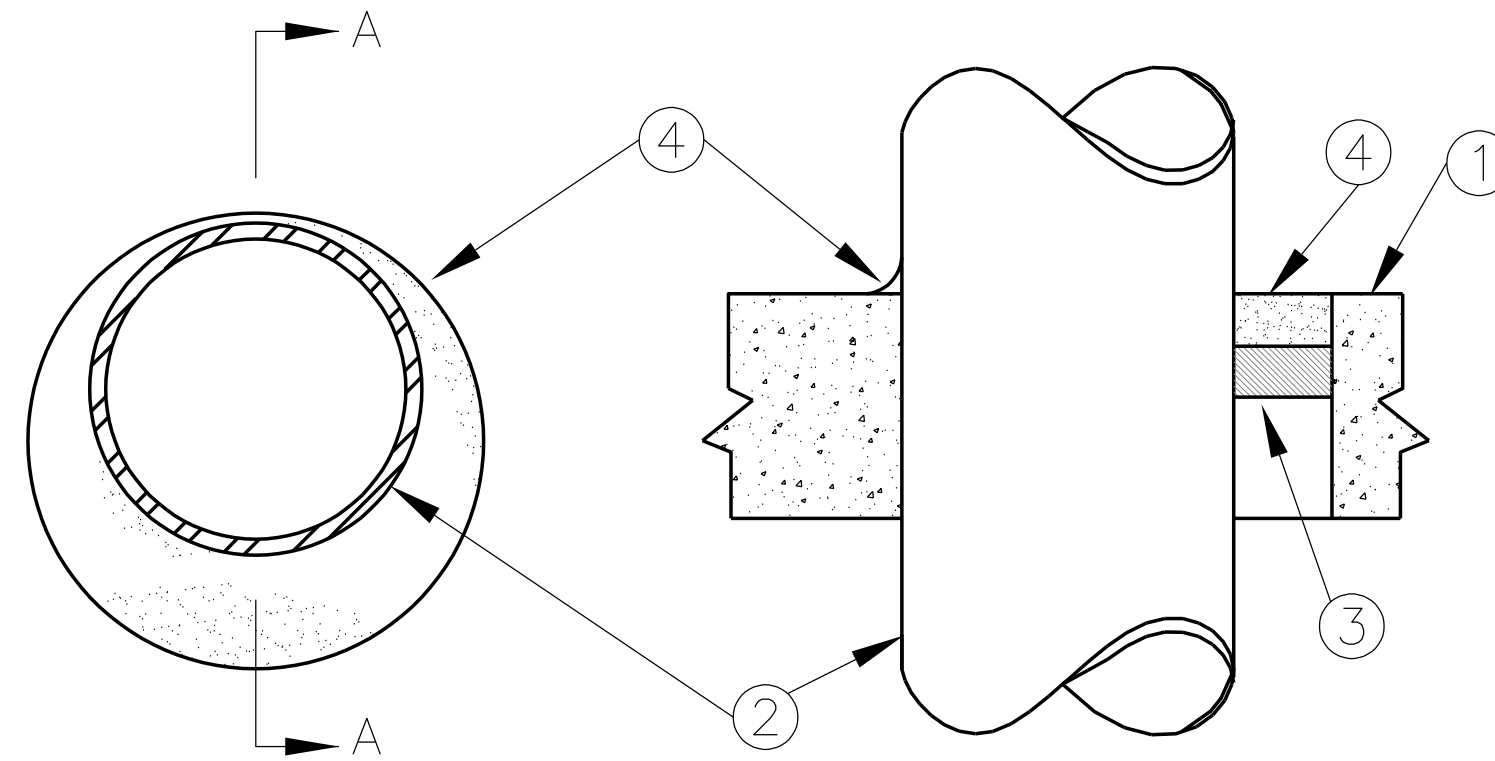
SCALE: NONE



SCALE: NONE

SYSTEM NO. C-AJ-1001
(FORMERLY SYSTEM NO. 49)

F RATINGS - 3 HR
T RATINGS - 0 HR



SECTION A-A

1. FLOOR OR WALL ASSEMBLY - MIN 4-1/2 IN. THICK
LIGHTWEIGHT OR NORMAL WEIGHT (100-150 PCF) CONCRETE.
WALL MAY ALSO BE CONSTRUCTED OF ANY UL CLASSIFIED
CONCRETE BLOCKS*. MAX DIAM OF CIRCULAR THROUGH
OPENING IS 22-1/2 IN.

SEE CONCRETE BLOCKS (CAZT) CATEGORY IN THE FIRE RESISTANCE DIRECTORY FOR NAMES OF MANUFACTURERS.

- 1A. STEEL SLEEVE - (OPTIONAL, NOT SHOWN) - NOM 12 IN. DIAM (OR SMALLER) SCHEDULE 40 (OR HEAVIER) STEEL PIPE SLEEVE CAST INTO CONCRETE FLOOR OR WALL. SLEEVE TO BE FLUSH WITH OR PROJECT MAX 2 IN. FROM TOP SURFACE OF FLOOR OR FROM BOTH SURFACES OF WALL.

2. PIPE OR CONDUIT - NOM 20 IN. DIAM (OR SMALLER)
SCHEDULE 10 (OR HEAVIER) STEEL PIPE, NOM 6 IN. DIAM (OR
SMALLER) RIGID CONDUIT OR TYPE L (OR HEAVIER) PIPE
COPPER TUBE, NOM 4 IN. DIAM (OR SMALLER) CAST IRON PIPE
OR STEEL EMT. MAX ONE PIPE OR CONDUIT PER THROUGH
OPENING. MAX ANNUAL SPACE BETWEEN PIPE OR CONDUIT
AND EDGE OF THROUGH OPENING NOT TO EXCEED 2-1/2 IN. MIN
ANNUAL SPACE BETWEEN PIPE OR CONDUIT AND EDGE OF
THROUGH OPENING IS ZERO IN. (POINT CONTACT). PIPE OR
CONDUIT TO BE RIGIDLY SUPPORTED ON BOTH SIDES OF FLOOR
OR WALL ASSEMBLY.

3. PACKING MATERIAL - POLYETHYLENE BACKER ROD OR NOM 1 IN. THICKNESS OF TIGHTLY-PACKED CERAMIC (ALUMINA SILICA) FIBER BLANKET, MINERAL WOOL BATT OR GLASS FIBER INSULATION MATERIAL USED AS A PERMANENT FORM. PACKING MATERIAL TO BE RECESSED FROM TOP SURFACE OF FLOOR OR FROM BOTH SURFACES OF SOLID CONCRETE OR CONCRETE BLOCK WALL AS REQUIRED TO ACCOMMODATE THE REQUIRED THICKNESS OF CAULK FILL MATERIAL (ITEM 4). AS AN ALTERNATE WHEN MAX PIPE SIZE IS 10 IN. DIAM AND WHEN MAX ANNULAR SPACE IS 1 IN. , A MIN 1 IN. THICKNESS OF TIGHTLY-PACKED CERAMIC FIBER BLANKET OF MINERAL WOOL BATT PACKING MATERIAL MAY BE RECESSED MIN 1/2 IN. FROM BOTTOM SURFACE OF FLOOR OR FROM EITHER SIDE OF SOLID CONCRETE WALL.

4. FILL, VOID OR CAVITY MATERIALS* - CAULK - APPLIED TO FILL THE ANNULAR SPACE TO THE MIN THICKNESS SHOWN IN THE FOLLOWING TABLE:

MAX PIPE DIAM IN	MAX ANNULAR SPACE IN.	PACKING MATERIAL TYPE (A)	MIN. CAULK THKNS IN
10	1	BR, CF, GF OR MW	1/2 (B)
10	1	CF OR MW	1/2 (C)
20	2-1/2	BR, CF, GF OR MW	1 (B)

(A) BR= POLYETHYLENE BACKER ROD.

CF= CERAMIC FIBER BLANKET.

GF= GLASS FIBER INSULATION.

MW= MINERAL WOOL BATT

- (B) CAULK INSTALLED FLUSH WITH TOP SURFACE OF FLOOR OR BOTH SURFACES OF WALL

- (C) CAULK INSTALLED FLUSH WITH BOTTOM SURFACE OF FLOOR OR ONE SURFACE OF SOLID (NON-CONCRETE BLOCK) WALL.

MINNESOTA MINING & MFG. CO.— TYPE CP-25 N/S

* BEARING THE UL CLASSIFICATION MARKING.

CAJ1001 (FLOOR OR

WALL PENETRATION DETAIL)

F7

SCALE: NONE

1	100% CD	07/26/2024
Revision Number	Revisions:	Date:

ARCHITECT/ENGINEER OF RECORD A/E: HARRELL DESIGN GROUP PC 10720 SIKES PLACE, SUITE 100 CHARLOTTE, NC 28227 704-814-1346 POC: LEE HARRELL, PE, LEED AP, QCxP PRESIDENT HDG PROJECT: 22018  HARRELL DESIGN GROUP, P.C. COPYRIGHT © 2024		STAMP 	OFFICE OF CONSTRUCTION AND FACILITIES MANAGEMENT  U.S. Department of Veterans Affairs	DRAWING TITLE DETAILS APPROVED	PHASE 100% CONSTRUCTION DOCUMENTS - ISSUED FOR CONSTRUCTION	PROJECT TITLE CHILLED WATER FROM B188 TO B3 CONTRACT NO. - 36C25219D0022 TASK ORDER NO. - 36C25223N0287	PROJECT NUMBER 556-23-104 BUILDING NUMBER -----
					LOCATION CAPTAIN JAMES A. LOVELL FEDERAL HEALTH CARE CENTER (FHCIC)	DRAWING NUMBER EE501	
					ISSUE DATE JULY 26, 2024	CHECKED LEH DRAWN CLE	